Qin Jiushao, Shuxue Jiuzhang, fascicles 1-9

English Translation

Modern LaTeX Editions of Public-Domain Mathematics Manuscripts

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数书九章

Mathematical Treatise in Nine Sections

Fascicle I – The Dayan Category

Bilingual Edition / 双语版

By **Qin Jiushao** (秦九韶)
c. 1202–1261 CE
Presented in the 7th year of Chunyou (淳祐七年), 1247 CE

Left side: Original Classical Chinese text
Right side: English translation with modern mathematical commentary

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1 Introduction / 导言

数书九章序

数书九章者，南宋秦九韶之所著也。秦氏字道古，鲁郡人。生于宋季，博学多能，尤精算学。此书成于淳祐七年（公元1247年），凡九卷，每卷

每问有问、答、术、草四目，体例谨严。大衍之类，尤推精妙，盖中国剩余定理之滥觞也。

Preface to the Shuxue Jiuzhang

The Shuxue Jiuzhang (Mathematical Treatise in Nine Sections), is the magnum opus of the Song dynasty mathematician Qin Jiushao (秦九韶, c. 1202–1261). Completed in 1247 CE during the Chunyou era (淳祐七年), it stands as one of the greatest achievements in the history of Chinese mathematics and indeed of world mathematics.

The work is organized into nine categories (类), each containing nine problems (问), for a total of eighty-one problems. Each problem follows a rigorous four-part structure: **Wen** (问) – the problem statement; **Da** (答) – the numerical answer; **Shu** (术) – the general method; **Cao** (草) – the detailed working.

1.1 The Nine Categories / 九类名目

九类之目：

1. **大衍** – 求一之术
2. **天时** – 天文历法
3. **田域** – 田地测量
4. **测望** – 勾股重差
5. **赋役** – 租税劳役
6. **钱谷** – 钱粮贸易
7. **营建** – 建筑工事
8. **军旅** – 兵家之事
9. **市易** – 市场交易

The nine categories are:

1. **Dayan** (大衍) – “Great Derivation”: Indeterminate analysis and the Chinese Remainder Theorem
2. **Tianshi** (天时) – Astronomical phenomena and calendar calculations
3. **Tianyu** (田域) – Land measurement and area calculation
4. **Cewang** (测望) – Surveying and observation
5. **Fuyi** (赋役) – Taxation and labor service
6. **Qian’gu** (钱谷) – Currency and grain commerce
7. **Yingjian** (营建) – Construction and engineering
8. **Junlv** (军旅) – Military affairs
9. **Shiyi** (市易) – Market transactions

2 Preface / 序

周教六艺，数实成之。学士大夫所从事尚矣。其用本太虚生一，而周流无穷。大则可以通神明、顺性命，小则可以经世务、类万物，讵容

Of the Six Arts taught by the Zhou, mathematics completes them. It has long been pursued by scholars and gentlemen. Its use originates from the One born of the Great Void, flowing endlessly. In the large, it can penetrate the divine and accord with nature; in the small, it can govern worldly affairs and classify all things. How could it be viewed as shallow or trivial?

若昔推策以迎日，定律而和气。髀距濬川，土圭度晷。天地之大，囿焉而不能外，况其间总总者乎？爰自河图洛书，闿发幽秘，八卦九畴

From ancient times, reckoning rods were used to predict the sun’ s course, and laws were established to harmonize the qi. The gnomon measured the rivers, the jade tablet measured the shadows. The greatness of heaven and earth is encompassed by it and cannot escape, how much more the multitude of things within?

极而至于大衍、皇极之用，而人事之变无不该，鬼神之情莫能隐矣。圣人神之言而遗其粗，常人昧之由而莫之觉。要其归，则数与道非二

Reaching to the applications of the Dayan and the Supreme Ultimate, there is no human affair that it does not encompass, no spirit’ s nature that it cannot reveal. The sages found its spirit and left its roughness; ordinary people are blind to its path. In essence, number and the Way are not two different roots.

独大衍法不载九章，未有能推之者。历家演法颇用之，以为方程者，误也。

Only the Dayan method is not recorded in the Nine Chapters, and none have been able to develop it. Calendarists have used it extensively in their computational methods, but those who take it as a system of equations are mistaken.

九韶愚陋，不闲于艺。然早岁侍亲中都，因得访习于太史，又尝从隐君子受数学。时际兵难，历岁遥塞，不自意全于矢石之间。更险离忧

I, Jiushao, am foolish and unrefined, not skilled in the arts. Yet in my youth I served my parents in the capital, and so had the opportunity to study with the Grand Astrologer; I also received instruction in mathematics from a gentleman recluse. When the time of military disaster came, I did not expect to survive between arrows and stones. Passing through danger and sorrow, ten years elapsed; my heart withered and my spirit fell. Truly I came to know that all things have number.

时淳祐七年九月，鲁郡秦九韶叙。

Written in the ninth month of the seventh year of Chunyou, by Qin Jiushao of Lu Commandery.

3 The Dayan General Method / 大衍总数术

3.1 Editor’s Commentary / 按语

按大衍术，以各分数之奇零求各分数之总数。大而天行，小而物数，皆可御之。其法有求元、求定、求衍、求奇、求乘、求用之目。大约

Editor’s note: The Dayan method takes the remainders (odd fragments) of each fractional divisor to find the total number. Whether for large celestial motions or small material quantities, it can be applied. The method has the following steps: finding the original numbers, finding the definitive numbers, finding the derivative numbers, finding the odd numbers, finding the multipliers, and finding the use numbers.

3.2 The Four Types of Numbers / 四种问数

大衍总数术曰：置诸问数，类名有四。

1. **元数** – 谓尾位见单零者
2. **收数** – 谓尾位见分厘者
3. **通数** – 谓诸数各有分子分母者
4. **复数** – 谓尾位见十或百及千以上者

The Dayan General Method states: Place the given numbers; there are four types.

1. **Original numbers** – Numbers ending in single units
2. **Collected numbers** – Numbers ending in fractional parts
3. **Common numbers** – Numbers each having numerator and denominator
4. **Repeated numbers** – Numbers ending in tens, hundreds, or thousands

3.3 Finding the Definitive Numbers / 求定数

元数者：先以两两连环求等，约奇弗约偶。或约得五而彼有十，乃约偶而弗约奇。或元数俱偶，约毕可存一位见偶。

For original numbers: First, pairwise in chain sequence find the GCD, reduce the odd one, not the even one. If reducing yields 5 while the other has 10, then reduce the even one, not the odd one. If all original numbers are even, after reduction one may keep one even number.

复数者：问数尾位见十以上者，以诸数求总等，存一位约众位，始得元数。两两连环求等，约奇弗约偶，复乘偶；或约偶弗约奇，弗乘奇。

For repeated numbers: When the given numbers end in tens or above, find the total GCD of all numbers, keep one position and reduce the rest. Then pairwise in chain sequence find GCDs; reduce the odd, not the even, and multiply back the even; or reduce the even, not the odd, and do not multiply back the odd.

3.4 Computing the Derivative Numbers / 求衍数

求定数勿使两位见偶，勿使见一太多。见一多则借用繁，不欲借则任得一。
以定相乘为衍母，以各定约衍母，得各衍数。
次以各定母满去衍数，各余名曰**奇数**。
以**大衍求一术**，得各**乘率**。
以乘衍数，各得**用数**。

Finding definitive numbers: Do not let two positions be even; do not let there be too many ones. Too many ones makes borrowing complicated.
Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain each **derivative number**.
Next, reduce each derivative number by its corresponding definitive number; the remainders are called the **odd numbers**.
Using the **Dayan Method for Finding Unity**, obtain each **multiplier**.
Multiplying each multiplier by its derivative number gives each **use number**.

3.5 The Dayan Method for Finding Unity / 大衍求一术

大衍求一术云：置奇右上，定居右下。立天元一于左上。先以右行上下两位，以少除多，所得商数，乃递互乘归左行。须使右上末后奇一

The Dayan Method for Finding Unity says: Place the odd number at the upper right, the definitive number at the lower right. Place the celestial element unity (1) at the upper left. First, divide the larger by the smaller of the two numbers in the right column; the quotient obtained is then used to multiply alternately into the left column. Continue until the upper right finally becomes one. Then check what is obtained at the upper left—this is the multiplier.

In modern mathematical terms: Given coprime integers a (the “odd number”) and m (the “definitive number"), find k such that:

$$k \cdot a \equiv 1 \pmod{m}$$

This is the **modular multiplicative inverse** of a modulo m, computed by the **extended Euclidean algorithm**.

4 Problem 1: Shigua Fawei / 蓍卦发微

蓍卦发微 – Divination by Yarrow Stalks

问曰:

《易》曰：大衍之数五十，其用四十有九。又曰：分而为二以象两，挂一以象三，揲之以四以象四时。三变而成爻，十有八变而成卦。欲知所衍

Question:

The Book of Changes says: "The number of the Great Derivation is fifty; of these, forty-nine are used." It also says: "Divide into two to symbolize the Two [heaven and earth]; hang one to symbolize the Three [heaven, earth, and man]; count off by fours to symbolize the Four Seasons." Three changes produce a line; eighteen changes produce a hexagram. The problem asks to find the computational method of this derivation and the numbers involved.
This problem applies the Dayan method to the classical yarrow-stalk divination of the I Ching.

答曰:

衍母十二，衍法三。
一元衍数二十四，二元衍数一十二，三元衍数八，四元衍数六。以上四位衍数，计五十一。
一揲用数一十二，二揲用数二十四，三揲用数四，四揲用数九。以上四位用数，计四十九。

Answer:

The derivative mother is 12, the derivative divisor is 3.
The first derivative number is 24, the second is 12, the third is 8, the fourth is 6. The sum of these four derivative numbers is 51.
The first manipulation use-number is 12, the second is 24, the third is 4, the fourth is 9. The sum of these four use-numbers is 49.[a]

[a] Trans. note: The sum 49 corresponds to "the number of the Great Derivation is fifty, of which forty-nine are used" from the Yijing.

术曰:

置诸元数，两两连环求等，约奇弗约偶。遍约毕，乃变元数皆曰定母，列右行。各立天元一为子，列左行。以诸定母互乘左行之子，各得名曰衍

Method:

Place the given original numbers; pairwise in a chain sequence find the GCD, reducing the odd and not the even. When reduction is complete everywhere, the original numbers become the definitive numbers; place them in the right column. Place the celestial element unity (1) as the seed in the left column. Cross-multiply the definitive numbers with the seeds in the left column; each result is called a derivative number.

次以各定母满去衍数，各余名曰奇数。以奇数与定母用大衍术求一，得各乘率。以乘衍数，各得用数。

Next, reduce each derivative number by its corresponding definitive number; each remainder is called the odd number. Using the Dayan Method for Finding Unity on the odd number and the definitive number, obtain each multiplier. Multiply each derivative number by its multiplier to obtain each use number.

5 Problem 2: Guli Huiji / 古历会积

古历会积 – Ancient Calendar Accumulation

问曰：

问古历会积，欲求积年。以历法积元为问，历过无考，但据近世演纪之法，推而上之。

Question:

This problem concerns the calculation of the accumulated years (积年) in ancient calendar systems. Given certain astronomical periodicities, one must find a number that simultaneously satisfies multiple congruence conditions. The epochs have passed without record; based on the method of computing epochs used in recent times, one extrapolates backward.

This was one of the primary applications of the Dayan method in traditional Chinese astronomy—finding a common epoch that aligns multiple celestial cycles.

术曰：

以大衍求之。置元历相关诸周期，连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以各定约衍母得衍数。各满定母去之，余为奇数。大衍求

Method:

Solve by the Dayan method. Place the relevant periods of the original calendar; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder is the sought accumulated years.

6 Problem 3: Tuiji Tugong / 推计土功

推计土功 – Estimating Earthworks

问曰：

四县（甲、乙、丙、丁）共同筑堤，根据各县的物力分配筑堤任务。

- 甲县物力：138,600贯
- 乙县物力：146,300贯
- 丙县物力：192,500贯
- 丁县物力：184,800贯

筑堤标准：每770贯物力分配一名劳动力，每人每天筑堤60平方尺。
进度情况：甲县和乙县已完成任务，丙县剩余51丈，丁县剩余18丈。

Question:

Four counties (A, B, C, D) jointly build a dike; the task is distributed according to each county’s resources.

- County A resources: 138,600 guan
- County B resources: 146,300 guan
- County C resources: 192,500 guan
- County D resources: 184,800 guan

Dike standard: for every 770 guan of resources, assign one laborer; each laborer builds 60 square feet of dike per day.
Progress: Counties A and B have completed their tasks; County C has 51 zhang remaining; County D has 18 zhang remaining.
This is a classic indeterminate problem solved by the Dayan method.

答曰：

堤的总长度为19里235步5尺。
各县所筑长度：

- 甲县：1,260丈
- 乙县：1,768步5尺6寸
- 丙县：4里328步5尺6寸
- 丁县：与前三县相应

Answer:

The total length of the dike is 19 li 235 bu 5 chi.
Each county’s contribution:

- County A: 1,260 zhang
- County B: 1,768 bu 5 chi 6 cun
- County C: 4 li 328 bu 5 chi 6 cun
- County D: corresponding to the first three

术曰：

以大衍求之。置各县物力，连环求等，约奇弗约偶，得定母。诸定相乘为衍母，以各定约衍母得衍数。各满定母去之，余为奇数。以大衍求一入之

Method:

Solve by the Dayan method. Place the resources of each county; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each county’s remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder is the sought answer.

7 Problem 4: Tuiku E Qian / 推库额钱

推库额钱 – Calculating Treasury Funds

问曰：

问有外邑七库，日纳息足钱适等，递年成贯整纳。近缘见钱希少，听各库照当处市陌准解旧会。其甲库有零钱一十文，丁、庚二库各零四文，戊

甲库所在市陌一十二文，递减一文至庚库而止。欲求诸库日息原纳足钱、展省及今纳旧会，并大小月分各几何？（七库依次为甲、乙、丙、丁、戊、己、庚，市陌分别为12, 11, 10, 9, 8, 7, 6文。甲库余10文，乙丙无余，丁余4文，戊余6文，己无余，庚余4文。）

Question:

There are seven treasuries in outlying districts; their daily interest payments in full cash are equal, rounded to whole guan annually. Recently, since circulating cash has become scarce, each treasury is permitted to convert its payment to old notes according to the local exchange rate (市陌).[a]

[a] Trans. note: The “market fraction” (市陌) was a local exchange rate between paper currency and full-value cash, varying by location.

The exchange rate at treasury A is 12 wen; it decreases by 1 wen for each subsequent treasury down to G. Find: the original daily interest in full cash for each treasury, the adjusted amount, and the current payment in old notes, for both long and short months.

(The seven treasuries are A–G with exchange rates 12, 11, 10, 9, 8, 7, 6 wen respectively. Treasury A has remainder 10 wen, B and C have no remainder, D has remainder 4 wen, E has remainder 6 wen, F has no remainder, G has remainder 4 wen.)

术曰：

以大衍求之。置甲库市陌，以递减数减之，各得诸库原陌。连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以定约衍母得衍数。各满定母去

Method:

Solve by the Dayan method. Place the exchange rate of treasury A; subtracting the decrement each time gives each treasury’s rate. Pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers.

8 Problem 5: Fentiao Tuiyuan / 分粜推原

分粜推原 – Grain Distribution

问曰:
有兄弟三人，各将收获之米送往不同的地方:
- 甲将米送往官场，余3斗2升
- 乙将米送往安吉乡，余7斗
- 丙将米送往平江，余3斗

官斛每石八斗三升，安吉斛每石一石一斗，平江斛每石一石三斗五升。欲求三人共收获米数以及各自分米数几何?

Question:
Three brothers each send their harvested rice to different places:
- A sends rice to the government office, remainder 3 dou 2 sheng
- B sends rice to Anji township, remainder 7 dou
- C sends rice to Pingjiang, remainder 3 dou

The official measure holds 8 dou 3 sheng per shi; the Anji measure holds 1 shi 1 dou; the Pingjiang measure holds 1 shi 3 dou 5 sheng. Find: the total amount of rice harvested by the three brothers, and the amount each brother received.

答曰:
三人共收获米738石。
各分米数:
- 甲: 296石，余3斗2升
- 乙: 223石，余7斗
- 丙: 182石，余3斗

Answer:
The three brothers harvested 738 shi of rice in total.
Each brother' s share:
- A: 296 shi, remainder 3 dou 2 sheng
- B: 223 shi, remainder 7 dou
- C: 182 shi, remainder 3 dou

术曰:
以大衍求之。置各斛率（单位: 升）: 官斛83升，安吉斛110升，平江斛135升。连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以各定约

Method:
Solve by the Dayan method. Place each measure's rate (in sheng): the official measure 83 sheng, the Anji measure 110 sheng, the Pingjiang measure 135 sheng. Pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers.

9 Problem 6: Chengxing Jidi / 程行计地

程行计地 – Journey Distance Calculation

问曰：

军师派遣三名急足（甲、乙、丙）分别前往都下报捷。

- 甲每天行300里
- 乙每天行240里
- 丙每天行180里

甲提前数日申未到，乙后数日未正到，丙今日辰末到。欲知从军前到都下的距离以及三人各自行走的天数几何?

Question:

A military commander sends three express messengers (A, B, C) to the capital to report victory.

- A travels 300 li per day
- B travels 240 li per day
- C travels 180 li per day

A arrives early (before shen hour); B arrives late (at wei hour); C arrives today (at chen hour). Find the distance from the army’s position to the capital, and the number of days each traveled.

答曰：

从军前到都下的距离为3,300里。

- 甲行11天
- 乙行13天4时
- 丙行18天2时

Answer:

The distance from the army to the capital is 3,300 li.

- A traveled 11 days
- B traveled 13 days 4 hours
- C traveled 18 days 2 hours

术曰：

以大衍求之。置三人日行里数，连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以各定约衍母得衍数。各满定母去之，余为奇数。以大衍求一

Method:

Solve by the Dayan method. Place the three persons' daily travel distances; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder is the sought distance.

10 Problem 7: Chengxiang Xiangji / 程行相及

程行相及 – Catching Up on a Journey

问曰:
甲、乙、丙三人行程相关。甲先行若干日，乙后行，丙又后行。各以不同速度行进。求某处三人相遇之时间及各地距离。

Question:
Three travelers A, B, and C are related in their journeys. A departs first, B departs later, and C departs still later. Each travels at a different speed. Find: the time when all three meet at a certain place, and the various distances.
This problem involves multiple travelers with different speeds and starting times, requiring the Dayan method to solve a system of congruences.

术曰:
以大衍求之。置各人行程速度及时间差，连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以各定约衍母得衍数。各满定母去之，余为奇数。

Method:
Solve by the Dayan method. Place each person's travel speed and time difference; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder is the sought answer.

11 Problem 8: Jichi Xunyuan / 积尺寻源

积尺寻源 – Finding Dimensions from Volume

问曰:

问欲砌基一段，见管大小方砖六门城砖四色。令匠取便，或平或侧，只用一色砖砌，须要适足。匠以砖量地计料，称：

- 大方：广多六寸，深少六寸
- 小方：广多二寸，深少三寸
- 城砖：长广多三寸，深少一寸
- 六门砖：长广多三寸，深多一寸

Question:

One wishes to build a foundation; available are large and small square bricks, six-portal bricks, and city-wall bricks—four types in all. The mason is instructed to use whichever is convenient, laying them flat or on edge, using only one type of brick, requiring them to fit exactly. The mason measured the ground and estimated materials, reporting:

- Large square: width excess 6 cun, depth deficit 6 cun
- Small square: width excess 2 cun, depth deficit 3 cun
- City brick: length and width excess 3 cun, depth deficit 1 cun
- Six-portal: length and width excess 3 cun, depth excess 1 cun

皆不合匝，未免修破砖料裨补。其四色砖尺寸：

- 大方：方一尺三寸
- 小方：方一尺一寸
- 城砖：长一尺二寸，阔六寸，厚二寸五分
- 六门砖：长一尺，阔五寸，厚二寸

欲知基深广几何?

All fail to fit exactly, so broken bricks must be used for patching. The four types of brick have dimensions:

- Large square: 1 chi 3 cun square
- Small square: 1 chi 1 cun square
- City brick: 1 chi 2 cun long, 6 cun wide, 2 cun 5 fen thick
- Six-portal: 1 chi long, 5 cun wide, 2 cun thick

Find: the depth and width of the foundation.
This is a classic problem of finding a common multiple with given remainders.

答曰:

深三丈七尺一寸，广一丈二尺三寸。

Answer:

Depth: 3 zhang 7 chi 1 cun; width: 1 zhang 2 chi 3 cun.

术曰:

以大衍求之。置四色砖尺寸及所余尺寸，连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以各定约衍母得衍数。各满定母去之，余为奇数。

Method:

Solve by the Dayan method. Place the four types of brick dimensions and the excess/deficit dimensions; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder gives the depth and width of the foundation.

12 Problem 9: Yumi Tuishu / 余米推数

余米推数 – Calculating Rice Remainders

问曰：

有米若干，以不同容量的容器量之，各余若干。欲求米之总数。

Question:

There is a quantity of rice; when measured with containers of different capacities, there are different remainders each time. Find the total amount of rice.

This is the classic “unknown quantity” problem. In modern terms: find N such that $N \equiv r_i \pmod{m_i}$ for given moduli m_i and remainders r_i.

术曰：

以大衍求之。置各容器容量为问数，连环求等，约奇弗约偶，得定母。诸定相乘为衍母，以各定约衍母得衍数。各满定母去之，余为奇数。大衍

Method:

Solve with the Dayan Method. Place the capacities of each container as the given numbers; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder is the sought amount of rice.

This problem is structurally identical to the famous “problem of the unknown quantity” from the Sunzi Suanjing (孙子算经), which asks: “We have things of which we do not know the number; if we count them by threes, the remainder is 2; if by fives, the remainder is 3; if by sevens, the remainder is 2. How many things are there?” Qin Jiushao’s general method solves all problems of this type systematically.

13 Mathematical Analysis / 数理分析

13.1 The Chinese Remainder Theorem / 中国剩余定理

The Dayan method solves systems of simultaneous linear congruences. Given a set of pairwise coprime moduli $m_1, m_2, \ldots, m_n$ and remainders $r_1, r_2, \ldots, r_n$, find N such that:

$$\begin{aligned} N &\equiv r_1 \pmod{m_1} \\ N &\equiv r_2 \pmod{m_2} \\ &\vdots \\ N &\equiv r_n \pmod{m_n} \end{aligned}$$

13.2 Qin Jiushao's Algorithm / 秦九韶算法

求定数 – 连环求等，约奇弗约偶
求衍母 – 诸定相乘
求衍数 – 以定约衍母
求奇数 – 各满定母去之
求乘率 – 大衍求一术
求用数 – 乘率乘衍数
求总数 – 余数乘用数并之

Step 1: Definitive Numbers. Given the original numbers (元数), reduce them pairwise using the GCD, keeping them coprime. The rule "reduce the odd, not the even" ensures unique factorization.
Step 2: Derivative Mother. Compute $M = m_1 \times m_2 \times \cdots \times m_n$.
Step 3: Derivative Numbers. Compute $M_i = M/m_i$ for each i.
Step 4: Odd Numbers. Compute $g_i = M_i \bmod m_i$.
Step 5: Finding Unity. For each i, find k_i such that $k_i \cdot g_i \equiv 1 \pmod{m_i}$. This is the modular multiplicative inverse, computed by the extended Euclidean algorithm.
Step 6: Use Numbers. Compute $u_i = k_i \cdot M_i$.
Step 7: Final Computation. The solution is: $N = \sum_{i=1}^{n} r_i \cdot u_i \pmod{M}$

13.3 Historical Significance / 历史意义

秦氏大衍术，超前欧洲数百年。其求一之术，即欧几里得之扩展算法也。后世郭守敬用于天文，李冶用于几何，西传欧洲，遂成所谓“中国剩余…
高斯于《算术探究》（1801年）首次在欧洲完整论述此定理，称之为“求一数，以各定数除之，各余已知数”之问题。今此定理为数论之基本…

Qin Jiushao's method predated similar European work by several centuries. The algorithm for finding modular inverses (大衍求一术) is essentially the extended Euclidean algorithm. The method was later applied by Guo Shoujing (郭守敬) to astronomy, by Li Ye (李冶) to geometry, and was transmitted to Europe where it became known as the "Chinese Remainder Theorem."
Gauss gave the first complete treatment in Europe in his Disquisitiones Arithmeticae (1801), calling it the "problem of finding a number that, when divided by given numbers, leaves given remainders." The theorem is now recognized as one of the fundamental results in number theory.

About This Edition / 关于本版

This bilingual edition presents Fascicle I of Qin Jiushao’s Shuxue Jiuzhang (Mathematical Treatise in Nine Sections), with the original classical Chinese text in the **left column** and an English translation with modern mathematical commentary in the **right column**.

Technical Notes / 技术说明

- Typeset with XeTeX using parallel `minipage` environments
- Font: Noto Sans CJK SC (supports both Chinese and Latin characters)
- The traditional structure of 问 (question), 答 (answer), 术 (method), and 草 (working) has been preserved
- Translator’s footnotes explain technical terminology and historical context

Translation Notes / 翻译说明

- “Dayan” (大衍) — translated as “Great Derivation,” preserving the I Ching reference
- “Qiuyi” (求一) — translated as “Finding Unity” (finding the modular inverse)
- “Yanshu” (衍数) — translated as “derivative numbers”
- “Yanmu” (衍母) — translated as “derivative mother”
- “Chengshu” (乘率) — translated as “multiplier”
- “Yongshu” (用数) — translated as “use number”

Further Reading / 参考文献

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- Qian, Baocong. Zhongguo Shuxue Shi (History of Chinese Mathematics). Science Press, 1964.
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Mathematical Treatise in Nine Sections

Shuxue Jiuzhang – 數學九章

Fascicle 2 / 卷二

天時類

Tian Shi Lei – Heavenly Patterns

Calendar, Astronomy and Weather / 曆法、天文與氣象

Bilingual Edition / 雙語版

Classical Chinese • English Translation

Original: Left / 原文：左欄　Translation: Right / 譯文：右欄

Qin Jiushao / 秦九韶

(1202–1261)

Song Dynasty, Chunyou 7th year (淳祐七年, 1247)

Bilingual edition based on the Siku Quanshu edition
雙語版據《四庫全書》本編排

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[0]*

天時類序 / 引言

[1]

Introduction to Fascicle 2

[0]* 天時類者，數學九章之第二卷也。凡九問，分為二類：上卷五問，論推步治曆、候星揆影之事；下卷四問，論天池圓罌測雨、峻積竹器驗雪之法。

秦九韶自序其書曰：「大则可以通神明，顺性命；小则可以经世务，类万物。」此卷九問，實以數術推天時，為農事所繫，其用至重。

[1] The **Tian Shi Lei** (天時類, "Heavenly Patterns") is the second of nine chapters in Qin Jiushao's *Shuxue Jiuzhang* (數學九章, Mathematical Treatise in Nine Sections). This fascicle contains **nine problems** divided into two parts, dealing with:

- Calendar computation and astronomical predictions (Problems 1–5)
- Measurement of rainfall and snowfall (Problems 6–9)

Qin Jiushao's preface to the *Shuxue Jiuzhang* (1247) explains his motivation:

> *"The art of numbers, rooted in the Great Void, generates One and circulates without end. In its large aspect it can penetrate the divine and comply with destiny; in its small aspect it can govern worldly affairs and classify the myriad things. How can it be viewed as shallow or narrow?"*

[0]* 此卷所用之法，大要有三：曰通積術，求諸周之公倍也；曰招差術，推日月五星之變行也；曰調日法，齊曆法之疏密也。三者皆中國古代天文數學之精髓。

[1] The nine problems of this fascicle demonstrate the practical application of mathematical techniques to calendar reform, planetary motion, and meteorological measurement—topics of vital importance to agricultural society.

The key methods employed are:

- **General accumulation method** (通積): Finding common periods
- **Interpolation** (招差): For irregular astronomical motions
- **Equalizing technique** (調日法): For calendrical adjustments

[0]*

卷二上

[1]

Volume 2, Part 1

[0]*

第一問：推氣治歷

問 太史觀天，慶元四年戊午歲冬至三十九日九十二刻四十五分，紹定三年庚寅歲冬至三十二日九十四刻一十二分。今欲知嘉泰甲子歲氣骨、歲餘、斗分各幾何?

按 按：紹定三年庚寅冬至，實紹定四年辛卯之歲首也。辛卯去戊午三十四年，即積三十三年。

答曰 氣骨一十一日三十八刻二十分八十一秒八十小分；歲餘五日二十四刻二十九分三十秒三十小分；斗分〇日二十四刻二十九分三十秒三十小分。

術曰 以兩測之年差為法。置前測之日刻分，以後測之日刻分減之，餘為率。(不滅，加紀策)。遞以紀策累加之，務令合天道，取五日以上為實。以法除實，得歲餘。去全分，餘為斗分。以歲餘乘前測至所求中積之年數，併入前測日刻分，滿紀策去之，餘為所求氣骨。

按 氣骨者，自冬至距夜半後甲子日之时刻也。歲餘者，歲周減六甲之餘也。斗分者，歲周減三百六十五日之餘也。此術尚未知歲實之確數，故以兩測氣骨之差為實，積年為法。

[1]

Problem 1: Calculating Qi and Regulating the Calendar

Question The Imperial Astronomer observed the heavenly way:

- In the 4th year of Qingyuan (慶元四年, 1198, year Wu-wu 戊午), the winter solstice was at 39 days, 92 *ke* (刻), 45 *fen* (分).
- In the 3rd year of Shaoding (紹定三年, 1230, year Geng-yin 庚寅), the winter solstice was at 32 days, 94 *ke*, 12 *fen*.

It is desired to find, for the intermediate year Jiatai Jiazi (嘉泰甲子, 1204): the *qi bone* (氣骨), the *year remainder* (歲餘), and the *dou fraction* (斗分).

Editor's Note The Shaoding 3rd year Geng-yin winter solstice actually marks the beginning of Shaoding 4th year Xin-mao (辛卯). Xin-mao is 34 years from Wu-wu, giving 33 accumulated years.

Answer

- Qi bone: 11 days, 38 *ke*, 20 *fen*, 81 *miao* (杪), 80 small fractions.
- Year remainder: 5 days, 24 *ke*, 29 *fen*, 30 *miao*, 30 small fractions.
- Dou fraction: 0 days, 24 *ke*, 29 *fen*, 30 *miao*, 30 small fractions.

Method First take the number of years between the two observations as the divisor. Place the earlier observation's days, *ke*, and *fen*, and subtract from the later observation's days, *ke*, and *fen*; the remainder is the rate. (If insufficient for subtraction, add the ji-ce 紀策). Accumulate by adding ji-ce repeatedly until the result is suitable for the heavenly way—using a number above 5 days as the dividend. Divide the dividend by the divisor to obtain the year remainder. Remove whole days; the remainder is the dou fraction. Multiply the year remainder by the number of years from the earlier observation to the desired intermediate year, add to the earlier observation's days, *ke*, and *fen*, and remove full ji-ce; the remainder is the desired qi bone.

Editor's Note The qi bone is the days and fractions from the winter solstice to the Jiazi day after midnight. The year remainder is the residue when the solar year length is reduced by six Jiazi cycles. The dou fraction is the residue when the solar year is reduced by 365 days. This method does not yet know the precise solar year length, hence it uses the difference between two observed qi bones as the dividend and accumulated years as divisor.

[0]*

第二問：治歷推閏

問　開禧曆，以嘉泰四年甲子歲天冬至一十一日，日辰乙亥，四十四刻六十一分五十四秒；十一月經朔一日，日辰乙丑，七十五刻五十五分六十二秒。問：閏骨、閏率各幾何？

答曰　閏骨九日六十九刻五分九十一秒，餘一百二十一（以一百六十九分為法）；閏骨率一十六萬三千七百七十一。

術曰　以日法通氣朔之日刻分秒，各為氣骨、朔骨之通分。氣骨通分以約率約之，適盡者可用；或不盡，圓徑以下在一刻內者，亦可用。乃以減朔骨通分，餘為閏骨率。以日法除之，得閏骨策。

按　若以朔骨通分徑減氣骨通分，餘為九日六十九刻五分九十二秒，即閏骨策。原草僅差四十八分之一百六十九分。其不徑減而以通分、約率、乘除往復者，蓋為後續諸算之便也。

[1]

Problem 2: Regulating the Calendar and Intercalary Months

Question　For the Kaixi calendar (開禧曆), taking the 4th year of Jiatai (嘉泰四年, 1204, year Jiazi 甲子):

- The celestial winter solstice is at 11 days, day-token Yi-hai (乙亥), 44 *ke*, 61 *fen*, 54 *miao*.
- The 11th-month new moon (經朔) is at 1 day, day-token Yi-chou (乙丑), 75 *ke*, 55 *fen*, 62 *miao*.

Question: What are the *run bone* (閏骨) and *run rate* (閏率)?

Answer

- Run bone: 9 days, 69 *ke*, 5 *fen*, 91 *miao*, with remainder 121/(169 fractions).
- Run bone rate: 163,771.

Method　Take the day-factor (日法) to convert the qi and new-moon days, *ke*, *fen*, and *miao* into qi-bone and new-moon-bone fractions respectively. The qi-bone fraction, when reduced by the approximation rate (約率), if exactly divisible, is usable; or if the remainder after rounding is below one *ke*, it is also usable. Then subtract from the new-moon-bone fraction; the remainder is the run bone rate. Divide by the day-factor to obtain the run bone *ce* (策).

Editor's Note　If one directly subtracts the new moon fraction from the winter solstice fraction, the remainder is 9 days, 69 *ke*, 5 *fen*, 92 *miao*, giving the run bone *ce*. The original working only differs by 48/(169 fractions). The reason for not directly subtracting but instead using common denominators, approximation, and repeated multiplication/division is to facilitate subsequent calculations.

[0]*

第三問：治歷演紀

問　開禧曆積年七百八十四萬一千一百八十三。今欲知演紀之術：調日法，求朔餘、朔率、斗分、歲率、歲閏、入元歲、入閏、朔定骨、閏泛骨、閏縮、紀率、氣元率、元閏、元數、氣等率、因率、蔀率、朔等數、因數、蔀數、朔積年，凡二十三事。各幾何？

答曰　（選錄要項）日法一萬六千九百；朔餘八千九百六十七；朔率四十九萬九千六十七；斗分即二十四刻二十九分三十杪三十小分；歲率六百一十七萬二千六百八；歲閏二萬七百二十八；入元歲九千二百四十；入閏三萬一千六十八。

術曰　以大衍術求氣分、朔分、紀分、衍母、正用。以衍母除氣分，得積年；除朔分，得積月；除紀分，得積紀。各以六十乘之，得積日。以氣骨乘紀正用，得紀總；以閏骨乘朔正用，得朔總。併二總，滿衍母去之，餘為所求率實。以氣分除之，得歷過之年；以減積年，得未至之年。

按　此問全用大衍求一術，以解同餘式組，施用於治曆。所算二十三率，備成開禧曆元之全體，自天文觀測而推得曆元諸數。

[1]

Problem 3: Deriving Calendar Epoch Calculations

Question　The Kaixi calendar has an accumulated year count of 7,848,183. It is desired to know the derivation of this computation: adjusting the day-factor, finding the new-moon remainder, new-moon rate, dou fraction, year rate, year intercalation, entry-year, entry-intercalation, new-moon fixed bone, intercalation floating bone, intercalation shrinkage, ji rate, qi yuan rate, yuan intercalation, yuan number, qi equal-rate, cause-rate, *bu* rate, new-moon equal-number, cause-number, *bu* number, and new-moon accumulated year—**23 items** in total. What is each?

Answer　(Selected key results:)

- Day-factor (日法): 16,900
- New-moon remainder (朔餘): 8,967
- New-moon rate (朔率): 499,067
- Dou fraction (斗分): equivalent to 24 *ke*, 29 *fen*, 30 *miao*, 30 small fractions
- Year rate (歲率): 6,172,608
- Year intercalation (歲閏): 20,728
- Entry-year (入元歲): 9,240
- Entry-intercalation (入閏): 31,068

Method　Use the method of Dayan (大衍術) to find the qi fraction, new-moon fraction, ji fraction, and derived mother (衍母), and all the proper use-numbers (正用). Divide the derived mother by the qi fraction to obtain the accumulated years; by the new-moon fraction to obtain the accumulated months; by the ji fraction to obtain the accumulated ji. Multiply by 60 to obtain accumulated days. Multiply the qi bone by the ji proper-use to obtain the ji total; multiply the run bone by the new-moon proper-use to obtain the new-moon total. Combine the two totals, remove full derived mothers, and the remainder is the sought rate-dividend. Divide by the qi fraction to obtain the calendar-passed (歷過) years; subtract from the accumulated years to obtain the years not yet reached.

Editor's Note　This problem demonstrates the full power of Qin Jiushao's **Dayan method** (大衍求一術) for solving systems of congruences, applied here to calendar computation. The 23 quantities computed form a complete system for deriving the epoch parameters of the Kaixi calendar from astronomical observations.

[0]*

第四問：綴術推星

問　歲星合伏段，一十六日九十分，行三度九十分，去日一十三度乃見。然後順行一百一十三日，一十七度八十三分，乃留。欲知：合伏段、晨疾初段之常度、初行率、末行率、平行率，各幾何？

術曰　（綴術）

1. 合伏段者，常度為總日數，初行率為總行度數，末行率等于初行率減日差乘（日數減一）。
2. 晨疾初段者，常度為所給日數，初行率為前段之末行率，末行率等于初行率減日差乘（日數減一）。
3. 平行率者，併初末二率而半之，即得平行率。

按　五星之行，有遲疾逆順，非均加速也。此術僅取合伏、見段之中數，而以遞加遞減求逐日之行率。古法之疏，於五星尤著。原草語多隱奧，今詳釋之，以見古今疏密之比。

草曰　合伏段：常度一十六日九十分；初行率二十二分七杪；末行率二十一分九十六杪；平行率約二十二分。晨疾初段：常度三十日；初行率二十一分九十六杪；日差一十一杪二十三分六十六小杪。

[1]

Problem 4: Calculating Planetary Motion

Question　The year-star (Jupiter, 歲星) in conjunction-hidden phase (合伏段): after 16 days 90 *fen*, it moves 3 degrees 90 *fen*, removing 13 degrees from the sun before becoming visible. Then it proceeds directly (順行) for 113 days, 17 degrees 83 *fen*, before halting (乃留). It is desired to know:

- The conjunction-hidden segment (合伏段)
- The morning-rapid initial segment (晨疾初段)
- The standard degree (常度)
- The initial motion-rate (初行率)
- The final motion-rate (末行率)
- The mean motion-rate (平行率)

What is each?

Method　(Method of *Zhui Shu*):

1. For the conjunction-hidden segment: The constant degree is the total number of days. The initial motion rate is the total degrees moved. The final motion rate equals the initial rate minus the daily difference (日差) multiplied by (days -1).
2. For the morning-rapid initial segment: The constant degree is the given number of days. The initial rate is the final rate of the previous segment. The final rate equals the initial rate minus the daily difference multiplied by (days -1).
3. For the mean motion rate: Add the initial and final rates and halve; this gives the parallel (mean) rate.

Editor's Note　The five planets' motions involve non-uniform acceleration and deceleration. The method here takes only the middle values of the conjunction-hidden and direct-visibility segments, then uses successive addition/subtraction to derive the daily motion. The ancient method's approximation is particularly evident for the five planets. The original text's language was mostly hidden and obscure; the following detailed explanation reveals the comparison between ancient and modern precision.

Working　**Key calculations:**

- Conjunction-hidden segment:
  - Constant degree: 16 days 90 *fen*
  - Initial rate: 22 *fen* 7 *miao*
  - Final rate: 21 *fen* 96 *miao*
  - Mean rate: approximately 22 *fen* (after rounding)
- Morning-rapid initial segment:
  - Constant degree: 30 days
  - Initial rate: 21 *fen* 96 *miao*

- Daily difference: 11 *miao* 23 small-*fen* 66 small-*miao*

[0]*

第五問：揆日究微

問　歷代測景，唯唐大衍曆最密。本朝崇天曆測陽城：冬至影一丈二尺七寸一分五十杪，夏至影一尺四寸七分七十杪，與大衍曆合。今開禧曆測臨安府：冬至影一丈八寸二分二十五杪，夏至影九寸一分。問：臨安夏至後幾日，其影與陽城夏至影等？又與大衍曆影較之，相差幾何？

答曰　大暑後五日午正，影長一尺四寸八分八十五杪（半）。影差六寸一分二十九杪（少）。

術曰　用乘法，以節率入之。

1. 置臨安所測冬至影長，減夏至影長，餘為景差，以為實。
2. 置象限度（九十一度三十一分四十四杪），加一十一度二十五分二十七秒半；以度為寸，得一百〇二寸五千六百七十一分五十杪為法。
3. 以法除實，得〇點九六六四……；棄餘數，自乘得節泛數九千四百分。
4. 以各節氣乘法率乘之，加夏至影冪，開方得各節氣影長。

按　各節氣之影長，當時實測而定，原不必算。此術故為繁難，以惑學者。細考之：以象限度加一十一度有餘為法，以景差除之而後自乘，得節率。每節氣有乘法率，以節率乘之，加夏至影冪，即為各節氣之影冪。故節率者，人為析出之數也。

[1]

Problem 5: Measuring the Sun's Shadow

Question　Among all historical shadow-measurements, only the Tang dynasty's Dayan calendar (大衍曆) was the most precise. This dynasty's Chongtian calendar (崇天曆) gives for Yangcheng (陽城):

- Winter solstice shadow: 1 *zhang* 2 *chi* 7 *cun* 1 *fen* 50 *miao*
- Summer solstice shadow: 1 *chi* 4 *cun* 7 *fen* 70 *miao*

These agree with the Dayan calendar. Now the Kaixi calendar (開禧曆) gives for Lin'an (臨安府):

- Winter solstice shadow: 1 *zhang* 8 *cun* 2 *fen* 25 *miao*
- Summer solstice shadow: 9 *cun* 1 *fen*

It is desired to find: after how many days past the summer solstice at Lin'an will the shadow equal the Yangcheng summer solstice shadow? And comparing with the Dayan calendar's shadow, by how much do they differ?

Answer　Five days after the Great Heat (大暑後五日) at noon; the shadow length is 1 *chi* 4 *cun* 8 *fen* 85 *miao* (half). The shadow difference is 6 *cun* 1 *fen* 29 *miao* (less).

Method　Use the multiplication-rate method, with the section-rate (節率) entering into it.

1. Place Lin'an's measured winter solstice shadow, subtract the summer solstice shadow; the remainder is the shadow-difference (景差), taken as dividend.
2. Place the 象限度 (91 degrees 31 *fen* 44 *miao*), add 11 degrees 25 *fen* 27.5 small-*fen*; treating degrees as *cun*, obtain 102 *cun* 5671.5 *fen* 50 *miao* as divisor.
3. Divide the dividend by the divisor, obtaining 0.9664...; discard the remainder and square to obtain the section floating-number (節泛數): 9,340 *fen*.
4. Multiply by the section multiplication-rates for each solar term, add the summer solstice shadow-squared, take the square root to obtain each solar term's shadow length.

Editor's Note　Each solar term's shadow length was determined by actual measurement at the time and does not need computation. The method presented here deliberately creates obscurity to puzzle the reader. Detailed examination shows: the 象限度 plus 11+ degrees is taken as divisor; the shadow-difference divided by this and then squared gives the section-rate. Each solar term has a multiplication-rate; multiplying the section-rate by this and adding the summer solstice shadow-power gives each term's shadow-power. Thus the section-rate is an artificially extracted number.

[0]*

卷二下

[1]

Volume 2, Part 2

[0]*

第六問：天池測雨

問　今州郡皆有天池盆以測雨，然人但知盆中之水即所受之雨，而不知器之圓斜各有所受，不可徑以盆測為平地之雨。設盆口徑二尺八寸，底徑一尺二寸，深一尺八寸，接雨水深九寸。問：平地雨深幾何?

答曰　平地雨深三寸。

術曰　用圓錐術：

1. 以口徑乘底徑，又各自乘，併三數。以半深乘之，又以十一乘、七除，得下積。
2. 以半深加雨深，以減全深，得上深。
3. 以底徑減口徑，以上深乘之。以半深乘之，加口徑乘半深，得面率。
4. 以面率除半深，得水面徑。
5. 續以乘除，求得雨水之積，以口冪（平方三倍）除之，得平地雨深。

按　天池盆者，圓錐之截頭也。以雨積除以口面積，即得平地雨深。其術以圓錐台體積之術推之，而歸於平地之算。

[1]

Problem 6: Measuring Rain with the Heavenly Pool Basin

Question Nowadays every prefecture and district has a “heavenly pool” basin (天池盆) for measuring rainwater. But people only know that the water in the basin equals the rain received, and do not understand that different vessel shapes receive different amounts of rain—so one cannot directly take the basin measurement as the flat-ground rain amount.

Suppose the basin has:

- Mouth diameter (口徑): 2 *chi* 8 *cun* (28 *cun*)
- Base diameter (底徑): 1 *chi* 2 *cun* (12 *cun*)
- Depth (深): 1 *chi* 8 *cun* (18 *cun*)
- Rainwater depth inside (接雨水深): 9 *cun*

What is the equivalent flat-ground rain depth?

Answer Flat-ground rain depth: **3 *cun***.

Method Use the method for circular cones (圓錐術):

1. Multiply the mouth diameter by the base diameter, and also square each; combine these three values. Multiply by half the depth; then multiply by 11 and divide by 7 to obtain the lower volume (下積).
2. Take half the depth plus the rain depth, subtract from the total depth to obtain the upper depth (上深).
3. Subtract the base diameter from the mouth diameter; multiply the remainder by the upper depth. Multiply by half the depth, add the result to the mouth-diameter times half-depth, to obtain the surface rate (面率).
4. Divide the surface rate by half the depth to obtain the water surface diameter.
5. Continue with further multiplications to isolate the rain volume, then divide by the mouth-area (squared and tripled) to obtain the flat-ground rain depth.

Editor’s Note The heavenly pool basin is a frustum of a cone. The rain volume collected is:

$$V_{\text{rain}} = \frac{\pi h_{\text{rain}}}{3}\left(R^2 + Rr + r^2\right)$$

where R is the top radius, r is the base radius, and h_{rain} is the rain depth. Dividing by the mouth area πR^2 gives the equivalent flat-ground rain depth.

[0]*

第七問：圓罌測雨

問　以圓罌受雨，口徑一尺五寸（一尺五分寸?），腹徑二尺四寸，底徑八寸，深一尺六寸，接雨水深一尺二寸。用密率入圓冪，問：平地雨深幾何?

答曰　平地雨深一尺八寸，又七百四十分之六百八十三寸。（校正值）

術曰　以底徑乘腹徑，又各自乘，併三數。以半器深乘之；以十一乘、四十二除，得下積。

以半器深加雨深，以減全深，得上深。以腹徑減口徑，以上深乘之。以半深乘之，加口徑乘半深，得面率。

以面率乘腹率，又各自乘併之；以十一乘、四十二除，得上率。併二率（上下）為雨積實。以口徑平方之三倍除之，得平地雨深。

按　原題云「密率入圓冪」，與求平地雨深無涉。若求器中雨積，則此言為當。原草有誤，今以正術列之。

[1]

Problem 7: Measuring Rain with a Round Vessel

Question　Using a round vessel (圓罌) to catch rain:

- Mouth diameter (口徑): 1 *chi* 5 *fen* (10.5 *cun*)
- Belly diameter (腹徑): 2 *chi* 4 *cun* (24 *cun*)
- Base diameter (底徑): 8 *cun*
- Depth (深): 1 *chi* 6 *cun* (16 *cun*)
- Rainwater depth inside: 1 *chi* 2 *cun* (12 *cun*)

Using the “dense rate” (密率, $\pi \approx 22/7$) for circle computations, what is the flat-ground rain depth?

Answer　Flat-ground rain depth: **1 *chi* 8 *cun* plus** $\frac{6483}{74088}$ ***cun***. (Editor’s corrected value.)

Method　Multiply the base diameter by the belly diameter, and also square each; combine these three values. Multiply by half the vessel depth; multiply by 11 and divide by 42 to obtain the lower volume.

Take half the vessel depth plus the rain depth, subtract from the total depth to obtain the upper depth. Subtract the belly diameter from the mouth diameter; multiply the remainder by the upper depth. Multiply by half the depth; add the mouth-diameter times half-depth to obtain the surface rate.

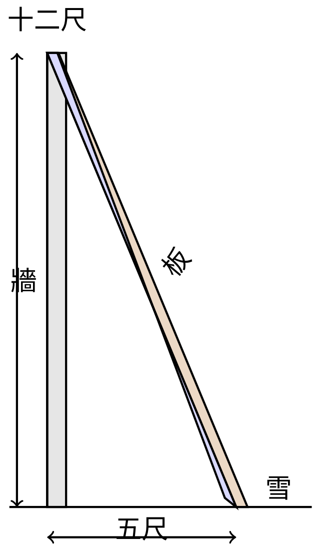
Multiply the surface rate by the belly rate; also square each and combine; multiply by 11 and divide by 42 to obtain the upper rate. Combine the two rates (upper and lower) to obtain the total rain volume as dividend. Divide by three times the squared mouth-diameter to obtain the flat-ground rain depth.

Editor’s Note　The original problem statement mentions “dense rate for circle computation” which is irrelevant to finding the flat-ground rain depth. If one were seeking the vessel’s rain volume, then this phrase would be appropriate. There were errors in the original computation; the above presents the corrected method.

[0]*

第八問：峻積驗雪

問　驗雪占年：牆高一丈二尺，以板倚之，去址五尺，板末與牆齊。雪積板上厚四寸，峻處薄而平處厚。問：平地雪厚幾何？



答曰　平地雪厚一尺四分。

術曰　用少廣術，連枝入之：
1. 以去址之數自乘，為隅。
2. 以牆高自乘，併入上隅。
3. 以雪厚自乘，乘上併數，為實。
4. （約之）開連枝平方，得平地雪厚。

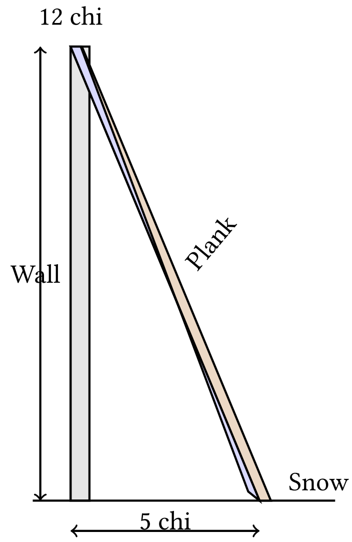
草曰　通分內子：去址五十寸，自乘得二千五百，為隅。牆高一百二十寸，自乘得一萬四千四百。併之得一萬六千九百。雪厚四寸，自乘得十六。乘之得二十七萬四百，為實。隅與實俱以百約之，得實二千七百〇四，隅二十五。開平方：$\sqrt{2704/25} = 52/5 = 10.4$ 寸，即一尺四分。

按　理與法俱正。實即勾股術，特不以此名之耳。「連枝」之號，隱其本理。蓋三角形相似，以比例求之也。

[1]

Problem 8: Verifying Snow Accumulation

Question　To verify snow for predicting the harvest: A wall is 1 *zhang* 2 *chi* (12 *chi*) high. A plank leans against it, the base 5 *chi* from the wall's foot, with the tip level with the wall top. Snow accumulates 4 *cun* thick on the plank. The steep accumulation is thin while the flat accumulation is thick. What is the flat-ground snow depth?



Answer Flat-ground snow depth: **1 *chi* 4 *fen***.

Method Use the method of Shao Guang (少廣), with the connected-branch (連枝) method:
1. Square the distance from the base (去址): this is the corner (隅).
2. Square the wall height; combine with the corner above.
3. Square the snow thickness; multiply by the combined value above to obtain the dividend.
4. (Reduce if possible.) Extract the connected-branch square root to obtain the flat-ground snow depth.

Working Convert all measurements to *cun*:
- Distance from base: 50 *cun*; squared = 2,500 (corner).
- Wall height: 120 *cun*; squared = 14,400.
- Combined: 14,400 + 2,500 = 16,900.
- Snow thickness: 4 *cun*; squared = 16.
- Multiply: $16{,}900 \times 16 = 270{,}400$ (dividend).

The corner and dividend share a factor of 100; reduce to get dividend 2,704 and corner 25. Extracting the square root: $\sqrt{2704/25} = 52/5 = 10.4$ *cun* = **1 *chi* 4 *fen***.

Editor's Note Both the principle and method are correct. In fact, this uses the right-triangle (勾股) method, though not named as such. The "connected-branch" terminology conceals the underlying principle. The diagram illustrates: AB is the flat-ground snow depth (equal to the wall-top snow depth), BC is the excess snow on the plank. Triangle ABC is similar to the triangle formed by the plank leaning against the wall. The wall height is the large leg, the distance from base is the large base, the plank is the large hypotenuse. The flat-ground snow depth is found by proportional reasoning via the Pythagorean theorem.

[0]*

第九問：竹器驗雪

問　以圓竹器驗雪，口徑一尺六寸，深一尺七寸，底徑一尺二寸。雪入其中，深一尺。器通風，雪多則平地淺。問：平地雪厚幾何?

答曰　平地雪厚九寸，又三千四百三十九分之七百六十四寸。

術曰　以底徑減口徑，以雪深乘之，半之，自乘，為隅。以器深自乘，乘雪深自乘之數；併入隅，以雪深自乘之數乘之，為實。開連枝三乘方，得平地雪厚。

草曰　通分內子：口徑十六寸，底徑十二寸。差四寸；雪深十寸。乘之得四十；半之得二十，為上廉。器深自乘：$17^2 = 289$。雪深自乘：$10^2 = 100$。乘之得二萬八千九百；併入隅，續以三乘方開法入之，得平地雪厚九寸，餘三千四百三十九分之七百六十四。

按　「通風」之語，不關算術。或謂器中之雪深與平地異，以風故也；然此問算理，與天池測雨之題同，以圓臺之體推之而已。

[1]

Problem 9: Verifying Snow with a Bamboo Container

Question　Using a round bamboo container (圓竹器) to verify snow:

- Mouth diameter (口徑): 1 *chi* 6 *cun* (16 *cun*)
- Depth (深): 1 *chi* 7 *cun* (17 *cun*)
- Base diameter (底徑): 1 *chi* 2 *cun* (12 *cun*)

Snow falls into it, filling to a depth of 1 *chi*. The container is permeable to wind; when much snow enters, the flat-ground depth is less. What is the flat-ground snow thickness?

Answer　Flat-ground snow thickness: **9 *cun* plus $\frac{764}{3439}$ *cun***.

Method　Subtract the base diameter from the mouth diameter; multiply by the snow depth, halve, and square to obtain the corner (隅). Square the container depth and multiply by the squared snow depth; combine with the corner and multiply by the squared snow depth to obtain the dividend. Extract the connected-branch cube root (三乘方) to obtain the flat-ground snow thickness.

Working　Convert all measurements to *cun*:

- Mouth diameter: 16 *cun*; base diameter: 12 *cun*.
- Difference: 4 *cun*; snow depth: 10 *cun*.
- Multiply: $4 \times 10 = 40$; halve = 20 (this is the upper-*lian* 上廉).
- Container depth squared: $17^2 = 289$.
- Snow depth squared: $10^2 = 100$.
- Multiply: $289 \times 100 = 28{,}900$; combine with the corner and continue the cube-root extraction.

The working proceeds through the Chinese "method of extracting cube roots with carrying" (三乘方開法), yielding the flat-ground snow depth of 9 *cun* with remainder 764/3439.

Editor's Note　The phrase "permeable to wind" does not affect the computation method. Some commentators suggest the snow depth in the container differs from the flat-ground depth due to wind effects; this problem uses the same geometric principle as the Heavenly Pool rain problem, adapted for a frustum-shaped bamboo container.

[0]*

天時類九問總表

[1]

Summary of Fascicle 2 Problems

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序	題名	術要
1	推氣治歷	兩測推氣骨、歲餘、斗分
2	治歷推閏	通分約率求閏骨、閏率
3	治歷演紀	大衍術推曆元二十三率
4	綴術推星	招差法推五星行率
5	揆日究微	乘法節率推各地日影
6	天池測雨	圓錐台體積求平地雨深
7	圓罂測雨	圓罂容積求平地雨深
8	峻積驗雪	少廣連枝開方求平地雪厚
9	竹器驗雪	三乘方開法求平地雪厚

[1]

No.	Title	Topic
1	推氣治歷 Tui Qi Zhi Li	Computing solar terms from two winter solstice observations
2	治歷推閏 Zhi Li Tui Run	Finding intercalation parameters for a calendar
3	治歷演紀 Zhi Li Yan Ji	Deriving 23 epoch parameters using Dayan method
4	綴術推星 Zhui Shu Tui Xing	Computing Jupiter’s motion rates
5	揆日究微 Kui Ri Jiu Wei	Comparing solar shadow lengths at two locations
6	天池測雨 Tian Chi Ce Yu	Measuring rain with a conical basin
7	圓罂測雨 Yuan Ying Ce Yu	Measuring rain with a round vessel
8	峻積驗雪 Jun Ji Yan Xue	Converting snow on a slanted plank to flat-ground depth
9	竹器驗雪 Zhu Qi Yan Xue	Converting snow in a bamboo container to flat-ground depth

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End of Fascicle 2 / 卷二終

“Seven planets revolve in the firmament; the affairs of men are tracked by pursuing and connecting [their motions].
Stars by night, gnomon by day—as ages pass, [methods] become coarse;
only the wise can reform them.”
— Qin Jiushao, Preface to *Shuxue Jiuzhang* / 秦九韶《數學九章序》

Siku Quanshu Edition / 四庫全書本

Shu Xue Jiu Zhang / 數學九章

Fascicles 3–4 / 卷三至四

Tian Yu (田域) + Ce Wang (測望)

Author: Song Dynasty, Qin Jiushao / 宋　秦九韶

Dates: c. 1202–1261

Topics: Juan 3 Shang/Xia: Tian Yu (Fields / 田域類)
Juan 4 Shang/Xia: Ce Wang (Surveying / 測望類)

Bilingual Edition / 雙語版

LEFT: Original Chinese / 原文　|　RIGHT: English Translation / 英譯

Typeset from the Siku Quanshu (四庫全書) edition
Bilingual edition with parallel Chinese text and English translation

Contents

Introduction / 緒言

數學九章為南宋秦九韶所撰，是中古數學之偉大著作。全書九章，每章一類。

本卷呈其第三、四卷：

- 卷三上：田域類 – 方田、圓田、三角田及組合圖形之面積
- 卷三下：田域類續 – 續田域問題，含籌算圖式
- 卷四上：測望類 – 隔河測高遠，用勾股重差之法
- 卷四下：測望類續 – 續測望問題，量不可至之遠近高下

文本依中算傳統格式：

- 問 – 問題陳述
- 答 – 答數
- 術 – 演算方法
- 草 – 詳細演草

The *Shu Xue Jiu Zhang* (Mathematical Treatise in Nine Chapters), written by Qin Jiushao during the Southern Song Dynasty, is one of the greatest masterpieces of medieval Chinese mathematics. The work contains nine chapters, each devoted to a different class of problems.

This volume presents Fascicles 3 and 4 of the work, covering:

- **Juan 3 Shang – Tian Yu (Fields, Part 1)**: Problems on area measurement, including squares, circles, triangles, and composite figures.
- **Juan 3 Xia – Tian Yu (Fields, Part 2)**: Continuation of field problems, including more complex configurations with counting rod diagrams.
- **Juan 4 Shang – Ce Wang (Surveying, Part 1)**: Problems on remote measurement and surveying, using similar triangles and the gou-gu method.
- **Juan 4 Xia – Ce Wang (Surveying, Part 2)**: Further surveying problems, including measurement of heights, distances, and inaccessible objects.

The text follows the classical format of Chinese mathematical literature:

- **Wen (問)**: The problem statement
- **Da Yue (答)**: The answer
- **Shu Yue (術)**: The method/algorithm
- **Cao Yue (草)**: The working/calculation

1 Juan 3 Shang – Tian Yu / 卷三上田域類

小序：此卷論田域面積之術。涉方圓、斜直、面冪、體積，皆相求之。其本則出於立天元一法。

Introductory note: This fascicle treats problems on area measurement (Tian Yu). The problems involve squares, circles, triangles, and composite figures. Qin Jiushao employs his "method of the celestial element" (Tian Yuan Shu) alongside traditional algorithmic techniques.

Problem 1: Fang Zhong Gu Yuan / 方中古圓池

問/ 第一問：方中古圓池

有方中古圓池，因陂毀於一角。從外方隅斜至內圓邊七尺六寸。欲依舊修之，求圓徑、方面、方斜各幾何。

Problem / Square with Inscribed Circular Pond

There is a square field with an ancient circular pond inscribed within it. The northern corner has partially collapsed. From the corner of the outer square to the edge of the inner circle, the diagonal distance is 7 chi 6 cun (7.6 chi). Wishing to repair it according to the original design, find: the diameter of the circle, the side of the square, and the diagonal of the square.

答/ 答曰：

池圓徑三丈六尺六寸，四百二十九分寸之四百一十二。

方面三丈六尺六寸，四百二十九分寸之四百一十二。

方斜五丈一尺八寸，四百二十九分寸之四百一十二。

Answer / Answer:

The pond's diameter is 36 chi 6 cun plus $\frac{412}{429}$ cun; the square's side is 36 chi 6 cun plus $\frac{412}{429}$ cun; the square's diagonal is 51 chi 8 cun plus $\frac{412}{429}$ cun.

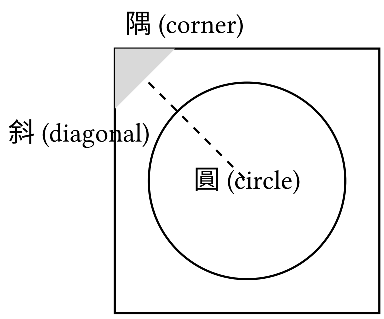
術/ 術曰：

以少廣求之，太朮入之。如積斜自乘倍之為實，倍斜為一方法，一半寸為從隅。開平方得徑。

Method / Method:

Using the "lesser breadth" (shao guang) method with the "transformation of the embryo" (tai tai) technique. Square the diagonal and double it to obtain the dividend (shi). Take twice the diagonal as the negative coefficient of the linear term (yi fang). Take half a cun as the coefficient of the quadratic term (cong yu). Extract the square root to find the diameter.

Tu (Diagram / 圖):



Problem 2: San Xie Qiu Ji / 三斜求積

問/ 第二問：三斜求積

有三斜形田，大斜一十三里，小斜一十四里，中斜一十五里。里法三百步，欲知為田積幾何。

Problem / Triangular Field Area

There is a triangular field. The large diagonal (da xie) is 13 li, the small diagonal (xiao xie) is 14 li, and the middle diagonal (zhong xie) is 15 li. With 300 bu per li, find the area.

答/ 答曰：

田積三百一十五頃。

Answer / **Answer**:

The area is 315 qing.

術/ 術曰：

以少廣求之。以小斜冪並大斜冪，減中斜冪，餘半之，自乘於上。以小斜冪乘大斜冪，減上，餘四約之為實。一為從隅，開平方得積。

Method / **Method**:

Using the “lesser breadth” method. Add the squares of the small and large diagonals, subtract the square of the middle diagonal. Halve the remainder and square it (placed above). Multiply the square of the small diagonal by the square of the large diagonal; subtract the result above; divide the remainder by 4 to get the dividend. Take 1 as the coefficient of the quadratic term; extract the square root to find the area.

In modern notation: If $a = 13$, $b = 14$, $c = 15$, then:

$$S = \sqrt{\frac{1}{4}\left[a^2b^2 - \left(\frac{a^2+b^2-c^2}{2}\right)^2\right]}$$

This is equivalent to **Heron’s formula**:

$$S = \sqrt{s(s-a)(s-b)(s-c)}, \quad s = \frac{a+b+c}{2}$$

Problem 3: Fang Tian Yuan Chi / 方田圓池

問/ 第三問：方田圓池

有方田一段，內有圓池水占之，外計地三千一百六十八步。只云從內池周至外田角斜二十一步半。內外方圓各幾何。

Problem / **Square Field with Central Circular Pond**

There is a square field with a circular pond inside it. The remaining land outside the pond amounts to 3168 bu. It is given that the diagonal distance from the edge of the inner pond to the corner of the outer square is 21.5 bu. Find the dimensions of both the inner circle and outer square.

答/ 答曰：

內池周四十二步，徑一十三步半。方田面六十步。

Answer / **Answer**:

The inner pond’s circumference is 42 bu, its diameter is 13.5 bu; the square field’s side is 60 bu.

術/ 術曰：

以少廣求之。置積以圓率三約之為實，倍斜為從方，斜冪為一負隅。開平方得內徑。加倍斜得方面。

Method / **Method**:

Using the “lesser breadth” method. Place the area and divide by the circular rate 3 to obtain the dividend. Twice the diagonal gives the linear coefficient. The square of the diagonal gives the negative quadratic coefficient. Extract the square root to find the inner diameter. Add twice the diagonal to obtain the square’s side.

Problem 4: Piao Tian / 漂田

問/ 第四問：漂田

有漂田一段，中斜一十六步，水止五步。減中斜一十六餘一十一步為元大斜。內減殘大斜二十步……

Problem / **The Drowned Field**

There is a “drowned field” (a triangular field partially submerged). The middle diagonal is 16 bu; the water depth is 5 bu. Subtracting 5 from 16 gives 11 bu as the

答/ 答曰：

田積若干。

術/ 術曰：

草曰：以水止五減中斜一十六餘一十一步為法。以中斜一十六乘大殘二十得三百二十為大斜實。以法除之得二十九步一十一分步之一為元大斜。內減殘大斜二十步，餘九步一十一分步之一……

base large diagonal.

Answer / **Answer**:

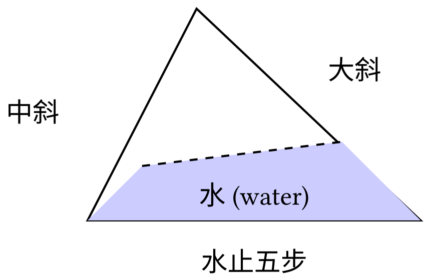
The field area is [to be determined].

Method / **Method**:

(Detailed calculation with specific numerical operations as recorded in the original text.)

Working: Take the water depth 5 subtracted from the middle diagonal 16, remainder 11 bu as the divisor. Multiply the middle diagonal 16 by the large remainder 20 to get 320 as the large diagonal dividend. Divide by the divisor to get 29 and $\frac{1}{11}$ bu as the original large diagonal. Subtract the large remainder 20 bu, remainder 9 and $\frac{1}{11}$ bu...

Tu (Diagram / 圖):



2 Juan 3 Xia – Tian Yu Xu / 卷三下田域類續

Problem 5: San Hu Fen Tian / 三戶分田

問/ 第五問：三戶分田

有田一段，欲分與甲乙丙三戶。甲多一百五十步，乙多一百二十步，丙多九十步。三戶共田若干，各得幾何。

答/ 答曰：

甲田若干，乙田若干，丙田若干。

術/ 術曰：

以少廣求之。併三戶多出之數以減總積，餘以三戶約之，各得本積。又各加多出之數即得各戶之田。

Problem / Division of Land Among Three Families

There is a field to be divided among three families jia, yi, and bing. Family jia receives 150 bu more than the base amount; family yi receives 120 bu more; family bing receives 90 bu more. If the total area is given, find each family's share.

Answer / Answer:

Family jia's share [is determined]; family yi's share [is determined]; family bing's share [is determined].

Method / Method:

Using the "lesser breadth" method. Add the excess amounts from all three families and subtract from the total area. Divide the remainder by 3 to get the base amount for each. Add each family's excess to the base amount to obtain each share.

Problem 6: Yuan Tian Fang Tian / 圓田方田

問/ 第六問：圓田方田

有方圓田各一段，共積一千二百八十步。只云方面如圓徑少四步，問方圓面徑各幾何。

答/ 答曰：

方面若干，圓徑若干。

術/ 術曰：

以少廣求之。置共積以圓率三、方率四各約之為實，方面少圓徑四步為從方。開平方得圓徑。減四得方面。

Problem / Circular and Square Fields

There is a square field and a circular field. Their total area is 1280 bu. It is given that the square's side is 4 bu less than the circle's diameter. Find the side and diameter.

Answer / Answer:

The square's side [and] the circle's diameter [are determined].

Method / Method:

Using the "lesser breadth" method. Place the combined area; divide by the circular rate 3 and square rate 4 to obtain the dividend. The difference of 4 bu between square side and circle diameter gives the linear coefficient. Extract the square root to find the diameter. Subtract 4 to obtain the square's side.

Problem 7: Hu Tian / 弧田

問/ 第七問：弧田

有弧田一段，弦五十步，矢二十五步，求積若干。

答/ 答曰：

田積若干。

術/ 術曰：

以弦矢求之。以弦乘矢，又以矢自乘，併之，半之為實。開平方得積。

Problem / Circular Segment Field

There is a circular segment field (hu tian) with chord (xian) 50 bu and arrow (shi) 25 bu. Find its area.

Answer / Answer:

The field area [is determined].

Method / Method:

Using the chord-arrow method. Multiply the chord by the arrow; add the square of the arrow; halve to obtain the dividend. Extract the square root to find the area.

Note: For a circular segment with chord c and sagitta s, Qin's formula gives:

$$A = \frac{cs + s^2}{2}$$

This is an approximation to the true area of a circular segment.

Counting Rod Diagram / 籌算圖式:

<table>
<tr><td>弦</td><td>矢</td><td>弦乘矢</td><td>矢冪</td></tr>
<tr><td>50</td><td>25</td><td>1250</td><td>625</td></tr>
<tr><td colspan="2">併</td><td>半之</td><td>積</td></tr>
<tr><td colspan="2">1875</td><td>937.5</td><td>≈ 30.6</td></tr>
</table>

3 Juan 4 Shang – Ce Wang / 卷四上測望類

小序：此卷論測望之術，量不可至之高遠。用勾股、重差之法，為中國古代測量學之基礎。

Introductory note: This fascicle treats problems on surveying and remote measurement (Ce Wang). These problems involve measuring inaccessible distances and heights using the "hook-and-gourd" (gou gu) method and similar triangles.

Problem 1: Yuan Cheng / 圓城

問/ 第一問：圓城

有圓城不知周徑，四門中開。北外三里有喬木，出南門折而東行九里乃見木。欲知城周徑各幾何。

答/ 答曰：

徑九里，周二十七里。

術/ 術曰：

以勾股夕桀求之。一為從隅，五因北里為從七廉，置北里冪八因為從五廉，以北里乘上廉為一從，三連負隅乘五廉為一上廉，以北里乘上廉為實。開玲瓏九乘方得數，自乘為徑。以三因徑得周。

Problem / **The Round City**

There is a circular city of unknown circumference and diameter. It has four gates at the cardinal points. Three li north of the north gate stands a tall tree. Going out the south gate and then turning east for 9 li, the tree becomes visible. Find the circumference and diameter of the city.

Answer / **Answer**:

The diameter is 9 li, the circumference is 27 li.

Method / **Method**:

Using the "hook-and-gourd evening-peak" method. Take 1 as the leading coefficient. Multiply the north distance by 5 for the seventh-degree coefficient. Place the square of the north distance, multiply by 8 for the fifth-degree coefficient. Multiply the north distance by the upper coefficient for the negative linear coefficient. Double the negative rate to form the fifth-degree coefficient as the negative upper coefficient. Multiply the north distance by the upper coefficient for the dividend. Extract the ninth-degree root (ling long kai fang) to obtain the number; square it for the diameter. Multiply the diameter by 3 for the circumference.

Problem 2: Ce Ta / 測塔

問/ 第二問：測塔

有塔高未知其數，去塔若干步立一表，人目上望見塔尖，退行若干步又立一表，人目上望仍見塔尖。求塔高及去塔遠近。

答/ 答曰：

塔高若干，去塔遠若干。

術/ 術曰：

以勾股求之。兩表退行之差為法，前表去塔遠乘表高，後表去塔遠乘表高，相減為實。實如法而一得塔高。又以法除前表去塔遠乘兩表高差加表高得塔高。

Problem / **Measuring a Tower**

A tower of unknown height stands at an unknown distance. At a certain distance from the tower, a pole is erected; looking upward from eye level, the tower's peak is visible. Retreating a certain distance and erecting another pole, the peak is again visible from eye level. Find the tower's height and distance.

Answer / **Answer**:

The tower height [and] distance to the tower [are determined].

Method / **Method**:

Using the hook-and-gourd method. The difference between the retreat distances of the two poles is the divisor. Multiply the front pole's distance to the tower by the pole height; multiply the rear pole's distance by the pole height; subtract to obtain the dividend. Divide to get the tower height. Alternatively, multiply the front distance by the difference of pole heights, divide by the divisor, and add the pole height.

In modern notation: Let d_1 = front distance, d_2 = rear distance, h = pole height. Then:

$$\text{Tower height} = \frac{d_2 \cdot h - d_1 \cdot h}{d_2 - d_1} = h$$

This uses the principle of similar triangles, where the ratio of heights equals the ratio of distances.

Problem 3: Fang Cheng / 方城

問/ 第三問：方城

有方城一座，不知大小。北門外有樹，出南門直行若干步折而西行若干步見樹。求城方面。

答/ 答曰：

城方面若干步。

術/ 術曰：

以勾股求之。兩行步相乘為實，併兩行步為從方。開平方得城方面。

Problem / The Square City

There is a square city of unknown size. Outside the north gate stands a tree. Going out the south gate and walking straight a certain number of steps, then turning west and walking a certain number of steps, the tree becomes visible. Find the side of the square city.

Answer / Answer:

The city's side is [determined] in bu.

Method / Method:

Using the hook-and-gourd method. Multiply the two walk distances for the dividend. Add the two walk distances for the linear coefficient. Extract the square root to obtain the city's side.

If a = southward steps and b = westward steps:

$$x^2+(a+b)x = ab \quad \Rightarrow \quad x = \frac{-(a+b)+\sqrt{(a+b)^2+4ab}}{2}$$

(
1em)

Problem 4: Chou Suan Kai Fang / 籌算開方

問/ 第四問：籌算開方

問以開方術解高次方程，其籌佈列之法若何。

術/ 術曰：

開方之法，先列實於中，次列隅於下，次列方於上。商除之，以上乘隅生廉，以上乘廉生方，以上乘方除實。實盡則止，不盡則倍方，三廉為方法，續商除之。

Problem / Counting Rod Root Extraction

On the method of extracting roots to solve higher-degree equations, how are the counting rods arranged?

Method / Method:

In the method of root extraction, first place the dividend (shi) in the middle, then place the unit coefficient (yu) below, and the linear coefficient (fang) above. Divide by estimation; multiply the estimate by the unit coefficient to generate the linear coefficient; multiply the estimate by the linear coefficient and subtract from the dividend. When the dividend is exhausted, stop. If not exhausted, double the linear coefficient and proceed to the next digit.

Counting Rod Layout / 籌式:

商	實	方	廉	隅
7	360	130	…	1
Iterative extraction / 逐次開方				

Note: In the counting rod system, vertical strokes represent units $(1, 2, 3, 4)$ and horizontal strokes represent fives. The symbol T represents 6 (5+1). The diagram shows the iterative process of root extraction.

4 Juan 4 Xia – Ce Wang Xu / 卷四下測望類續

Problem 5: Ce Shui Shen / 測水深

問/ 第五問：測水深

有井不知其深，以繩測之，繩長倍於井深餘三尺；三摺而測之適足。求井深及繩長。

答/ 答曰：

井深三尺，繩長九尺。

術/ 術曰：

以盈不足求之。倍繩餘三尺為盈，三摺適足為適足。以盈乘適足為實，併盈適足為法，除之得井深。倍之加盈得繩長。

Problem / Measuring Depth of Water

There is a well of unknown depth. Measuring with a rope, the rope's length is twice the well's depth with 3 chi to spare. Folding the rope in three and measuring, it fits exactly. Find the depth of the well and the length of the rope.

Answer / Answer:

The well is 3 chi deep; the rope is 9 chi long.

Method / Method:

Using the "surplus and deficit" (ying bu zu) method. The 3 chi excess when doubled is the surplus; the exact fit when tripled is the exact amount. Multiply the surplus by the exact amount for the dividend; add surplus and exact amount for the divisor; divide to obtain the well depth. Double and add the excess for the rope length.

In modern notation: Let d = well depth, L = rope length.

$$L = 2d + 3, \qquad \frac{L}{3} = d$$

Substituting: $2d + 3 = 3d \Rightarrow d = 3$, $L = 9$.

Problem 6: Ce Ta Gao / 測塔高

問/ 第六問：測塔高

有塔高未知，去塔百步立一表高五尺，人目去表一尺八寸，上望塔尖與表端參合。退行六十步，人目去表端仍參合塔尖。求塔高及塔心去表遠近。

答/ 答曰：

塔身高三丈，塔高一十一丈七尺，與八丈七尺為塔心目長。

術/ 術曰：

此皆大小形同勢相求法也。人目去塔為總股，人目去杆為分小股，人目上塔尖高塔身節為總股高。相論高為分大勾，數一杆去塔分大勾於小角杆去塔為分大勾。

Problem / Measuring the Height of a Pagoda

A pagoda of unknown height stands at an unknown distance. 100 bu from the pagoda, a 5-chi pole is erected. The observer's eye is 1.8 chi from the ground; looking up, the pagoda's peak aligns with the pole's tip. Retreating 60 bu, the observer's eye again aligns with both the pole's tip and the pagoda's peak. Find the pagoda's height and its distance from the pole.

Answer / Answer:

The pagoda's body is 3 zhang high; the total height (from ground to peak) is 11 zhang 7 chi; the distance from the pagoda's center to the measuring point is 8 zhang 7 chi.

Method / Method:

These are all applications of similar triangles (da xiao xing tong shi). The distance from eye to pagoda is the total height line; the distance from eye to pole is the small height line. The height from eye level to pagoda peak is the total height; the height from eye level to pole top is the small height. The method uses proportional relationships between these quantities.

This problem uses the principle of similar triangles:

$$\frac{\text{Pagoda height} - \text{eye height}}{\text{Distance to pagoda}} = \frac{\text{Pole height} - \text{eye height}}{\text{Distance to pole}}$$

(
1em)

Problem 7: Ge He Ce Yuan / 隔河測遠

問/ 第七問：隔河測遠

問隔河望一高岸，不知其遠。平地立一表，人目上望見岸頂，又退立一表望之仍見岸頂。求河闊及岸高。

答/ 答曰：

河闊若干步，岸高若干尺。

術/ 術曰：

以重差求之。兩表退行步數之差為法，前表去岸遠乘表高為實，實如法而一得岸高。又以法除後表去岸遠乘表高亦得岸高，兩者相驗。

Problem / Measuring Distance Across a River

Across a river, a high bank is visible at an unknown distance. A pole is erected on level ground; looking up from eye level, the bank's top is visible. Retreating and erecting another pole, the bank's top is again visible. Find the width of the river and the height of the bank.

Answer / Answer:

The river width [and] the bank height [are determined].

Method / Method:

Using the "double difference" (chong cha) method. The difference between the two retreat distances is the divisor. Multiply the front pole's distance to the bank by the pole height for the dividend; divide to obtain the bank height. Similarly for the rear pole; the two results should agree (providing verification).

(
1em)

Problem 8: Jiu Cheng Fang / 九乘方

問/ 第八問：九乘方

問以開九乘方術解方程，其立法之原若何。

術/ 術曰：

開方之術，至於九乘方，其理一也。先列實於中，次定隅位，次立方廉法。商除之際，以上乘隅生廉，以上乘廉生方，以上乘方除實。每進一位，則方退一，廉退二，隅退三。如是疊進，直至實盡為止。

Problem / The Nine-Degree Root Method

On the method of extracting ninth-degree roots to solve equations — what is the fundamental principle of this method?

Method / Method:

The method of root extraction, up to the ninth degree, follows the same principle. First place the dividend in the middle; then determine the unit position; then establish the linear and intermediate coefficients. During division-by-estimation: multiply the estimate by the unit coefficient to generate the intermediate coefficients; multiply the estimate by the intermediate coefficients to generate the linear coefficient; multiply the estimate by the linear coefficient and subtract from the dividend. At each decimal shift, the linear coefficient descends one place, the intermediate descend two places, and the unit descends three. Continue iteratively until the dividend is exhausted.

Note: This describes **Horner's method** for solving polynomial equations, which Qin Jiushao developed independently centuries before Horner. For a polynomial:

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = 0$$

The method systematically reduces the polynomial by factoring out $(x - r)$ where r is the estimated root at each step.

Layout for Ninth-Degree Root Extraction / 九乘方籌式:

商	實	方	一廉	二廉	三廉	四廉	五廉	六廉	隅
7	–	153	…	…	…	…	…	…	1
Iterative computation with place-value shifts / 逐位遞推									

Appendix / 附錄: Mathematical Notes

A.1 籌算 / The Counting Rod System

籌算為中國古代主要計算工具。以竹籌佈列十進位制：

- 一至四：縱式 |, ||, |||, ||||
- 五：橫式 ≡
- 六至九：組合，如 T（五加一）
- 零：以空位或圓 ◯ 表之

籌列於算板，每列一位。此位值制助成高深代數運算，如開方、解方程。

A.1 The Counting Rod System (Chou Suan)

The counting rod system (Chou Suan) was the primary computational tool in ancient Chinese mathematics. Numbers were represented by bamboo sticks arranged in a decimal place-value system:

- **Units (1–4)**: Vertical strokes |, ||, |||, ||||
- **Fives**: A horizontal stroke = placed over the units
- **6–9**: Combinations, e.g., T (5+1), ≡ (3 horizontal over vertical)
- **Zero**: Represented by a blank space or circle ◯

The rods were arranged on a counting board in columns, with each column representing a decimal place. This positional system enabled sophisticated algebraic computations including root extraction and solution of polynomial equations.

A.2 天元術 / The Celestial Element Method

秦九韶精於天元術，為早期代數符號。未知數謂之「天元」，方程以籌式佈列。此法可解十次方程，早霍納五百餘年。

A.2 The "Celestial Element" Method (Tian Yuan Shu)

Qin Jiushao was a master of the "celestial element" method, an early form of algebraic symbolism. The unknown quantity was called "celestial element" (Tian Yuan, tian yuan), and equations were set up using counting rod arrangements. This method could solve polynomial equations of degree up to 10, anticipating Horner's method by over 500 years.

A.3 三斜求積公式 / Qin's Formula for Triangular Area

三角形三邊 a, b, c，秦九韶公式：

$$S = \sqrt{\frac{1}{4}\left[c^2a^2 - \left(\frac{c^2+a^2-b^2}{2}\right)^2\right]}$$

此與海龍公式等價，顯十三世紀非凡代數造詣。

A.3 Qin Jiushao's Formula for Triangular Area

For a triangle with sides a, b, c, Qin gave the formula:

$$S = \sqrt{\frac{1}{4}\left[c^2a^2 - \left(\frac{c^2+a^2-b^2}{2}\right)^2\right]}$$

This is equivalent to Heron's formula and demonstrates remarkable algebraic insight for the 13th century.

A.4 盈不足術 / The Surplus and Deficit Method

盈不足術為通用線性問題解法。設兩試值得盈、不足，則真值：

$$x = \frac{a_2b_1 + a_1b_2}{a_1 + a_2}$$

其中 a_1, a_2 為試入之數，b_1, b_2 為對應盈不足之數。

A.4 The "Surplus and Deficit" Method (Ying Bu Zu)

The method of surplus and deficit (Ying Bu Zu Shu) is a general technique for solving linear problems. Given two trial values yielding surplus and deficit results, the true value is found by:

$$x = \frac{a_2b_1 + a_1b_2}{a_1 + a_2}$$

where a_1, a_2 are the trial inputs and b_1, b_2 are the corresponding surplus/deficit amounts. This method can be applied to a wide variety of practical problems.

End of Fascicles 3–4 / 卷三至四終

Shu Xue Jiu Zhang – Juan 3 Shang, Juan 3 Xia, Juan 4 Shang, Juan 4 Xia / 數學九章

數學九章

卷第五、卷第六

卷五：賦役類 | 卷六：錢穀類

宋　秦九韶　撰

（宋）秦九韶（約 1202–1261）著

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Fascicles Five & Six

Fasc. 5: **Taxation and Corvée (Fuyi)** | Fasc. 6: **Money and Grain (Qiangu)**

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Qin Jiushao (c. 1202–1261)

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LEFT: Original Chinese | RIGHT: English Translation

Traditional structure: Wen (Problem) / Da (Answer) / Shu (Method) / Cao (Working) preserved

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Contents

卷五上　賦役類

按：賦役一章，即古九章中差分粟米諸法。但從來算書設題之數未有如此卷之多者。如首題答數至一百七十五條，每條步算之數至十餘位，而得數皆無不合，亦可謂能廣其用矣。且諸問皆足以明貴賤廩稅及交質變易之同異，使公私各得其平，誠治民之要務也。但題語多晦，術草中間有與所問不相應者，亦於各條下指出，俾觀者無惑焉。

［按］此卷凡九問：一曰復邑修賦，二曰圍田租畝，三曰戶稅移割，四曰均科綿稅，五曰均定勸分，六曰均定合解，七曰寬減屯租，八曰戶稅均寬，九曰米穀粒分。皆賦役之屬也。首問最繁，術用衰分，蓋以三差九等之衰分佈官物，自漢以來均輸之正法也。

Fascicle 5A: Taxation and Corvée (Fuyi)

Editorial: This fascicle on Taxation and Corvée applies the ancient Nine Chapters' methods of difference distribution and grain conversion. Never before has a mathematical text contained problems with data as extensive as this fascicle. For example, the first problem's answer contains 175 items, each with computed values of more than ten digits, yet all results are consistent—a remarkable expansion of mathematical application. These problems illuminate the variations in tax assessment and exchange procedures, ensuring fairness between public and private interests, which is essential to governance. However, the problem statements are often obscure, and some methods do not correspond to the questions asked; such discrepancies are noted below to aid readers.

[Ed.] This fascicle contains nine problems: (1) Restoring the District and Assessing Levies, (2) Polder Field Rent, (3) Household Tax Transfer, (4) Equalizing Silk Tax, (5) Equalizing Voluntary Contributions, (6) Equalizing Combined Remittances, (7) Reducing Garrison Rent, (8) Equalizing Household Tax Relief, (9) Rice and Grain Granule Count. All concern taxation and labor service. The first problem is the most complex, using graded distribution to allocate official commodities across three gradations and nine field grades—the orthodox method of equalized transport (jun shu) since the Han dynasty.

復邑修賦

問：有海坍縣地，今有復漲，歲火鄉井再成，申諸剏邑。稱土排到六鄉，以附郭為甲，最遠為已，各有田九等，開具下項：

甲鄉共計田十四萬一百九十三畝三角一十二步。上等：上田五千六百七十八畝一角四十八步；中田四千八百九十二畝三十步；下田六千六百二十一畝五十四步。中等：上田八千二百二十五畝二十四步；中田一萬三十五畝六步；下田一萬六千五百三十畝。下等：上田二萬一千九十畝二十四步；中田三萬二千六十畝三步；下田三萬五千六十一畝三角三步。

乙鄉田共計八萬四千一十畝二角二步。上等：上田六千七百八十九畝一角三十六步；中田五千九百八十七畝二角；下田八千一十畝三角三步。中等：上田七千五百四十一畝；中田九千一百二十一畝二角一十二步；下田一萬九千六十步。下等：上田一萬八千三十七畝一角六步；中田九千四百五十六畝三角五十九步；下田無。

丙鄉田共計一十二萬九百三十五畝五十八步五分。上等：上田四千八百六十八畝二角三步；中田五千九百七十九畝三角六步；下田六千八百八十八畝

Restoring the District and Assessing Levies

Question: There is a coastal county whose land had collapsed into the sea, but has now silted back up. Over many years the villages and wells have re-formed, and an application has been made to establish a new district. A land survey has been conducted across six townships: Jia (A) being adjacent to the city walls, and Ji (F) the farthest away. Each township has fields of nine grades. The details are listed below:

Township A: Total fields 140,193 mu 3 jiao 12 bu. Upper grade: upper fields 5,678 mu 1 jiao 48 bu; middle fields 4,892 mu 30 bu; lower fields 6,621 mu 54 bu. Middle grade: upper fields 8,225 mu 24 bu; middle fields 10,035 mu 6 bu; lower fields 16,530 mu. Lower grade: upper fields 21,090 mu 24 bu; middle fields 32,060 mu 3 bu; lower fields 35,061 mu 3 jiao 3 bu.

Township B: Total fields 84,010 mu 2 jiao 2 bu. Upper grade: upper fields 6,789 mu 1 jiao 36 bu; middle fields 5,987 mu 2 jiao; lower fields 8,010 mu 3 jiao 3 bu. Middle grade: upper fields 7,541 mu; middle fields 9,121 mu 2 jiao 12 bu; lower fields 19,060 bu. Lower grade: upper fields 18,037 mu 1 jiao 6 bu; middle fields 9,456 mu 3 jiao 59 bu; lower fields none.

Township C: Total fields 120,935 mu 58 bu 5 fen. Upper grade: upper fields 4,868 mu 2 jiao 3 bu; middle fields 5,979 mu 3 jiao 6 bu; lower fields 6,888 mu 2 jiao 6 bu. Middle grade: upper fields 7,984 mu 1 jiao; middle fields 14,056 mu 12 bu; lower fields 23,333 mu 12 bu. Lower grade: upper fields 27,755 mu 16 bu 5 fen; middle fields none; lower fields 30,070 mu 3 bu.

Township D: Total fields 89,066 mu 2 bu 3 fen. Upper grade: upper fields 11,102 mu 1 bu; middle fields 9,876 mu 1 jiao; lower fields 8,765 mu 1 jiao 30 bu. Middle grade: upper fields 7,539 mu 34 bu 3 fen; middle fields none; lower fields 12,987 mu 42 bu. Lower grade: upper fields none; middle fields 5,432 mu 1 jiao 6 bu; lower fields 33,363 mu 3 jiao 9 bu.

Township E: Total fields 204,474 mu 1 jiao 24 bu 4 fen. Upper grade: upper fields 24,632 mu 39 bu; middle fields 13,521 mu 27 bu; lower fields 9,988 mu 3 jiao 3 bu. Middle grade: upper fields 8,877 mu 56 bu 4 fen; middle fields 11,333 mu 3 jiao; lower fields 27,067 mu. Lower grade: upper fields 19,876 mu 3 jiao 6 bu; middle fields 79,135 mu 3 jiao 43 bu; lower fields 10,041 mu 2 jiao 30 bu.

Township F: Total fields 158,460 mu 3 jiao 18 bu 2 fen. Upper grade: upper fields none; middle fields 7,788 mu 3 jiao 51 bu; lower fields none. Middle grade: upper fields 9,999 mu 2 jiao 3 bu; middle fields 10,836 mu 56 bu; lower fields none. Lower grade: upper fields 32,089 mu 1 jiao 45 bu 6 fen; middle fields 43,678 mu 2 jiao 57 bu; lower fields 54,067 mu 3 jiao 45 bu 6 fen.

It is known that this county previously assessed seedling rice at 103,567 shi 8 dou 4 sheng 4 ge 3 shao, harmonized purchase at 13,498 bolts 1 zhang 7 chi 3 cun 7 fen 6 li, and summer tax at 9,876 bolts 3 zhang 2 chi 6 cun 5 fen 8 li. The six townships' fields are of three colors: Jia is upper, Yi and Bing are middle, and Ding, Wu, and Ji are lower. First, the official commodities are divided into three gradations, with upper to middle and middle to lower each differing by 1 part in 10. Then each township's nine grades also differ by 1 part in 10, and the assessments are distributed according to the total quotas. Township B's fields are the most fertile, followed by Ding, Jia, Bing, Ji, and Wu. Find the assessment rate per mu for each of the three colors and nine grades, and the total assessment for each township.

[Ed.] The problem says the fields are of three colors, with Jia as upper, etc., and also says Township B's fields are the most fertile, etc., which seems contradictory. The former refers to the collective assessment grades of all six townships, while the latter refers to the difference in yield

二角六步。中等：上田七千九百八十四畝一角；中田一萬四千五十六畝一十二步；下田二萬三千三百三十三畝一十二步。下等：上田二萬七千七百五十五畝一十六步五分；中田無；下田三萬七十畝三步。

丁鄉田共計八萬九千六十六畝二步三分。上等：上田一萬一千一百二畝一步；中田九千八百七十六畝一角；下田八千七百六十五畝一角三十步。中等：上田七千五百三十九畝三十四步三分；中田無；下田一萬二千九百八十七畝四十二步。下等：上田無；中田五千四百三十二畝一角六步；下田三萬三千三百六十三畝三角九步。

戊鄉田共計二十萬四千四百七十四畝一角二十四步四分。上等：上田二萬四千六百三十二畝三十九步；中田一萬三千五百二十一畝二十七步；下田九千九百八十八畝三角三步。中等：上田八千八百七十七畝五十六步四分；中田一萬一千三百三十三畝三角；下田二萬七千六十七畝。下等：上田一萬九千八百七十六畝三角六步；中等田七萬九千一百三十五畝三角四十三步；下田一萬四十一畝二角三十步。

已鄉田共計一十五萬八千四百六十畝三角一十八步二分。上等：上田無；中田七千七百八十八畝三角五十一步；下田無。中等：上田九千九百九十九畝二角三步；中田一萬八百三十六畝五十六步；下田無。下等：上田三萬二千八十九畝一角四十五步六分；中田四萬三千六百七十八畝二角五十七步；下田五萬四千六十七畝三角四十五步六分。

照得昨來本縣原科苗米一十萬三千五百六十七石八斗四升四合三勺，和買一萬三千四百九十八匹一丈七尺三寸七分六釐，夏税九千八百七十六匹三丈二尺六寸五分八釐。其六鄉田係三色，甲為上，乙丙為次，丁戊己又為次。先令官物為三差，使上比中、中比下皆十分外差一，次令各鄉九等皆於十分內差一，拋科用合租額。其乙鄉田最肥，次丁，次甲，次丙，次己，次戊。欲知三色九等每畝等則及共科數各鄉幾何？

［按］題内言田有三色，甲為上云云，又言乙鄉田最肥云云，語若相左。葢前言三色者六鄉共科之等，後言田肥者每畝所出之異也。若易田係三色為科有三差，則語意明矣。

答曰：甲鄉：上等上田苗三斗二升二合四勺，和一尺六寸九分，稅一尺二寸三分；中田苗二斗九升九勺，和一尺五寸二分，稅一尺一寸一分；下田苗二斗五升八合六勺，和一尺三寸五分，稅九寸九分。中等上田苗二斗二升六合二勺，和一尺一寸八分，稅八寸六分；中田苗一斗九升三合九勺，和一尺一分，稅七寸四分；下田苗一斗六升一合六勺，和八寸四分，稅六寸二分。下等上田苗一斗二升九合三勺，和六寸七分，稅四寸九分；中田苗九升六合九勺，和五寸一分，稅三寸七分；下田苗六升四合六勺，和三寸四分，稅二寸五分。

乙鄉：上等上田苗三斗六升三合三勺，和一尺八寸九分，稅一尺三寸九分；中田苗三斗二升六合九勺，和一尺七寸，稅一尺二寸五分；下田苗二斗九升六勺，和一尺五寸二分，稅一尺一寸一分。中等上田苗二斗五升四合三勺，和一尺三寸三分，稅九寸七分；中田苗二斗一升八合，和一尺一寸四分，稅八寸三分；下田苗一斗八升一合六勺，和九寸五分，稅六寸九分。下等上田苗一斗四升五合三勺，和七寸六分，稅五寸五分；中田苗一斗九合，和五寸七分，稅四寸二分；下田苗七升二合七勺，和三寸八分，稅二寸八分。

丙鄉：上等上田苗三斗三合五勺，和一尺五寸八分，稅一尺一寸六分；中田苗二斗七升三合一勺，和一尺四寸二分，稅一尺四分；下田苗二斗四升二合八勺，和一尺二寸七分，稅九寸三分。中等上田苗二斗一升二合四勺，和一尺一寸一分，稅八寸一分；中田苗一斗八升二合一勺，和九寸五分，稅六寸九分；下田苗一斗五升一合七勺，和七寸九分，稅五寸八分。下等上田苗一斗二升一合四勺，和六寸三分，稅四寸六分；中田苗九升一合，和四寸七分，稅三寸五分；下田苗六升七勺，和三寸二分，稅二寸三分。

丁鄉：上等上田苗三斗四升三合二勺，和一尺七寸九分，稅一尺三寸一分；中田苗三斗八合九勺，和一尺六寸一分，稅一尺一寸八分；下田苗二斗七升四合六勺，和一尺四寸三分，稅一尺五分。中等上田苗二斗四升二勺，和一尺二寸五分，稅九寸二分；中田苗二斗五合九勺，和一尺七分，稅七寸九分；下田苗一斗七升一合六勺，和八寸九分，稅六寸五分。下等上田苗一斗三升七合三勺，和七寸二分，稅五寸二分；中田苗一斗三合，和五寸四分，稅三寸九分；下田苗六升八合六勺，和三寸六分，稅二寸六分。

戊鄉：上等上田苗一斗五升三合八勺，和八寸，稅五寸九分；中田苗一斗三升八合四勺，和七寸二分，稅五寸三分；下田苗一斗二升三合一勺，和六寸四分，稅四寸七分。中等上田苗一斗七合七勺，和五寸六分，稅四寸一分；中田苗九升二合三勺，和四寸八分，稅三寸五分；下田苗七升六合九勺，和四寸，稅二寸九分。下等上田苗六升一合五勺，和三寸二

per mu. If "fields of three colors" is changed to "assessments of three gradations," the meaning becomes clear.

Answer: **Township A:** Upper-upper fields: seedling rice 3 dou 2 sheng 2 ge 4 shao, harmonized purchase 1 chi 6 cun 9 fen, summer tax 1 chi 2 cun 3 fen; upper-middle fields: seedling rice 2 dou 9 sheng 9 shao, harmonized 1 chi 5 cun 2 fen, tax 1 chi 1 cun 1 fen; upper-lower fields: seedling rice 2 dou 5 sheng 8 ge 6 shao, harmonized 1 chi 3 cun 5 fen, tax 9 cun 9 fen.

Middle-upper fields: seedling rice 2 dou 2 sheng 6 ge 2 shao, harmonized 1 chi 1 cun 8 fen, tax 8 cun 6 fen; middle-middle fields: seedling rice 1 dou 9 sheng 3 ge 9 shao, harmonized 1 chi 1 fen, tax 7 cun 4 fen; middle-lower fields: seedling rice 1 dou 6 sheng 1 ge 6 shao, harmonized 8 cun 4 fen, tax 6 cun 2 fen.

Lower-upper fields: seedling rice 1 dou 2 sheng 9 ge 3 shao, harmonized 6 cun 7 fen, tax 4 cun 9 fen; lower-middle fields: seedling rice 9 sheng 6 ge 9 shao, harmonized 5 cun 1 fen, tax 3 cun 7 fen; lower-lower fields: seedling rice 6 sheng 4 ge 6 shao, harmonized 3 cun 4 fen, tax 2 cun 5 fen.

[Similar detailed rates for Townships B–F follow the same structure of nine grades each.]

Total assessments per township: Township A: seedling rice 19,550 shi 2 dou 4 sheng 8 ge 3 shao; harmonized purchase 10,192 zhang 2 chi 6 cun 5 fen 6 li; summer tax 7,457 zhang 6 chi 8 cun 9 fen 8 li. Township B: seedling rice 17,772 shi 9 dou 5 sheng 3 ge; harmonized purchase 9,265 zhang 6 chi 9 cun 6 fen; summer tax 6,779 zhang 7 chi 1 cun 8 fen. Township C: same as B. Township D: seedling rice 16,157 shi 2 dou 3 sheng; harmonized purchase 8,423 zhang 3 chi 6 cun; summer tax 6,163 zhang 3 chi 8 cun. Townships E and F: same as D.

分，稅二寸三分；中田苗四升六合一勺，和二寸四分，稅一寸八分；下田苗三升八勺，和一寸六分，稅一寸二分。

已鄉：上等上田苗二斗八升二合二勺，和一尺四寸七分，稅一尺八分；中田苗二斗五升三合九勺，和一尺三寸二分，稅九寸七分；下田苗二斗二升五合七勺，和一尺一寸八分，稅八寸六分。中等上田苗一斗九升七合五勺，和一尺三分，稅七寸五分；中田苗一斗六升九合三勺，和八寸八分，稅六寸五分；下田苗一斗四升一合一勺，和七寸四分，稅五寸四分。下等上田苗一斗一升二合九勺，和五寸九分，稅四寸三分；中田苗八升四合六勺，和四寸四分，稅三寸二分；下田苗五升六合四勺，和二寸九分，稅二寸二分。

各鄉共科數：甲鄉苗米一萬九千五百五十石二斗四升八合三勺；和買一萬一百九十二丈二尺六寸五分六釐；夏稅七千四百五十七丈六尺八寸九分八釐。乙鄉苗米一萬七千七百七十二石九斗五升三合；和買九千二百六十五丈六尺九寸六分；夏稅六千七百七十九丈七尺一寸八分。丙鄉苗米一萬七千七百七十二石九斗五升三合；和買九千二百六十五丈六尺九寸六分；夏稅六千七百七十九丈七尺一寸八分。丁鄉苗米一萬六千一百五十七石二斗三升；和買八千四百二十三丈三尺六寸；夏稅六千一百六十三丈三尺八寸。戊鄉苗米一萬六千一百五十七石二斗三升；和買八千四百二十三丈三尺六寸；夏稅六千一百六十三丈三尺八寸。已鄉苗米一萬六千一百五十七石二斗三升；和買八千四百二十三丈三尺六寸；夏稅六千一百六十三丈三尺八寸。

Method: Solve by the method of graded distribution (cui fen). First list the county's color grades, arranged in a descending cone (zhui) pattern from the bottom; then multiply by the corresponding township counts. Add these together to form the divisor (fa). Divide the official commodity quantities by this to obtain the rate for one portion. Multiply this rate by the uncombined numbers to obtain each township's amount. Next, list each township's nine grades, beginning with 10 fen for the upper grade, and successively applying a 9/10 reduction for each lower grade down to the ninth grade. Multiply each by the mu and bu, add together to form the township divisor, and divide each township's official commodity amount by this to obtain the rate for one grade. Multiply by the uncombined numbers to obtain the tax rate per mu for each grade.

術曰：以衰分求之。先列本縣色位，自下錐行列之，又以鄉數對列而乘之，副併為法，以除官物數，得一分之率。以率數乘未併者，各得諸鄉之數。次列各鄉等位，自上等置十分，每以內分錐行九折之，至九等止，又各以畝步乘之，副併為鄉法，以除諸各鄉所得官物數，所得為一分之率，以乘未併者，各得每畝稅色。

Working: Set up the three color grades for the county. From the bottom grade taking 11/10 of the standard, with upper to middle adding 1/10, we obtain upper = 121, middle = 110, lower = 100 as the three rates, placed in the right column. Then place Jia = 1 paired with the upper rate, Yi and Bing together = 2 paired with the middle rate, and Ding, Wu, and Ji together = 3 paired with the

草曰：列本縣色位三，自下色例十分中十一分，上比中身外加一，上得一百二十一，中得一百一十，下得一百，為上中下三率，列之右行。乃列甲一對上率，乙丙共二對中率，丁戊己共三對下率，列左行。乃以左行列數各相對乘右行率數，其上得一百二十一，其十得二百二十，其下得三百，乃副置而併之，得六百四十一為法。置元科苗米一萬三千五百六十七石八

斗四升四合三勺為寔，以法除之，得一百六十一石五斗七升二合三勺為一分之率。以未併下率一百乘率，得一萬六千一百五十七石二斗三升為丁戊己三鄉各科數。以於身下加一，得一萬七千七百七十二石九斗五升三合為乙丙二鄉各科數。又於身下加一，得一萬九千五百五十石二斗四升八合三勺為甲合科數。求和買亦用六百四十一為法，置元科和買一萬三千四百九十八匹，以匹法四丈通之，得五萬三千九百九十二丈，內零一丈七尺三寸七分六釐，得五萬三千九百九十三丈七尺三寸七分六釐為寔，以法除之，得八十四丈二尺三寸三分六釐為一分之率。亦以未併下率一百乘之，得八千四百二十三丈三尺六寸為丁戊己三鄉各科數。次於身下加一，得九千二百六十五丈六尺九寸六分為乙丙二鄉各科數。又於身下加一，得一萬一百九十二丈二尺六寸五分六釐為甲鄉合科數。求夏税亦置六百四十一為法，置元科九千八百七十六匹，以匹法四丈通之，得三萬九千五百四丈，內零三丈二尺六寸五分八釐，得三萬九千五百七丈二尺六寸五分八釐為寔，以法除之，得六十一丈六尺三寸三分八釐為一分率。亦以未併下率一百乘之，得六千一百六十三丈三尺八寸為丁戊己三鄉各科數。次於身下加一，得六千七百七十九丈七尺一寸八分为乙丙二鄉各科數。又於身下加一，得七千四百五十七丈六尺八寸九分八釐為甲鄉數。

lower rate, in the left column. Multiply the left column numbers by their corresponding right column rates: upper yields 121, middle yields 220, lower yields 300. Add these together to obtain 641 as the divisor. Take the original seedling rice assessment of 103,567 shi 8 dou 4 sheng 4 ge 3 shao as the dividend (shi). Divide by the divisor to obtain 161 shi 5 dou 7 sheng 2 ge 3 shao as the rate for one portion. Multiply this by the uncombined lower rate 100 to obtain 16,157 shi 2 dou 3 sheng as the combined assessment for townships D, E, and F. Increase this by 1/10 to obtain 17,772 shi 9 dou 5 sheng 3 ge as the combined assessment for townships B and C. Increase again by 1/10 to obtain 19,550 shi 2 dou 4 sheng 8 ge 3 shao as the assessment for township A.

For the harmonized purchase, also use 641 as divisor. Take the original harmonized purchase of 13,498 bolts, convert using the bolt standard of 4 zhang to obtain 53,992 zhang, plus the remainder of 1 zhang 7 chi 3 cun 7 fen 6 li, giving 53,993 zhang 7 chi 3 cun 7 fen 6 li as dividend. Divide by the divisor to obtain 84 zhang 2 chi 3 cun 3 fen 6 li as the rate for one portion. Multiply by the uncombined lower rate 100 to obtain 8,423 zhang 3 chi 6 cun for townships D, E, F. Increase by 1/10 to obtain 9,265 zhang 6 chi 9 cun 6 fen for B and C, and again to obtain 10,192 zhang 2 chi 6 cun 5 fen 6 li for A.

For the summer tax, again use 641 as divisor. Take the original 9,876 bolts, convert at 4 zhang per bolt to obtain 39,504 zhang, plus the remainder 3 zhang 2 chi 6 cun 5 fen 8 li, giving 39,507 zhang 2 chi 6 cun 5 fen 8 li as dividend. Divide to obtain 61 zhang 6 chi 3 cun 3 fen 8 li as the rate. Multiply by 100 to obtain 6,163 zhang 3 chi 8 cun for D, E, F. Increase by 1/10 to obtain 6,779 zhang 7 chi 1 cun 8 fen for B and C, and again to obtain 7,457 zhang 6 chi 8 cun 9 fen 8 li for A.

圍田租畝

問：有興復圍田已成，共計三千二十一頃五十一畝一十五步，分三等。其上等每畞起租六斗，中等四斗五升，下等四斗。中田多上田弱半，不及下田太半。欲知三色田畝及各租幾何?

［按］題言中田多上田弱半，不及下田太半。蓋以弱半為三分之一，太半為三分之二也。多上田弱半者，即比上田多三分之一也。不及下田太半者，即比下田少三分之二也。是上田加三分之一為中田，即下田三分之一也。轉言之，則中田為下田三分之一，上田為中田四分之三也。

Polder Field Rent Assessment

Question: A reclaimed polder field has been completed, totaling 3,021 qing 51 mu 15 bu, divided into three grades. Upper-grade fields pay rent of 6 dou per mu, middle-grade 4 dou 5 sheng, and lower-grade 4 dou. The middle fields exceed the upper fields by one weak half (1/3), and fall short of the lower fields by one great half (2/3). Find the mu of each of the three grades of fields and the rent for each.

[Ed.] The problem says the middle fields exceed the upper fields by a weak half and fall short of the lower fields by a great half. Here, weak half means 1/3 and great half means 2/3. Exceeding the upper fields by a weak half means exceeding by 1/3. Falling short of the lower fields by a great half means falling short by 2/3. Thus upper

fields plus 1/3 equals middle fields, which is 1/3 of the lower fields. In other words, middle fields are 1/3 of lower fields, and upper fields are 3/4 of middle fields.

Answer: Upper fields: 477 qing 8 mu 15 bu, rice 28,624 shi 8 dou 3 sheng 7 ge 5 shao; Middle fields: 636 qing 10 mu 3 jiao, rice 28,624 shi 8 dou 3 sheng 7 ge 5 shao; Lower fields: 1,908 qing 32 mu 1 jiao, rice 76,332 shi 9 dou.

答曰：上田四百七十七頃八畝一十五步，米二萬八千六百二十四石八斗三升七合五勺；中田六百三十六頃一十畝三角，米二萬八千六百二十四石八斗三升七合五勺；下田一千九百八頃三十二畝一角，米七萬六千三百三十二石九斗。

Method: Solve by the method of graded distribution (cui fen). Set up the numerator and denominator to find the field rates. Add them together to form the divisor. Take the total fields as dividend. Divide to obtain the rate for one portion. Multiply this throughout by the uncombined numbers to obtain the three grades of fields. Multiply each by the base rent to obtain the rice amounts.

術曰：以衰分求之。列母子求田率，副併為法，以共田為寔，寔如法而一，得一分之率。以徧乘未併者，得三等田。各以起租乘之，各得米。

Working: Set the weak-half denominator 4 as the middle rate and numerator 3 as the upper rate. Subtract the great-half numerator 2 from denominator 3, remainder 1. Multiply by the middle rate 4, obtaining 4 as the middle general (fan). Also multiply the remainder 1 by the upper rate 3, obtaining 3 as the upper general. Next multiply the middle general 4 by the great-half denominator 3, obtaining 12 as the lower general. Add the three generals to obtain 19 as the divisor. Take the fields 3,021 qing 51 mu, convert using the mu standard of 240 bu to obtain 72,516,240 bu, plus the numerator 15 bu, giving 72,516,255 bu as dividend. Divide by the divisor 19 to obtain 3,816,645 bu as the amount for one portion. Multiply by the upper general 3 to obtain 11,449,935 bu as the upper product. Multiply the one-portion amount by the middle general 4 to obtain 15,266,580 bu as the middle product. Multiply the one-portion amount by the lower general 12 to obtain 45,799,740 bu as the lower product. Convert each product to mu using 240 bu per mu: upper fields = 477 qing 8 mu 15 bu; middle fields = 636 qing 10 mu 3 jiao; lower fields = 1,908 qing 32 mu 1 jiao. For the rents: multiply the upper product 11,449,935 bu by upper rent 6 dou, obtaining 6,869,961 shi as dividend, divided by 240 bu to obtain 28,624 shi 8 dou 3 sheng 7 ge 5 shao as upper-field rent. Multiply the middle product by middle rent 4 dou 5 sheng, obtaining 6,869,961 shi, divided by 240 to obtain 28,624 shi 8 dou 3 sheng 7 ge 5 shao. Multiply the lower product by lower rent 4 dou, obtaining 18,319,896 shi, divided by 240 to obtain 76,332 shi 9 dou as lower-field rice.

草曰：置弱半母四為中率，子三為上率。以太半子二減母三，餘一，以乘中率四，只得四為中泛。又以餘一乘上率三，只得三為上泛。次以太半母三乘中泛四，得一十二為下泛。副併三泛，得一十九為法。置田三千二十一頃五十一畆，以畝法二百四十通之，得七千二百五十一萬六千二百四十步，內子一十五步，得七千二百五十一萬六千二百五十五步為寔。以法一十九除之，得三百八十一萬六千六百四十五步為一分之數。以上泛三因之，得一千一百四十四萬九千九百三十五步為上積。又以中泛四因之一分數，得一千五百二十六萬六千五百八十步為中積。又以下泛一十二乘一分數，得四千五百七十九萬九千七百四十步為下積。其三積各以畝法二百四十約之為畆。其上田得四百七十七頃八畝一十五步，其中田得六百三十六頃一十畝三角，其下積得一千九百八頃三十二畝一角。各以起租，三積為三寔。其上積一千一百四十四萬九千九百三十五步乘上積六斗，得六百八十六萬九千九百六十一石為寔，以畝法二百四十除之，得二萬八千六百二十四石八斗三升七合五勺為上田租。其中積一千五百二十六萬六千五百八十步乘中田租四斗五升，得六百八十六萬九千九百六十一石為寔，以畝法二百四十除之，得二萬八千六百二十四石八斗三升七合五勺。其下積四千五百七十九萬九千七百四十步乘下租四斗，得一千八百三十一萬九千八百九十六石為寔，以畝法二百四十除之，得七萬六千三百三十二石九斗為下田米。

[Ed.] In the working, the intention in finding each graded rate is: taking 3 fen as the upper-field rate, adding the weak half (1/3) to obtain 4 fen as the middle-field rate, and tripling the middle-field rate to obtain 12 fen as the lower-field rate, which sum to 19 fen as the total graded rate. The method is essentially clear, but the terminology

［按］草内求各衰數，意謂以三分為上田衰數，加弱半三分之一得四分為中田衰數，三因中田衰數得一十二分為下田衰數，併之得一十九分為總衰數。其法本屬顯明，但語中參入母子副泛等名目，其意反晦矣。

of numerators, denominators, and generals makes it more obscure.

營田造租

問：有營田五十頃六十二畝五分，係官出租。三等出租：上田每畝五斗，中田四斗二升，下田三斗五升。其田上中下相半，上田不及中田頃畝六分半，下田比中田多三分半。欲知三色田頃畝及各租幾何？

［按］此題用衰分法，以上中下衰率求之。題言上田不及中田六分半，下田比中田多三分半，皆以中田為率而損益之也。

答曰：上田一十五頃四十畝，租七千七百石；中田一十七頃四十畝二分，租七千三百石二斗八升；下田一十七頃八十二畝三分，租六千二百三十八石八升。

術曰：以衰分求之。置中田為率，以六分半減之為上田衰，以三分半加之為下田衰。各徧乘共畝為實，副併衰數為法。實如法而一，各得每衰之畝。以各租乘之，得各租米。

草曰：置中田一十畝為率，以六分半減之，得九十三分半為上田衰。以三分半加中田率，得一十三分半為下田衰。并三衰，得三十七分为法。置營田五十頃六十二畝五分，通為五千六十二畝五分，以各衰遍乘之：以上田衰九十三分半乘，得四十七萬三千四百三十七畝半為上實；以中田衰一十畝乘，得五萬六百二十五畝為中實；以下田衰一十三分半乘，得六萬八千三百四十三畝七分五厘為下實。三實皆如法三十七而一：上田得一萬五千四百畝，展為一十五頃四十畝；中田得一萬七千四百畝二分，展為一十七頃四十畝二分；下田得一萬七千八百二十三畝三分，展為一十七頃八十二畝三分。各以起租乘之：上田一十五頃四十畝乘五斗，得七千七百石；中田一十七頃四十畝二分乘四斗二升，得七千三百石二斗八升；下田一十七頃八十二畝三分乘三斗五升，得六千二百三十八石八升。

Garrison Field Rent Assessment

Question: There are 50 qing 62 mu 5 fen of garrison fields rented by the government, divided into three grades: upper fields at 5 dou per mu, middle fields at 4 dou 2 sheng, lower fields at 3 dou 5 sheng. The fields are divided equally among the three grades. The upper fields fall short of the middle fields by 6.5 fen in qing-mu measure, and the lower fields exceed the middle fields by 3.5 fen. Find the qing and mu of each of the three grades of fields and their respective rents.

[Ed.] This problem uses the method of graded distribution, seeking the rates of upper, middle, and lower fields. The problem states that upper fields fall short of middle fields by 6.5 fen, and lower fields exceed middle fields by 3.5 fen—both taking the middle fields as the base and applying decrease or increase.

Answer: Upper fields: 15 qing 40 mu, rent 7,700 shi; Middle fields: 17 qing 40 mu 2 fen, rent 7,300 shi 2 dou 8 sheng; Lower fields: 17 qing 82 mu 3 fen, rent 6,238 shi 8 sheng.

Method: Solve by the method of graded distribution. Take the middle fields as the base rate. Subtract 6.5 fen to obtain the upper-field rate, and add 3.5 fen to obtain the lower-field rate. Multiply the total mu by each rate to form the dividends. Add the rates together to form the divisor. Divide each dividend by the divisor to obtain the mu per rate. Multiply each by the respective rent to obtain the rice rents.

Working: Set 10 mu of middle fields as the base. Subtract 6.5 fen to obtain 93.5 fen as the upper-field rate. Add 3.5 fen to the middle rate to obtain 13.5 fen as the lower-field rate. Add the three rates to obtain 37 as the divisor. Take the garrison fields of 50 qing 62 mu 5 fen, convert to 5,062.5 mu. Multiply by each rate: upper-field rate 93.5 gives 473,437.5 mu as upper dividend; middle-field rate 10 gives 50,625 mu as middle dividend; lower-field rate 13.5 gives 68,343.75 mu as lower dividend. Divide each of the three dividends by the divisor 37: upper fields obtain 15,400 mu = 15 qing 40 mu; middle fields obtain 17,400 mu 2 fen = 17 qing 40 mu 2 fen; lower fields obtain 17,823 mu 3 fen = 17 qing 82 mu 3 fen. Multiply each by the base rent: upper fields 15 qing 40 mu at 5 dou = 7,700 shi; middle fields 17 qing 40 mu 2 fen at 4 dou 2 sheng = 7,300 shi 2 dou 8 sheng; lower fields 17 qing 82 mu 3 fen at 3 dou 5 sheng = 6,238 shi 8 sheng.

Fascicle 5B: Taxation and Corvée (continued)

卷五下　賦役類（續）

Household Tax Transfer

Question: A certain county received a petition from household A stating: This household's fields originally paid seedling rice of 35 shi 7 dou; harmonized purchase in kind of 11 bolts 2 zhang 2 chi 9 cun 4 fen 8 li 7 hao 5 si; converted silk of 27 bolts 2 cun 1 fen 3 li 7 hao 5 si; coarse silk converted silk of 8 bolts 3 zhang 9 chi 7 cun 3 fen 9 li; coarse silk in kind of 20 bolts 2 zhang 8 chi 9 cun 3 fen 9 li. The household has already transferred 407 mu of fields to B and 516 mu to C. The petition requests that the taxes on the transferred fields be moved and consolidated with households B and C for payment. It was found that B originally had 375 mu of fields and C originally had 463 mu, all in the same township and same grade. Per mu: seedling rice 3 sheng 5 ge, tax silk 1 chi 1 cun 5 fen, property value 1 guan 200 wen; for this grade of fields, coarse silk is 1 chi 3 cun 4 fen, property value 900. For every 32 guan of property value, 1 bolt of harmonized purchase is assessed: 3/10 in kind, 7/10 converted. Summer tax: 7/10 in kind, 3/10 converted. Coarse silk: 5/10 in kind, 5/10 converted, combined with silk. Find household A's remaining fields, and the respective amounts of seedling rice, harmonized purchase, summer tax, converted silk, in-kind silk, and remainders that each of A, B, and C should pay after the division.

[Ed.] This problem's tax assessment divides into fields (tian) and land (di) categories. Fields have three components: rice assessment, silk tax, and property value; land has only two: coarse-silk tax and property value. The conversion rates for silk and coarse silk differ, and the harmonized purchase based on property value combines both fields and land. Also, household A has both fields and land, while households B and C have fields only. These distinctions should have been clearly stated in the problem but were not.

Answer: Household A originally had 1,020 mu of fields and 11 mu 2 jiao 48 bu of land.

Household A now has 97 mu of fields: seedling rice 3 shi 3 dou 9 sheng 5 ge; summer tax converted silk 3 zhang 3 chi 4 cun 6 fen 5 li; in-kind silk 1 bolt remainder 3 zhang 8 chi 8 fen 5 li; property value 116 guan 400 wen. Land of 11 mu 2 jiao 48 bu: coarse-silk converted tax 7 chi 8 cun 3 fen 9 li; coarse-silk remainder 7 chi 8 cun 3 fen 9 li; land property value 10 guan 530 wen; combined fields and land property value 126 guan 930 wen. Harmonized purchase: converted silk 2 bolts 3 zhang 1 chi 6 fen 3 li 7 hao 5 si; in-kind 1 bolt remainder 7 chi 5 cun 9 fen 8 li 7

戶稅移割

問：某縣據甲稱：本户田地原納苗三十五石七斗，和買本色一十一匹二丈二尺九寸四分八釐七毫五絲，折帛二十七匹二寸一分三釐七毫五絲，紬絹折帛八匹三丈九尺七寸三分九釐，紬絹本色二十匹二丈八尺九寸三分九釐。已将田四百七畆出與乙，五百十六畆出與丙。訖乞移割本户所出田上稅賦，歸併乙丙兩户送納。會到乙原有田三百七十五畆，丙原有田四百六十三畆，並係本鄉本等。每畆苗三升五合，稅一尺一寸五分，物力一貫二百；本等田紬一尺三寸四分，物力九百；物力及三十二貫數和買一匹，内三分本色七分折帛；夏稅七分本色三分折帛；紬五分本色五分折帛，併絹紬。欲知甲田地及甲乙丙分割合納苗米、和買、夏稅、折帛、本色、畸零各幾何？

［按］此題科稅分田地二項。田有科米、稅絹、物力三色，地只有稅紬、物力二色，絹與紬折色不同，物力和買合田地總計之。又甲户有田者有地，乙丙二户有田無地。此等皆不可不明言者，題中殊未分晰。

答曰：甲原有田一千二十畆，地一十一畆二角四十八步。

甲今有田九十七畆，苗米三石三斗九升五合，夏稅折帛三丈三尺四寸六分五釐，本色一匹畸零三丈八尺八分五釐，物力一百一十六貫四百文。地一十一畆二角四十八步，稅紬折帛七尺八寸三分九釐，稅紬畸零七尺八寸三分九釐，物力一十貫五百三十文，田地共物力一百二十六貫九百三十文。和買折帛二匹三丈一尺六分三釐七毫五絲，本色一疋畸零七尺五寸九分八釐七毫五絲。

乙今有田七百八十二畆，苗米二十七石三斗七升，夏稅折帛六匹二丈九尺七寸九分，本色一十五匹畸

零二丈九尺五寸一分，物力九百三十八貫四百文，和買折帛二十匹二丈一尺一寸，本色八匹畸零三丈一尺九寸。

丙今有田九百七十九畝，苗米三十四石二斗六升五合，夏税折帛八匹一丈七尺七寸五分五釐，本色一十九匹畸零二丈八尺九分五釐，物力一千一百七十四貫八百文，和買折帛二十五匹二丈七尺九寸五分，本色一十一匹畸零五寸五分。

hao 5 si.

Household B now has 782 mu of fields: seedling rice 27 shi 3 dou 7 sheng; summer tax converted silk 6 bolts 2 zhang 9 chi 7 cun 9 fen; in-kind 15 bolts remainder 2 zhang 9 chi 5 cun 1 fen; property value 938 guan 400 wen. Harmonized purchase: converted silk 20 bolts 2 zhang 1 chi 1 cun; in-kind 8 bolts remainder 3 zhang 1 chi 9 cun.

Household C now has 979 mu of fields: seedling rice 34 shi 2 dou 6 sheng 5 ge; summer tax converted silk 8 bolts 1 zhang 7 chi 7 cun 5 fen 5 li; in-kind 19 bolts remainder 2 zhang 8 chi 9 fen 5 li; property value 1,174 guan 800 wen. Harmonized purchase: converted silk 25 bolts 2 zhang 7 chi 9 cun 5 fen; in-kind 11 bolts remainder 5 cun 5 fen.

術曰：以粟米及衰分求之。置甲原納米為實，以每畝苗為法除之，得甲原有田。以乘每畝税，得田絹。次以甲紬絹本色折帛併之，內減田絹，餘為地紬。以每紬約之，得甲原有地。次以甲出田併乙丙原有田，各得三户今有田地。各以等則乘之，各得物力苗税。次以每畝物力率約各户共物力，得和買。副之，三分因之退位為和本，七分因之退位為和折；夏税反其分而因之退之；紬以半之，併入税，各得。

Method: Solve by the methods of grain conversion (su mi) and graded distribution (cui fen). Take household A's original rice payment as dividend and divide by the per-mu seedling rate to obtain A's original fields. Multiply by the per-mu tax to obtain the field silk. Then combine A's coarse silk and silk in-kind and converted, subtract the field silk, and the remainder is the land coarse silk. Divide by the per-unit coarse silk to obtain A's original land. Next, combine the transferred fields with B and C's original fields to obtain each household's current fields. Multiply each by the grade rates to obtain property values and seedling taxes. Then divide each household's total property value by the per-mu property rate to obtain the harmonized purchase. Split it: 3/10 as in-kind harmonized purchase, 7/10 as converted; for summer tax reverse the ratios; for coarse silk halve it and combine with silk tax to obtain the results.

草曰：置甲原納苗三十五石七斗為實，以每畝苗三升五合為法除之，得甲田一千二十畝。以每畝税一尺一寸五分乘之，得一百一十七丈三尺為田絹。以甲原紬絹本色二十匹二丈八尺九寸三分九釐、折帛八匹三丈九尺七寸三分九釐併之，共二十八匹三丈七尺六寸七分八釐。內減田絹一百一十七丈三尺，餘為地紬。以每地紬一尺三寸四分約之，得甲地一十一畝二角四十八步。以甲出田四百七畝併乙田三百七十五畝，得七百八十二畝為乙户今有田。以甲出田五百十六畝併丙田四百六十三畝，得九百七十九畝為丙户今有田。甲今有田一千二十畝減出田九百二十三畝，餘九十七畝為甲今有田。各以每畝物力率乘之：乙田七百八十二畝乘一貫二百，得九百三十八貫四百文；丙田九百七十九畝乘一貫二百，得一千一百七十四貫八百文；甲田九十七畝乘一貫二百，得一百一十六貫四百文。地一十一畝二角四十八步乘九百，得十貫五百三十文。共甲物力一百二十六貫九百三十文。各以物力三十二貫約之：乙得二十四匹餘十六貫四百文，三分本色得七匹餘八分，七分折帛得一十六匹二丈一尺一寸；丙得三十六匹餘二十六貫八百文，

Working: Take household A's original seedling rice of 35 shi 7 dou as dividend; divide by the per-mu seedling rate of 3 sheng 5 ge to obtain A's fields of 1,020 mu. Multiply by per-mu tax of 1 chi 1 cun 5 fen to obtain 117 zhang 3 chi of field silk. Combine A's original coarse silk and silk: in-kind 20 bolts 2 zhang 8 chi 9 cun 3 fen 9 li and converted 8 bolts 3 zhang 9 chi 7 cun 3 fen 9 li, totaling 28 bolts 3 zhang 7 chi 6 cun 7 fen 8 li. Subtract field silk 117 zhang 3 chi; the remainder is land coarse silk. Divide by per-unit coarse silk of 1 chi 3 cun 4 fen to obtain A's land of 11 mu 2 jiao 48 bu. Combine A's transferred fields 407 mu with B's 375 mu to obtain 782 mu as B's current fields. Combine A's transferred 516 mu with C's 463 mu to obtain 979 mu as C's current fields. A's original 1,020 mu minus transferred 923 mu leaves 97 mu as A's current fields. Multiply each by the per-mu property rate: B's 782 mu at 1 guan 200 = 938 guan 400 wen; C's 979 mu at 1 guan 200 = 1,174 guan 800 wen; A's 97 mu at 1 guan 200 = 116 guan 400 wen. Land 11 mu 2 jiao 48 bu at 900 = 10 guan 530 wen. Total A property value = 126 guan 930 wen. Divide each property value by 32 guan: B obtains 24 bolts remainder 16 guan 400 wen, 3/10 in-kind = 7 bolts remainder 8 fen, 7/10 converted = 16 bolts 2 zhang 1 chi 1 cun. C obtains 36 bolts remainder 26 guan 800 wen, 3/10 in-kind = 11 bolts remainder 8 fen, 7/10 converted = 25 bolts 2 zhang 7 chi 9 cun 5 fen. A obtains 3 bolts remainder 30 guan 930 wen, 3/10 in-kind = 1 bolt remainder chi, 7/10 converted = 2 bolts 3 zhang 1 chi 6 fen 3 li. Summer tax uses 7:3 ratio on property value: B 7/10 in-kind = 15 bolts remainder 2 zhang 9 chi 5 cun 1 fen, 3/10 converted = 6 bolts 2 zhang 9 chi 7 cun 9 fen. C 7/10 in-kind = 19 bolts remainder 2 zhang 8 chi 9 fen 5 li, 3/10 converted = 8 bolts 1 zhang 7 chi 7 cun 5 fen 5 li. A 7/10 in-kind = 1 bolt remainder 3 zhang 8 chi 8 fen 5 li, 3/10 converted = 3 zhang 3 chi 4 cun 6 fen 5 li.

三分本色得一十一匹餘八分，七分折帛二十五匹二丈七尺九寸五分；甲得三匹餘三十貫九百三十文，三分本色得一匹餘尺，七分折帛二匹三丈一尺六分三釐。夏税以物力七三因之：乙七分本色一十五匹餘二丈九尺五寸一分，三分折帛六匹二丈九尺七寸九分；丙七分本色一十九匹餘二丈八尺九分五釐，三分折帛八匹一丈七尺七寸五分五釐；甲七分本色一匹餘三丈八尺八分五釐，三分折帛三丈三尺四寸六分五釐。

Equalizing Silk Tax Assessment

均科綿税

Question: A county assesses silk floss across five grades of households, totaling 11,033 households, with total silk assessment of 88,337 liang 6 qian. Upper grade: 12 households; second grade: 87 households; middle grade: 464 households; fourth grade: 2,035 households; lower grade: 8,435 households. It is desired that the upper three grades differ by halving (upper:second:middle = 4:2:1), and the lower two grades differ from the middle grade by a 6:4 ratio (fourth:lower = 6:4 of middle). Using the assessment rate, find the amount per household and the total for each grade.

問：縣科綿有五等户，共一萬一千三十三户，共科綿八萬八千三百三十七兩六錢。上等一十二户，副等八十七户，中等四百六十四户，次等二千三十五户，下等八千四百三十五户。欲令上三等折半差，下二等比中等六四折差。科率求之，問各户納及各等幾何？

[Ed.] The problem states that the lower two grades differ from the middle grade by a 6:4 ratio, meaning the fourth grade is 4/6 of the middle grade, and the lower grade is 4/6 of the fourth grade. However, in the method and working, both are reduced by 4/10, which is a 4-tenths reduction rather than a 6:4 ratio. The collection contains many such cases where the problem and method do not correspond—could it be that the author was unable to thoroughly revise the work after its completion?

［按］題言下二等比中等六四折差，是次等為中等六分之四，下等又為次等六分之四也。乃術草中皆以十分之四收之，是四分折差而非四六折差矣。集中題與術不相應者多類此，豈成書之後未能重加校正歟？

Answer: Upper grade: 124 liang per household, 12 households totaling 1,488 liang. Second grade: 62 liang per household, 87 households totaling 5,394 liang. Middle grade: 31 liang per household, 464 households totaling 14,384 liang. Fourth grade: 12 liang 4 qian per household, 2,035 households totaling 25,234 liang. Lower grade: 4 liang 9 fen 6 li per household, 8,435 households totaling 41,837 liang 6 qian.

答曰：上等一户一百二十四兩，一十二户計一千四百八十八兩；副等一户六十二兩，八十七户計五千三百九十四兩；中等一户三十一兩，四百六十四户計一萬四千三百八十四兩；次等一户一十二兩四錢，二千 035 户計二萬五千二百三十四兩；下等一户四兩九錢六分，八千四百三十五户計四萬一千八百三十七兩六錢。

Method: List the five grades of households. First apply a 4/10 reduction to the lower-grade count, add to the fourth-grade households, apply a 4/10 reduction again, add to the middle-grade count, then apply a 1/2 reduction, add to the second-grade count, apply a 1/2 reduction again, and add

術曰：列五等户數，先以四折下等數，加次等户，又以四折之，加中等户數，却以半折之，加二等户，又以半折之，加上等户數，不折便為法。除科綿，得上等一户之綿。復半之為副等，又半之為中等，又四折之為次等，又四折之為下等。各一户綿，却各以户數

乘之，各得五等共出綿。

to the upper-grade count. Do not reduce this final sum; it becomes the divisor. Divide the total silk assessment by this to obtain the silk per upper-grade household. Halve this for the second grade, halve again for the middle grade, apply a 4/10 reduction for the fourth grade, and another 4/10 reduction for the lower grade. Multiply each per-household amount by the respective household count to obtain the total silk for each grade.

草曰：先置下等户八千四百三十五，以四折之，得三千三百七十四。乃以得數折入次户二千三十五内，得五千四百九户訖。又以四折次數五千四百九户，得二千一百六十三户六分。乃以得次數併入中户四百六十四内，共得二千六百二十七户六分。次以五分折中數二千六百二十七户六分，得數。次以得數一千三百一十三户八分併副户八十七，得一千四百户八分。又以五折副數一千四百户八分，得七百户四分，既得數。乃以副數七百户四分併上户一十二户，得七百一十二户四分，不折便為法。

累折至等，共得七百一十二户四分以為總法。乃置科綿八萬八千三百三十七兩六錢為實，法七百一十二户四分而一，得一百二十四兩為上等一户所出綿。以五折之，得六十二兩為副等一户所出綿。又五折之，得三十一兩為中等一户所出綿。乃以四折之，得一十二兩四錢為次等一户所出綿。又以四折之，得四兩九錢六分為下等一户所出綿。乃以各等户數各乘一户所出，即各得每等共數之綿。上等一十二户乘一百二十四兩，得一千四百八十八兩；副等八十七户乘六十二兩，得五千三百九十四兩；中等四百六十四户乘三十一兩，得一萬四千三百八十四兩；次等二千三十五户乘一十二兩四錢，得二萬五千二百三十四兩；下等八千四百三十五户乘四兩九錢六分，得四萬一千八百三十七兩六錢。併五等之綿，得八萬八千三百三十七兩六錢，與原科之數相合。

Working: Set the lower-grade households 8,435, apply 4/10 reduction to obtain 3,374. Add this to the fourth-grade households 2,035 to obtain 5,409. Apply 4/10 reduction to 5,409 to obtain 2,163.6. Add this to the middle-grade households 464 to obtain 2,627.6. Apply 1/2 reduction to 2,627.6 to obtain 1,313.8. Add to the second-grade households 87 to obtain 1,400.8. Apply 1/2 reduction to 1,400.8 to obtain 700.4. Add to the upper-grade households 12 to obtain 712.4, which becomes the divisor without further reduction.

The accumulated reduction yields 712.4 as the total divisor. Take the silk assessment 88,337 liang 6 qian as dividend. Divide by divisor 712.4 to obtain 124 liang as the silk per upper-grade household. Apply 1/2 reduction to obtain 62 liang for the second grade. Apply 1/2 reduction again to obtain 31 liang for the middle grade. Apply 4/10 reduction to obtain 12 liang 4 qian for the fourth grade. Apply 4/10 reduction again to obtain 4 liang 9 fen 6 li for the lower grade.

Multiply each per-household amount by the respective household count: upper grade 12 x 124 = 1,488 liang; second grade 87 x 62 = 5,394 liang; middle grade 464 x 31 = 14,384 liang; fourth grade 2,035 x 12.4 = 25,234 liang; lower grade 8,435 x 4.96 = 41,837.6 liang. The sum of all five grades equals 88,337.6 liang, matching the original assessment.

均定勸分

Equalizing Voluntary Contribution Levies

問：欲勸糴賑濟，據甲民户物力畆步排定，共計一百六十二户，作九等：上等三户，第二等五户，第三等七户，第四等八户，第五等十三户，第六等二十一户，第七等二十六户，第八等三十四户，第九等四十五户。今先觀諭，第一等上户願糴五千石，第九等户願糴二百石。欲知各等拋差石數併數認米數各幾何?

Question: It is desired to encourage households to sell grain for famine relief. Based on the property-value and field-size ranking of township A households, there are 162 households in total, divided into nine grades: upper grade 3 households, second grade 5 households, third grade 7 households, fourth grade 8 households, fifth grade 13 households, sixth grade 21 households, seventh grade 26 households, eighth grade 34 households, ninth grade 45 households. After consultation, the first-grade upper households are willing to contribute 5,000 shi, and the ninth-grade households are willing to contribute 200 shi. Find the incremental difference (pao cha) in shi between grades and the total committed rice for each grade.

答曰：認該米二十三萬七千六百石。上等一户米五

Answer: Total committed rice: 237,600 shi. Upper grade:

5,000 shi per household, 3 households = 15,000 shi. Second grade: 4,400 shi per household, 5 households = 22,000 shi. Third grade: 3,800 shi per household, 7 households = 26,600 shi. Fourth grade: 3,200 shi per household, 8 households = 25,600 shi. Fifth grade: 2,600 shi per household, 13 households = 33,800 shi. Sixth grade: 2,000 shi per household, 21 households = 42,000 shi. Seventh grade: 1,400 shi per household, 26 households = 36,400 shi. Eighth grade: 800 shi per household, 34 households = 27,200 shi. Ninth grade: 200 shi per household, 45 households = 9,000 shi.

千石，三户計一萬五千石；二等一户米四千四百石，五户計二萬二千石；三等一户米三千八百石，七户計二萬六千六百石；四等一户米三千二百石，八户計二萬五千六百石；五等一户米二千六百石，一十三户計三萬三千八百石；六等一户米二千石，二十一户計四萬二千石；七等一户米一千四百石，二十六户計三萬六千四百石；八等一户米八百石，三十四户計二萬七千二百石；九等一户米二百石，四十五户計九千石。

Method: Solve by the method of graded distribution. Set the rice amounts of the upper and lower households, subtract to obtain the dividend. List the number of grades, subtract one, and use the remainder as divisor. Divide to obtain the incremental difference in shi. Successively subtract this difference from the upper-grade amount to obtain each grade's amount. Multiply each by the respective household count and add together for the total.

術曰：以衰分求之。置上下户米，減餘為實。列等數減一，餘為法。除之，得抛差石數。以差累減上等米，各得諸等米。以各等户乘之，併之為總數。

Working: Set upper-grade household rice at 5,000 shi, subtract lower-grade 200 shi, remainder 4,800 shi as dividend. Subtract 1 from 9 grades, remainder 8 as divisor. Divide to obtain 600 shi as the incremental difference per grade. Subtract from the upper grade to obtain 4,400 shi as the second-grade amount. Subtract 600 again to obtain 3,800 shi as third grade. Continue subtracting 600: 3,200 shi (fourth), 2,600 shi (fifth), 2,000 shi (sixth), 1,400 shi (seventh), 800 shi (eighth), and 200 shi (ninth). Multiply each by the respective household count and add to obtain the total committed rice of 237,600 shi.

草曰：置上等户米五千石，減下等户二百石，餘四千八百石為實。以九等減一，餘八為法。除實，得六百石為每等抛差。用減上等，餘四千四百石為二等米。又減六百，得三千八百石為三等米。又減六百，得三千二百石為四等米。又減六百，得二千六百石為五等米。又減六百，得二千石為六等米。又減六百，得一千四百石為七等米。又減六百，得八百石為八等米。又減六百，得二百石為九等米。又各乘户數，併之，得總認米石數二十三萬七千六百石。

Equalizing Combined Remittances

均定合解

Question: A prefecture must remit named funds (keming qian) to various offices: Ministry of Revenue 965,421 guan, General Office 643,614 guan, Transport Office 16,090 guan 350 wen. Now, of all the named funds, 9,253 guan 620 wen has been collected in advance. It is desired to distribute this proportionally according to the original quotas for deposit and remittance. How much should each office receive?

問：州郡合解諸司窠名錢：户部九十六萬五千四百二十一貫文，總所六十四萬三千六百一十四貫文，運司一萬六千九十貫三百五十文。今諸窠名先催到九千二百五十三貫六百二十文，欲照原額分數均定樁收解，合各幾何?

Answer: Ministry of Revenue: 5,497 guan 200 wen; General Office: 3,664 guan 800 wen; Transport Office: 91 guan 620 wen.

答曰：户部五千四百九十七貫二百文；總所三千六百六十四貫八百文；運司九十一貫六百二十文。

Method: Solve by the method of graded distribution. Set up the original rates, reducing by common factors where possible. Add together to form the divisor. Multiply the collected amount by each uncombined rate to form the dividends. Divide each dividend by the divisor.

術曰：以衰分求之。置諸原率，可約約之。副併為法。以催到錢乘未併者，各為實。實如法而一。

Working: List Ministry of Revenue 965,421 guan, General Office 643,614 guan, Transport Office 16,090 guan

草曰：列户部九十六萬五千四百二十一貫，總所六十四萬三千六百一十四貫，運司一萬六千九十貫三

百五十文，各為原率。今原率可約，求等得一萬六千九十貫三百五十為等數，俱約之：户部得六十，總所得四十，運司得一，各為率。副併得一百一為法。以催到錢九千二百五十三貫六百二十文乘未併數：六十、四十及一，畢得五十五萬五千二百一十七貫二百文為户部之實，三十七萬一百四十四貫八百文為總所之實，九千二百五十三貫六百二十文為運司之實。三實皆如法一百一而一：其户部得五千四百九十七貫二百文，總所得三千六百六十四貫八百文，運司得九十一貫六百二十，各為候解錢。

350 wen as the original rates. These can be reduced; finding the common measure of 16,090 guan 350 wen, we reduce all: Ministry obtains 60, General Office 40, Transport Office 1 as the rates. Add together to obtain 101 as the divisor. Multiply the collected amount 9,253 guan 620 wen by each uncombined rate (60, 40, and 1): Ministry dividend = 555,217 guan 200 wen; General Office dividend = 370,144 guan 800 wen; Transport Office dividend = 9,253 guan 620 wen. Divide each by the divisor 101: Ministry obtains 5,497 guan 200 wen; General Office obtains 3,664 guan 800 wen; Transport Office obtains 91 guan 620 wen, as the amounts awaiting remittance.

寬減屯租

Reducing Garrison Field Rent

問：屯租欲議寬減，仍聽以夏麥折納分數。官牛種者與減二分，私牛種者與減四分。每歲租穀以三分之一許夏折二麥，内四分大六分小折色。每大麥三石折小麥二石，小麥二石折穀三石五斗。屯租舊額官種一石納租五石，私種一石納租三石。今某州屯田去年計官私種共九千七百八十二石，共合收租穀三萬九千五百八十六石。欲知官私種各數、原額、今減、合催、成年夏麥秋穀幾何？

Question: Garrison field rents are to be reduced, with the option to pay in summer wheat according to specified conversion fractions. For fields cultivated with official oxen, a reduction of 2/10 is granted; for private oxen, a reduction of 4/10. Each year's grain rent allows 1/3 to be paid in summer wheat (two types): 4/10 great barley and 6/10 small wheat as converted payment. The conversion rates are: 3 shi great barley = 2 shi small wheat; 2 shi small wheat = 3.5 shi grain. The old garrison rent quotas are: official cultivation 1 shi seed pays 5 shi rent; private cultivation 1 shi seed pays 3 shi rent. Now a certain prefecture's garrison fields last year had combined official and private seed of 9,782 shi, with total rent grain due of 39,586 shi. Find: the official and private seed amounts, original quotas, current reductions, amounts collectable, and the mature-year summer wheat and autumn grain totals.

［按］此題有官私共種數、租數，有種一石租數，求各種數租數，古謂之差分，今謂之和較比例。折色亦差分也，今謂之和數比例。大小麥與穀相易，古謂之異乘同乘，今謂之和率比列。

[Ed.] This problem gives the combined official and private seed amounts and rent amounts, with the rent per shi of seed, to find each seed and rent amount. Anciently called "difference distribution" (cha fen), now called "sum-difference proportion." The converted payment is also difference distribution, now called "sum-ratio proportion." The exchange of great and small wheat for grain, anciently called "different multiplication, same division," now called "combined-rate proportion."

答曰：官種五千一百二十石，私出種四千六百六十二石。原租額共三萬九千五百八十六石：官種二萬五千六百石，私種一萬三千九百八十六石。今減一萬七百一十四石四斗：官種五千一百二十石，私種五千五百九十四石四斗。合催二萬八千八百七十一石六斗。官種租二萬四千石：一分折麥計穀八千石；四分折大麥三千二百石（係折小麥二千一百三十三石三斗三升三分升之一，復折穀三千七百三十三石三斗三升三分升之二）；六分折小麥四千八百石（係折穀八千四百石）；正色穀一萬六千石。私種租一萬五千八百七十一石六斗：一分折麥計穀五千二百九

Answer: Official seed: 5,120 shi; private seed: 4,662 shi. Original rent quota: 39,586 shi total—official cultivation 25,600 shi, private cultivation 13,986 shi. Current reduction: 10,714 shi 4 dou—official 5,120 shi, private 5,594 shi 4 dou. Amount collectable: 28,871 shi 6 dou. Official cultivation rent 24,000 shi: 1/3 converted to wheat = 8,000 shi grain equivalent; 4/10 great barley = 3,200 shi (equals 2,133 shi 3 dou 3 sheng + 1/3 small wheat, which equals 3,733 shi 3 dou 3 sheng + 2/3 grain); 6/10 small wheat = 4,800 shi (equals 8,400 shi grain); in-kind grain 16,000 shi. Private cultivation rent 15,871 shi

6 dou: 1/3 converted to wheat = 5,290 shi 5 dou + 2/3 grain equivalent; 4/10 great barley = 2,146 shi 8 dou 8 sheng (equals 1,431 shi 2 sheng + 2/3 small wheat, which equals 2,504 shi 3 sheng + 1/3 grain); 6/10 small wheat = 3,220 shi 2 dou 8 sheng (equals 5,635 shi 4 dou 9 sheng grain); 2/10 in-kind grain 10,714 shi 4 dou. Total mature-year receipts: Summer wheat 10,266 shi 6 dou 6 sheng + 2/3, comprising great barley 5,346 shi 8 dou 8 sheng and small wheat 5,919 shi 7 dou 8 sheng + 2/3; autumn in-kind grain 22,667 shi 3 dou 3 sheng + 1/3.

十石五斗三分升之二；四分折大麥二千一百四十六石八斗八升（係折小麥一千四百三十一石二升三分升之二，復折穀二千五百四石三升三分升之一）；六分折小麥三千二百二十石二斗八升（係折穀五千六百三十五石四斗九升）；二分正色穀一萬七百一十四石四斗。成年共收：夏折大小麥一萬二百六十六石六斗六升三分升之二，內大麥五千三百四十六石八斗八升，小麥五千九百一十九石七斗八升三分升之二；秋正穀二萬二千六百六十七石三斗三升三分升之一。

Method: Solve by the method of difference distribution. Set official rent at 5 shi and private rent at 3 shi per shi of seed. Apply the reductions of 2/10 and 4/10: official retains 3.8 (3 shi 8 dou), private retains 1.8 (1 shi 8 dou). Multiply each by 1 shi of seed: official obtains 3.8 shi as the official rate, private obtains 1.8 shi as the private rate. Add together to form the divisor. Take the total rent as dividend. Divide to obtain the amount for one portion. Multiply by each rate to obtain the collectable rent for each category. Divide each rate by the respective rent per shi of seed to obtain the seed amounts. For conversions: 1/3 is the total conversion rate, 4/10 great barley and 6/10 small wheat are the component rates. Multiply each rent by the total rate to obtain the grain-equivalent of converted wheat. Using great barley rate 3 and small wheat rate 2, convert between them to obtain the wheat amounts. Then convert using 2 shi small wheat = 3.5 shi grain to obtain the grain equivalents. Subtract from the collectable rent; the remainder is in-kind autumn grain.

術曰：以差分求之。置官租五石、私租三石，以所減二分、四分各減之，官存三分八釐，私存一分八釐。以各對乘官私種一石，官得三石八斗為官率，私得一石八斗為私率。副併為法。以共租為實。實如法而一，得一分之數。以官私各率乘之，得各合催租。又以種一石各對除官私率，得各合種。其折色以三分之一為總率，四分大、六分小各為子率。以各租乘總率，得折麥計穀之數。以大麥率三、小麥率二通轉相易，各得大小麥。又各以小麥二石折穀三石五斗約之，得大小麥各折穀之數。以減各合催租，餘為正色秋穀。

Equalizing Household Tax Relief

戶稅均寬

Question: A prefecture has granted relief by remitting the autumn tax arrears of the lower three grades of taxable households in a certain county: total remitted 1,355 guan 706 wen in cash and 5,272 shi 1 dou 9 sheng in rice. The property value of the lower three grades in this county is 37,658 guan 500 wen. Now officials have pointed out that many households in this county paid their taxes willingly without arrears, and by remitting only the tax arrears, it seems to favor those who defaulted, which is unfair. It has been proposed to grant the willing-payer households of the three lower grades a reduction in next year's two taxes according to the same pattern. The property value of the three lower grades of non-arrears households is found to be 228,015 guan 321 wen. Find the cash and rice remission per 100 wen of property value, and the total reduction for each.

問：州郡寬恤，近將某縣下三等稅户秋科餘欠錢米，已與蠲放：共錢一千三百五十五貫七百六文，米五千二百七十二石一斗九升。某本縣下三等物力計三萬七千六百五十八貫五百文。今來官員陳述：本縣多有樂輸無欠之户，今蒙蠲放稅尾，似反寬潤頑輸之户，於理未均。遂議將樂輸三等户於明年兩稅與照昨來體例減免。契勘得三等無欠户物力二十二萬八千一十五貫三百二十一文。欲知每百文合減免錢米及共減各幾何？

Answer: Per 100 wen of property value: remit 3.6 wen cash, remit 1.4 ge rice. Next year's two-tax remission: cash 7,949 guan 351 wen 5 fen 5 li 6 hao; rice 30,914 shi 1 dou 4 sheng 4 ge 9 shao 4 chao.

答曰：每物力一百文，放錢三文六分，放米一升四合。明年兩稅放免：錢七千九百四十九貫三百五十一文五分五釐六毫；米三萬九百一十四石一斗四升四合九勺四抄。

術曰：以粟米衰分求之。置原物力為法，除原放錢米，得每百文物力所放錢米率。以各率乘今來物力，各得錢米，為明年兩稅合放數。

Method: Solve by the methods of grain conversion and graded distribution. Take the original property value as divisor. Divide the original remitted cash and rice by this to obtain the cash and rice remission rates per 100 wen of property value. Multiply each rate by the current property value to obtain the cash and rice amounts for next year's two-tax remission.

草曰：以原物力三萬七千六百五十八貫五百文為法。先除原放錢一千三百五十五貫七百六文，定法百文上得一文為商，除得三文六分為物力百文所放錢率。次除原放米五千二百七十二石一斗九升，得一升四合為物力百文所放米率。以今三等户物力二十二萬八千一十五貫三百二十一文徧乘所放錢米二率，得錢七千九百四十九貫三百五十一文五分五釐六毫，得米三萬九百一十四石一斗四升四合九勺四抄，為明年兩稅合放數。

Working: Take the original property value 37,658 guan 500 wen as divisor. First divide the original remitted cash 1,355 guan 706 wen by this: at the 100-wen position, 1 wen as quotient, obtaining 3.6 wen as the cash remission rate per 100 wen of property value. Next divide the original remitted rice 5,272 shi 1 dou 9 sheng, obtaining 1.4 ge as the rice remission rate per 100 wen of property value. Multiply both rates by the current three-grades property value 228,015 guan 321 wen: cash = 7,949 guan 351 wen 5 fen 5 li 6 hao; rice = 30,914 shi 1 dou 4 sheng 4 ge 9 shao 4 chao, as the two-tax remission for next year.

［按］此題之法至為簡明，以物力為法，以所放錢米為實，除之得每百文物力合放之率，又以今來物力乘之，得合放之數。蓋以物力之多寡為差，物力多者放多，物力少者放少，斯為均矣。

[Ed.] The method of this problem is extremely clear: taking property value as divisor and remitted cash and rice as dividend, dividing obtains the remission rate per 100 wen of property value; then multiplying by the current property value obtains the remission amount. Since property value serves as the measure of difference, those with more property value receive more remission, and those with less receive less—this is equitable.

米穀粒分

Rice and Grain Granule Count

問：開倉受約，有甲户米一千五百三十四石。到廊驗得米内夾穀，乃於樣内取米一捻，數計二百五十四粒，内有穀二十八顆。凡粒米率每勺三百。今欲知米内雜穀多少，以折米數科責及粒各幾何？

Question: When opening the granary to accept delivery, household A delivered 1,534 shi of rice. Upon inspection at the corridor, the rice was found to be mixed with unhusked grain. A pinch of the sample was taken, containing 254 granules, of which 28 were unhusked grain. The standard rate is 300 granules per shao. Now it is desired to know how much unhusked grain is mixed in the rice, the amount of rice to be deducted, and the total granule count.

答曰：米一千三百六十四石八斗九升七合六勺一百二十七分勺之四十八；穀一百六十九石一斗二合三勺一百二十七分勺之七十九。合折米八十四石五斗五升一合一勺一百二十七分勺之一百三。原米折米共計四十三億四千八百三十四萬六千四百五十六粒。

Answer: Rice: 1,364 shi 8 dou 9 sheng 7 ge 6 shao + 48/127 shao; Grain: 169 shi 1 dou 2 ge 3 shao + 79/127 shao. Converted rice deduction: 84 shi 5 dou 5 sheng 1 ge 1 shao + 103/127 shao. Total granules of original and converted rice: 4,383,464,456.

術曰：以粟米求之，衰分入之。置樣米粒數為法，以常穀顆數減之，餘與穀為列衰。可約約之。以共米乘列衰為各實，實如法而一，各得米數穀數。置穀數以粟率折之，為穀所折米。次以勺率遍乘米數折米，得粒數。

Method: Solve by the grain conversion method, with graded distribution applied. Set the sample granule count as divisor. Subtract the unhusked grain count; the remainder and the grain count form the graded rates. Reduce by common factors if possible. Multiply the total rice by each graded rate to form the dividends. Divide each by the divisor to obtain the rice and grain amounts. Convert the grain amount using the grain conversion rate to obtain the rice equivalent of the grain. Then multiply both rice amounts by the shao rate to obtain the granule count.

Working: Set the sample pinch count 254 as divisor; unhusked grain 28 as the grain rate; subtract from divisor, remainder 226 as the rice rate. These two rates and the divisor can all be reduced; finding common factor 2: divisor = 127, rice rate = 113, grain rate = 14. Multiply total rice 1,534 shi by each rate: rice dividend = 173,342 shi; grain dividend = 21,476 shi. Divide each by divisor 127: rice = 1,364 shi 8 dou 9 sheng 7 ge 6 shao + 48/127 shao; grain = 169 shi 1 dou 2 ge 3 shao + 79/127 shao. Convert grain at 50% (1 shi grain = 5 dou rice) to obtain 84 shi 5 dou 5 sheng 1 ge 1 shao + 103/127 shao as the rice equivalent of grain. Combine both rice amounts: 1,449 shi 4 dou 4 sheng 8 ge 8 shao + 24/127 shao. Convert to improper fraction: 184,080 shi numerator. Multiply by shao rate 300 granules to obtain 55,224,000,000 granules as dividend. Divide by denominator 127 to obtain 434,834,6456 granules, with remainder 88 discarded.

草曰：置一捻樣粒數二百五十四為法，以帶穀二十八顆為穀衰，以減法，餘二百二十六為米衰。此二衰與法皆可約，求等得二，俱以約之：法得一百二十七，米衰得一百一十三，穀衰得一十四。以米一千五百三十四石遍乘二衰，得一十七萬三千三百四十二石為米實，得二萬一千四百七十六石為穀實。皆如法一百二十七而一：米得一千三百六十四石八斗九升七合六勺一百二十七分勺之四十八，穀得一百六十九石一斗二合三勺一百二十七分勺之七十九。以粟率五十折之，得八十四石五斗五升一合一勺一百二十七分勺之一百三為穀折納米數。併二米，得一千四百四十九石四斗四升八合八勺一百二十七分勺之二十四。先通分納子一十八萬四千八十石，以勺率三百粒乘之，得五千五百二十二億四千萬粒為實，以母一百二十七除之，得四十三億四千八百三十四萬六千四百五十六粒，不盡八十八棄之。

[Ed.] This problem uses a pinch sample to determine the ratio of rice granules to unhusked grain, and thereby ascertains the rice and grain in the full amount—this is the ancient method of "inspecting provisions." With 28 unhusked grains out of 254 granules, the impurity rate is evident. The grain conversion rate of 50% means 1 shi of grain converts to 5 dou of rice. The shao rate of 300 means 300 granules per shao. This method uses graded distribution as the primary technique and grain conversion secondarily—quite ingenious.

［按］按此題以取樣一捻之數，驗米粒與穀粒之率，因以知全數之米穀，古所謂驗食之法也。二百五十四粒中穀二十八顆，其不純率可知矣。以粟率五十折之者，謂穀一石折米五斗也。勺率三百者，謂每勺有三百粒也。此法以衰分為主，而粟米次之，可謂巧矣。

Household Salt Ration Assessment

戶口食鹽

Question: A certain prefecture administers nine counties, which according to regulations distribute salt annually: County A has 143,500 households, County B 97,400, County C 76,500, County D 54,320, County E 43,600, County F 32,480, County G 26,750, County H 19,320, County I 12,560. The prefecture has determined the nine counties' total households to be 486,820. The salt tax assessment is 1 guan wen in full-value coin, to be paid in huizi notes. At that time, old huizi traded at 54 wen full-value per guan, and new huizi at 270 wen full-value per guan. Find the salt tax each county should pay.

問：某州管下九縣，依例歲敷食鹽：甲縣户一十四萬三千五百，乙縣户九萬七千四百，丙縣户七萬六千五百，丁縣户五萬四千三百二十，戊縣户四萬三千六百，已縣户三萬二千四百八十，庚縣户二萬六千七百五十，辛縣户一萬九千三百二十，壬縣户一萬二千五百六十。今來州司根括到九縣共計户四十八萬六千八百二十，每科食鹽一貫文足，係用會子納。其時舊會每貫五十四文足，新會每貫二百七十文足。欲知各縣合納鹽錢幾何?

Answer: County A: 3,189 guan 300 wen; County B: 2,165 guan 200 wen; County C: 1,700 guan wen; County D: 1,206 guan 640 wen; County E: 968 guan wen; County F: 721 guan 600 wen; County G: 594 guan 400 wen; County H: 428 guan 960 wen; County I: 279 guan 120 wen.

答曰：甲縣三千一百八十九貫三百文；乙縣二千一百六十五貫二百文；丙縣一千七百文；丁縣一千二百六貫六百四十文；戊縣九百六十八貫文；已縣七百二十一貫六百文；庚縣五百九十四貫四百文；辛縣四百二十八貫九百六十文；壬縣二百七十九貫一百二十文。

Method: Solve by the method of graded distribution. List

術曰：以衰分求之。列各縣户數為列衰，副併為法。

以共科鹽錢乘未併者，各為實。實如法而一，各得縣合納鹽錢。

each county’s household count as graded rates. Add together to form the divisor. Multiply the total salt tax by each uncombined rate to form the dividends. Divide each dividend by the divisor to obtain each county’s salt tax payment.

草曰：列九縣户數為列衰：甲一十四萬三千五百，乙九萬七千四百，丙七萬六千五百，丁五萬四千三百二十，戊四萬三千六百，已三萬二千四百八十，庚二萬六千七百五十，辛一萬九千三百二十，壬一萬二千五百六十。副併九縣之數，得四十八萬六千八百二十為法。置共科鹽錢一萬八千五百貫文為實。以法四十八萬六千八百二十除之，得每贯率數。以未併各縣户數徧乘之：甲一十四萬三千五百乘率數，得三千一百八十九貫三百文；乙九萬七千四百乘率數，得二千一百六十五貫二百文；丙七萬六千五百乘率數，得一千七百貫文；丁五萬四千三百二十乘率數，得一千二百六貫六百四十文；戊四萬三千六百乘率數，得九百六十八貫文；已三萬二千四百八十乘率數，得七百二十一貫六百文；庚二萬六千七百五十乘率數，得五百九十四貫四百文；辛一萬九千三百二十乘率數，得四百二十八貫九百六十文；壬一萬二千五百六十乘率數，得二百七十九貫一百二十文。各為縣合納鹽錢之數。

Working: List the nine counties’ household counts as graded rates: A 143,500, B 97,400, C 76,500, D 54,320, E 43,600, F 32,480, G 26,750, H 19,320, I 12,560. Add all nine counties to obtain 486,820 as divisor. Take the total salt tax assessment 18,500 guan wen as dividend. Divide by 486,820 to obtain the rate per unit. Multiply each uncombined county household count by this rate: A 143,500 x rate = 3,189 guan 300 wen; B 97,400 x rate = 2,165 guan 200 wen; C 76,500 x rate = 1,700 guan wen; D 54,320 x rate = 1,206 guan 640 wen; E 43,600 x rate = 968 guan wen; F 32,480 x rate = 721 guan 600 wen; G 26,750 x rate = 594 guan 400 wen; H 19,320 x rate = 428 guan 960 wen; I 12,560 x rate = 279 guan 120 wen. Each is the respective county’s salt tax payment.

Fascicle 6A: Money and Grain (Qiangu)

卷六上　錢穀類

Editorial: This fascicle on Money and Grain specializes in grain conversion, light-cargo converted payments, and principal-interest calculations. The ancient Nine Chapters had a chapter on Grain Conversion (su mi) for managing exchange and barter, and a chapter on Construction Calculation (shang gong) for managing labor and volumes. This fascicle combines and expands these topics, comprehensively covering the receipts and disbursements of money and grain, the rise and fall of commodity prices, the light and heavy burden of transport fees, and the complexity of conversion procedures. Its methods take grain conversion as fundamental, supplemented by graded distribution (cui fen), equalized transport (jun shu), and excess-deficit (ying nao) methods. Thus the problems ask about the relative costliness of commodities, the amount of transport fees, the correctness of converted payments, and the increase of principal and interest. All of these matters concern state finance and people's livelihood, and therefore are classified as a separate category.

按：錢穀一章，專主粟米互換、輕齎折納、本息推求之事。蓋古者九章有粟米一章，以御交質變易；又有商功一章，以御功程積實。此則合而廣之，於錢穀之出入、物價之低昂、運脚之輕重、折變之煩簡，無不備載。其為術也，以粟米為本，而以衰分、均輸、盈朒諸法參之。故其題也，或問物價之貴賤，或問運費之多寡，或問折色之正偏，或問本息之增減。凡此數端，皆國用民生之所繫，故列為專類焉。

[Ed.] This fascicle contains nine problems: (1) Assessing Grain Purchase, (2) Converted Remittance of Light Cargo, (3) Tracing Rent to its Origin, (4) Calculating Principal and Interest, (5) Finding the Original from Converted Certificates, (6) Grain and Rice Exchange, (7) Calculating Rice for Yeast, (8) Recovered Transport Fees, (9) Three-Color Average Price. All concern money and grain. The methods mostly use grain conversion techniques, taking commodity price rates as the standard and using the mutual determination of quantities as the application.

［按］此卷凡九問：一曰課糴，二曰折解輕齎，三曰僦直推原，四曰推求本息，五曰易牒知原，六曰粟米交易，七曰計米易麴，八曰囘運費，九曰三色均價。皆錢穀之屬也。其術多用粟米互換之法，蓋以物價之率為準，而以多寡之相求為用也。

Assessing Grain Purchase

課糴

Question: Officials are dispatched to five circuits for state grain purchases. The data are: Zhexi Pingjiang Prefecture—price 35 guan per shi, 135 ge (per dou), water transport to Zhenjiang 900 wen per shi; Anji Prefecture—price 29 guan 500 wen, 110 ge, water transport to Zhenjiang 1 guan 200 wen per shi; Jiangxi Longxing Prefecture—price 28 guan 100 wen, 115 ge, water transport to Jiankang 1 guan 700 wen per shi; Jizhou—price 25 guan 850 wen, 120 ge, water transport to Jiankang 2 guan 900 wen per shi; Hunan Tanzhou—price 27 guan 300 wen, 118 ge, water transport to Ezhou 1 guan 700 wen per shi. All payments are in 17th-series official huizi notes; all rice is measured using the Wensi Bureau hu (1 hu = 83 ge) for conversion. Find the price per official-hu shi for each circuit and compare their relative costliness.

問：差人五路和糴。據浙西平江府石價三十五貫文，一百三十五合，至鎮江水脚錢每石九百文；安吉州石價二十九貫五百文，一百一十合，至鎮江水脚錢每石一貫二百文；江西隆興府石價二十八貫一百文，一百一十五合，至建康水脚錢每石一貫七百文；吉州石價二十五貫八百五十文，一百二十合，至建康水脚錢每石二貫九百文；湖南潭州石價二十七貫三百文，一百一十八合，至鄂州水脚錢每石一貫七百文。其錢並十七界官會，其米並用文思院斛交量紐數。欲皆以官解計石錢相比貴賤幾何？

［按］按草中係二貫一百文，此訛。文思院解每斗八十三合。

[Ed.] Note: The working mentions 2 guan 100 wen for Tanzhou water transport, which appears to be an error. The Wensi Bureau hu equals 83 ge per dou.

答曰：文思院斛石錢：安吉州二十三貫一百六十四文一十一分文之六；平江府二十二貫七十一文二十七分文之二十三；隆興府二十一貫五百七文二十三分文之一十九；潭州二十貫六百七十九文五十九分文之四十九；吉州一十九貫八百八十五文十二分文之五。

Answer: Wensi Bureau hu price per shi: Anji Prefecture 23 guan 164 wen + 6/11 wen; Pingjiang Prefecture 22 guan 71 wen + 23/27 wen; Longxing Prefecture 21 guan 507 wen + 19/23 wen; Tanzhou 20 guan 679 wen + 49/59 wen; Jizhou 19 guan 885 wen + 5/12 wen.

［按］按三十九訛四十九。

[Ed.] Note: 49 is a scribal error for 39.

術曰：以粟米互換求之。置石價併水脚，乘石數，又乘官斗合數為實。各如本州合數而一，各得官解石錢。以課貴賤。

Method: Solve by the grain conversion method. Set the shi price combined with water transport fee, multiply by the shi count, then multiply by the official dou ge count to form the dividend. Divide each by the respective circuit's dou ge count to obtain the price per official-hu shi. Compare to determine relative costliness.

草曰：置安吉州石價二十九貫五百文，平江石價三十五貫文，隆興石價二十八貫一百文，吉州石價二十五貫八百五十文，潭州石價二十七貫三百文，列右行。次置水脚：安吉一貫二百文，平江九百文，隆興一貫七百文，吉州二貫九百文，潭州二貫一百文，列左行，各對本州石價。以兩行數併之，得數：安吉三十貫七百文，平江三十五貫九百，隆興二十九貫八百，潭州二十九貫四百，吉州二十八貫七百五十，仍於右行。次以文思院官斗八十三合遍乘之：安吉州得二千五百四十八貫一百文，平江府得二千九百七十九貫七百文，江西隆興得二千四百七十三貫四百文，湖南潭州得二千四百四十貫二百文，江南吉州得二千三百八十六貫二百五十文，各為實於右。得次列安吉斗一百一十合，平江斗一百三十五合，隆興斗一百一十五合，潭州斗一百一十八合，吉州斗一百二十合於左行為法。以对除右行之實：安吉得二十三貫一百六十四文一十一分之六，平江得二十三貫七十一文二十七分文之二十三，隆興得二十一貫五百七文二十三分文之一十九，潭州得二十貫六百七十九文五十九文分之四十九，吉州得一十九貫八百八十五文一十二分文之五。相課石價，其安吉州最貴，平江次之，隆興又次之，潭州又次之，吉州最賤。

Working: Set the shi prices in the right column: Anji 29 guan 500 wen, Pingjiang 35 guan, Longxing 28 guan 100 wen, Jizhou 25 guan 850 wen, Tanzhou 27 guan 300 wen. Set the water transport fees in the left column: Anji 1 guan 200 wen, Pingjiang 900 wen, Longxing 1 guan 700 wen, Jizhou 2 guan 900 wen, Tanzhou 2 guan 100 wen, each paired with its circuit's shi price. Combine the two columns: Anji 30 guan 700 wen, Pingjiang 35 guan 900 wen, Longxing 29 guan 800 wen, Tanzhou 29 guan 400 wen, Jizhou 28 guan 750 wen, still in the right column. Next multiply each by the Wensi Bureau standard of 83 ge: Anji = 2,548 guan 100 wen, Pingjiang = 2,979 guan 700 wen, Longxing = 2,473 guan 400 wen, Tanzhou = 2,440 guan 200 wen, Jizhou = 2,386 guan 250 wen, as dividends in the right column. Place each circuit's dou measure in the left column as divisor: Anji 110 ge, Pingjiang 135 ge, Longxing 115 ge, Tanzhou 118 ge, Jizhou 120 ge. Divide the right-column dividends by the left-column divisors: Anji = 23 guan 164 wen + 6/11 wen; Pingjiang = 22 guan 71 wen + 23/27 wen; Longxing = 21 guan 507 wen + 19/23 wen; Tanzhou = 20 guan 679 wen + 49/59 wen; Jizhou = 19 guan 885 wen + 5/12 wen. Comparing the prices: Anji is most expensive, followed by Pingjiang, then Longxing, then Tanzhou, with Jizhou the cheapest.

［按］按此題以各路米價併水脚錢，以官斛八十三合為準，約為官斛石錢，以相比較貴賤。其法以每路斗合數除之，使歸於一律，然後可以相課。此所謂異乘同除之法也。

[Ed.] This problem combines each circuit's rice price with water transport fees, using the standard official hu of 83 ge to convert to price per official-hu shi for comparison. The method divides by each circuit's dou ge measure to normalize to a common standard before comparing. This is the so-called "different multiplication, same division" method.

Fascicle 6B: Money and Grain (continued)

卷六下　錢穀類（續）

Tracing Rent to its Origin

Question: Among the shop-rent records, one household currently pays 156 wen 8 fen in full-value coin per day as the standard. The regulations state: those never reduced receive a 3/10 reduction; those already reduced by 3/10 receive a further 2/10 reduction; those already reduced by 2/10 receive yet another 2/10 reduction. Now this household has been subject to cumulative reductions. Find the original rent amount.

Answer: Original amount: 350 wen.

Method: Solve by the method of graded distribution. Place two rows of 10, each with three positions. List the reduction fractions and subtract from the right row. Multiply the remainders together to form the divisor. Multiply the original entries in the left row together, then multiply by the current payment to form the dividend. Divide to obtain the original rent.

Working: Place three 10s in the left row and three 10s in the right row. From the upper right subtract the first reduction of 3/10 (10 – 3 = 7), from the middle right subtract the second reduction of 2/10 (10 – 2 = 8), from the lower right subtract the further reduction of 2/10 (10 – 2 = 8). The right row remainders 7, 8, 8 multiply together to give 448 as the divisor. The left row three 10s multiply to give 1,000 as the multiplier. Multiply by the current daily payment 156 wen 8 fen to obtain 156 guan 800 wen as dividend. Divide by 448 to obtain 350 wen as the household's original rent.

[Ed.] This problem uses cumulative reductions to trace back to the original amount; the method is entirely clear. First reduction of 3/10 leaves 7/10; second reduction of 2/10 leaves 8/10; further reduction of 2/10 leaves 8/10. These three numbers multiplied together give 448 parts. Taking the payment 156.8 wen as the dividend for 448 parts, the dividend for 1,000 parts (the original) must be 350 wen.

僦直推原

問：房廊數内，一户日納一百五十六文八分足為準。指揮未曾經減者，減三分；已曾經減三分者，減二分；已曾經減二分者，更減二分。今本户累經減者，欲知原額房錢幾何？

答曰：原額三百五十文。

術曰：以衰分求之。列一十分兩行，各三位；列減分，對減右行。以餘者相乘為法。以左行原列相乘，得納錢為實。實如法而一，得原額錢。

草曰：列一十三位於左行，又列一十分三位於右行。其右上減去初減三分，右中減去次減二分，右下減去更減二分。右行餘七八八，以相乘得四百四十八為法。乃以左行三位一十分相乘，得一千為乘率。以乘見日納錢一百五十六文八分，得一百五十六貫八百文為實。實如法而一，得三百五十文，為本户原額戸錢。

［按］此題以累減之法推原額，術意至明。初減三分存七分，次減二分存八分，更減二分存八分。三數相連乘得四百四十八分，以納錢一百五十六文八分為四百四十八分之實，則原額一千分之實必為三百五十文矣。

Calculating Principal and Interest

Question: Three treasuries have the following interest rules: above 10,000 guan pays 1 li (0.001), above 1,000 guan pays 2.5 li (0.0025), above 100 guan pays 3 li (0.003). Treasury A has principal 493,800 guan; Treasury B 370,300 guan; Treasury C 246,800 guan. Now the three treasuries together have collected 25,644 guan 200 wen in interest. The assessment rates are: A uses inverted-cone difference, B uses square-cone difference, and C uses puncture-

推求本息

問：三庫息例：萬貫以上一釐，千貫以上二釐五毫，百貫以上三釐。甲庫本四十九萬三千八百貫，乙庫本三十七萬三百貫，丙庫本二十四萬六千八百貫。今三庫共約到息錢二萬五千六百四十四貫二百文。其典率：甲反錐差，乙方錐差，丙蒺藜差。欲知原典三例本息各幾何?

vine difference. Find the original principal and interest for each treasury under the three rates.

［按］此即衰分題也。其差有反錐、方錐、蒺藜之名，蓋以一二三遞減如立錐為反錐，以一四九平方遞加為方錐，以一三六三數遞加為蒺藜。是必古有其名也。至以各差求各本，則因各本原依各差入之也。

[Ed.] This is a graded-distribution problem. The differences have names: inverted cone (fan zhui), square cone (fang zhui), and puncture vine (ji li). The inverted cone decreases by 1, 2, 3 like an upright cone; the square cone increases by squares 1, 4, 9; the puncture vine increases by 1, 3, 6. These must have been ancient terms. The use of each difference to find each principal reflects how the principals were originally assigned according to these differences.

答曰：甲庫共納息九千五十三貫文：一釐息二千四百六十九貫文，二釐半息四千一百一十五貫文，三釐息二千四百六十九貫文。

乙庫共納息一萬五十一貫文：一釐息二百六十四貫五百文，二釐半息二千六百四十五貫文，三釐息七千一百四十一貫五百文。

丙庫共納息六千五百四十貫二百文：一釐息二百四十六貫八百文，二釐半息一千八百五十一貫文，三釐息四千四百四十二貫四百文。

Answer: **Treasury A** total interest 9,053 guan: 1-li rate 2,469 guan; 2.5-li rate 4,115 guan; 3-li rate 2,469 guan.

Treasury B total interest 10,051 guan: 1-li rate 264 guan 500 wen; 2.5-li rate 2,645 guan; 3-li rate 7,141 guan 500 wen.

Treasury C total interest 6,540 guan 200 wen: 1-li rate 246 guan 800 wen; 2.5-li rate 1,851 guan; 3-li rate 4,442 guan 400 wen.

術曰：置諸庫諸色之差，照釐率為三行。縱併之為約率，橫命之為乘率。以約率各約自庫之本，各得。以遍乘未併乘率，然後各以釐率橫乘之，次以縱併之，為各庫共息。

Method: Set up the differences for each treasury's categories, arranged as three rows according to the li rates. Add vertically to obtain the reduction rates; assign horizontally as multiplication rates. Divide each treasury's principal by its reduction rate to obtain the quotients. Multiply these throughout by the uncombined multiplication rates, then multiply horizontally by each li rate, and finally add vertically to obtain each treasury's total interest.

草曰：置甲庫反錐差，自下置三二一於右行。次置乙庫方錐差，自上置一四九於中行。次置丙庫蒺藜差，自上置一三六於左行，各為三庫上中下三等乘率。乃縱併甲差三二一，得六為甲約率。縱併乙差一四九，得一十四為乙約率。縱併丙差一三六，得一十為丙約率。直命九位數各為上中下乘率。乃先以約率各約自庫之本：乃以甲約率六約甲本四十九萬三千八百貫，得八萬二千三百貫為甲得。次以乙約率一十四約乙本三十七萬三百貫，得二萬六千四百五十貫為乙得。次以丙約率一十約丙本二十四萬六千八百貫，得二萬四千六百八十貫為丙得。以各得乘未併乘率：其甲所得八萬二千三百貫，乘反錐乘率三二一，得二十四萬六千九百貫為上率，得一十六萬四千六百貫為中率，得八萬二千三百貫為下率。其乙所得二萬六千四百五十貫，以乘方錐差一四九，得二萬六千四百五十貫為上率，得一十萬五千八百貫為中率，得二十三萬八千五十貫為下率。其丙所得二萬四千六百八十貫，以乘蒺藜差一三六，得二萬四千六百八十貫為上率，得七萬四千四十貫為中率，得一十四萬八千八十貫為下率。然後各以息釐數乘各庫三乘：其甲以一釐乘上率二十四萬六千九百貫，得二千四百六十九貫為上息；以二釐五毫乘中率一十六萬四千六百貫，得四千一百一十五貫為中息；以三釐乘

Working: Set Treasury A inverted-cone difference, place 3, 2, 1 from bottom in the right row. Set Treasury B square-cone difference, place 1, 4, 9 from top in the middle row. Set Treasury C puncture-vine difference, place 1, 3, 6 from top in the left row. These are the three treasuries' upper-middle-lower multiplication rates. Add A's differences vertically: 3+2+1 = 6 as A's reduction rate. Add B's: 1+4+9 = 14 as B's reduction rate. Add C's: 1+3+6 = 10 as C's reduction rate. The nine numbers are assigned as upper-middle-lower multiplication rates.

Divide each treasury's principal by its reduction rate: A principal 493,800 / 6 = 82,300 guan as A's quotient. B principal 370,300 / 14 = 26,450 guan as B's quotient. C principal 246,800 / 10 = 24,680 guan as C's quotient.

Multiply each quotient by the uncombined multiplication rates: A's 82,300 x inverted-cone rates 3, 2, 1 = upper 246,900 guan, middle 164,600 guan, lower 82,300 guan. B's 26,450 x square-cone rates 1, 4, 9 = upper 26,450 guan, middle 105,800 guan, lower 238,050 guan. C's 24,680 x puncture-vine rates 1, 3, 6 = upper 24,680 guan, middle 74,040 guan, lower 148,080 guan.

Then multiply each treasury's three rates by the interest li: A: 1 li x upper 246,900 = 2,469 guan upper interest;

2.5 li x middle 164,600 = 4,115 guan middle interest; 3 li x lower 82,300 = 2,469 guan lower interest. Total A interest = 9,053 guan.

B: 1 li x 26,450 = 264.5 guan upper; 2.5 li x 105,800 = 2,645 guan middle; 3 li x 238,050 = 7,141.5 guan lower. Total B = 10,051 guan.

C: 1 li x 24,680 = 246.8 guan upper; 2.5 li x 74,040 = 1,851 guan middle; 3 li x 148,080 = 4,442.4 guan lower. Combined total interest = 25,644 guan 200 wen.

下率八萬二千三百貫，得二千四百六十九貫為下息。併上中下三息，得九千五十三貫文為甲庫共息。其乙庫以一釐乘上率二萬六千四百五十貫，得二百六十四貫五百文為上息；以二釐五毫乘中率一十萬五千八百貫，得二千六百四十五貫為中息；以三釐乘下率二十三萬八千五十貫，得七千一百四十一貫五百為下息。併上中下三息，得一萬五十一貫文為乙庫共息。其丙庫以一釐乘上率二萬四千六百八十貫，得二百四十六貫八百為上息；以二釐五毫乘中率七萬四千四十貫，得一千八百五十一貫為中息；以三釐乘下率一十四萬八千 080 貫，得四千四百四十二貫四百文為下息。併三庫共息，得二萬五千六百四十四貫二百文為總息。

Finding the Original from Converted Certificates

[Ed.] Note: The original edition lacked this problem statement; it has been restored here.

Question: Official certificates (du die) were issued for dispatching agents to conduct trade: every 3 certificates exchange for 13 bags of salt; 2 bags of salt exchange for 84 bolts of cloth; 15 bolts of cloth exchange for 3.5 bolts of silk; 6 bolts of silk exchange for 7.2 liang of silver. Now 9,172.8 liang of silver has been obtained. Find the original number of certificates issued.

Answer: Certificates: 180.

Method: Solve by the grain conversion method using mutual multiplication for exchange. List each pair of numbers with their original commodities facing each other, like goose wings in flight. Multiply the numbers with one more item in the chain to form the dividend; multiply those with one fewer item to form the divisor. Divide to obtain the answer.

Working: First multiply certificates 3 by salt 2 bags = 6. Multiply by cloth 15 = 90. Multiply by silk 6 bolts = 540. Multiply by silver 91,728 qian (9,172.8 liang converted) = 49,533,120 qian as dividend. Next multiply salt 13 bags by cloth 84 = 1,992. Multiply by silk 3.5 bolts = 6,972. Multiply by silver 7.2 liang = 501,984 qian as divisor. Divide: 49,533,120 / 501,984 = 180 certificates as the original number issued.

[Ed.] This problem involves four successive exchanges: certificates to salt, salt to cloth, cloth to silk, silk to silver. The method multiplies the chain with one more item as dividend and the chain with one fewer item as divisor—this is the mutual multiplication method. The "goose wing" arrangement places original commodities facing each other, like geese flying in symmetric formation.

易牒知原

［按］按舊本此問無題，今增入。

問：出度牒差人營運：每三道易鹽一十三袋，鹽二袋易布八十四疋，布一十五疋易絹三疋半，絹六疋易銀七兩二錢。今趂到銀九千一百七十二兩八錢。欲知原闕度牒道數幾何？

答曰：度牒一百八十道。

術曰：以粟米互乘易法求之。列各數以本色相對，如鴈翅。以多一事者相乘為實，以少一事者相乘為法，除之。

草曰：先以度牒三道乘鹽二袋，得六。以乘布一十五，得九十。又乘絹六疋，得五百四十。乃乘銀九萬一千七百二十八錢，得四千九百五十三萬三千一百二十錢為實。次以鹽一十三袋乘布八十四，得一千九百九十二。以乘絹三疋五分，得六千九百七十二。乃乘銀七兩二錢，得五十萬一千九百八十四錢為法。除實，得一百八十道為原闕度牒。

［按］按此題以度牒易鹽、鹽易布、布易絹、絹易銀，凡四變。術以多一事者相乘為實，少一事者相乘為法，此互乘之法也。鴈翅列者，以本色對列，如鴈翅飛行，左右對稱之義也。

Grain and Rice Exchange

粟米交易

［按］按舊本此問無題，今增。

[Ed.] Note: The original edition lacked this problem statement; it has been restored.

問：菽三升易小麥二升，小麥一升五合易油麻八合，油麻一升二合易粳米一升八合。今將菽一十四石四斗欲易油麻，又將小麥二十一石六斗欲易粳米。問各幾何？

Question: 3 sheng of beans exchange for 2 sheng of small wheat; 1.5 sheng of small wheat exchanges for 8 ge of sesame; 1.2 sheng of sesame exchanges for 1.8 sheng of polished rice. Now 14 shi 4 dou of beans are to be exchanged for sesame, and 21 shi 6 dou of small wheat are to be exchanged for polished rice. Find each amount.

答曰：油麻五石一斗二升；粳米一十七石二斗八升。

Answer: Sesame: 5 shi 1 dou 2 sheng; Polished rice: 17 shi 2 dou 8 sheng.

術曰：以粟米換易求之。置原易率，本色對列，如鴈翅。以多一事者相乘為實，以少一事者相乘為法，除之。各得或問數，不干其率者不置。

Method: Solve by the grain conversion exchange method. Set up the original exchange rates with original commodities facing each other, like goose wings. Multiply the numbers with one more item in the chain to form the dividend; multiply those with one fewer item to form the divisor. Divide to obtain the answers. Items not in the relevant chain are not used.

草曰：置四色六數，列六位率，如鴈翅，皆化為合。先將菽一十四石四斗化作一萬四千四百合，乃對前二句率數四位，如鴈翅，至欲易油麻止，共五事為上圖。次將小麥二十一石六斗化為二萬一千六百合，乃對後兩句率四位，鴈翅，至欲粳米止，共五事為下圖。

下圖：其上圖以菽一萬四千四百合乘麥二十，得二十八萬八千，又乘油麻八合，得二百三十萬四千合為油麻實。次以菽三十合乘麥一十五合，得四百五十合為法。除之，得五千一百二十合，展為五石一斗二升為油麻。

其下圖以小麥二萬一千六百合乘油麻八合，得一十七萬二千八百合，又乘粳米一十八合，得三百一十一萬四百合為粳米實。以小麥一升五合乘油麻一十二合，得一百八十合為法。除之，得一萬七千二百八十，展作一十七石二斗八升為粳米。

Working: Set up four commodities with six numbers, arranged in six positions like goose wings, all converted to ge. First convert beans 14 shi 4 dou to 14,400 ge. Pair with the first two exchange rates (4 numbers) for the upper diagram, ending at the sesame exchange, 5 items total. Next convert small wheat 21 shi 6 dou to 21,600 ge. Pair with the last two exchange rates (4 numbers) for the lower diagram, ending at the polished rice exchange, 5 items total.

Upper diagram: Multiply beans 14,400 ge by wheat rate 20 (the cross-rate) = 288,000; then multiply by sesame 8 ge = 2,304,000 ge as sesame dividend. Multiply beans 30 ge by wheat 15 ge = 450 ge as divisor. Divide: 2,304,000 / 450 = 5,120 ge = 5 shi 1 dou 2 sheng of sesame.

Lower diagram: Multiply small wheat 21,600 ge by sesame 8 ge = 172,800 ge; then multiply by polished rice 18 ge = 3,114,000 ge as polished rice dividend. Multiply small wheat 15 ge by sesame 12 ge = 180 ge as divisor. Divide: 3,114,000 / 180 = 17,280 ge = 17 shi 2 dou 8 sheng of polished rice.

計米易麴

Calculating Rice for Yeast Production

［按］按舊本此問無題，今增。

[Ed.] Note: The original edition lacked this problem statement; it has been restored.

問：庫率：粳穀七石出米三石，糯米一斗易小麥一斗七升，小麥五升踏麴二斤四兩，麴一百一十斤醞糯米一石三斗。今有糯穀一千七百五十九石三斗八升，欲出穀做米，易麥踏麴，還自醞，餘穀之米須令適足。各合幾何？

Question: Granary rates: 7 shi of glutinous grain yields 3 shi of rice; 1 dou of glutinous rice exchanges for 1 dou 7 sheng of small wheat; 5 sheng of small wheat produces 2 jin 4 liang of yeast; 110 jin of yeast ferments 1 shi 3 dou of glutinous rice. Now there are 1,759 shi 3 dou 8 sheng of glutinous grain. It is desired to hull the grain to make rice, exchange the rice for wheat, use the wheat to produce yeast, and ferment the remaining grain's rice with this yeast so that it is exactly sufficient. Find each amount.

Answer: Total grain: 1,759 shi 3 dou 8 sheng. Grain hulled: 924 shi, yielding rice 396 shi. Wheat obtained: 673 shi 2 dou. Yeast produced: 30,294 jin. Remaining grain: 835 shi 3 dou 8 sheng. Fermented rice: 358 shi 2 sheng.

答曰：共穀一千七百五十九石三斗八升。出穀九百二十四石，得米三百九十六石。易麥六百七十三石二斗。踏麴三萬二百九十四斤。餘穀八百三十五石三斗八升。醞米三百五十八石二升。

Method: Solve by the grain conversion exchange method. Set up the rates with original commodities facing each other, like goose wings. Where there are fractions, convert them; where commodities differ, transform them. Multiply from top to bottom (forward multiplication) and from bottom to top (backward multiplication) to obtain combined numbers. Where pairs face each other, multiply them; where there is no pair, assign directly—these become the rates. Add the unmatched top and bottom rates to form the divisor rate. Multiply the given amount by all the rates (excluding the divisor rate) to form dividends. Divide all dividends by the divisor to obtain the converted amounts, then perform mutual exchanges and divisions to obtain the final answers.

術曰：以粟米換易求之。置諸率，隨本色對列，如鴈翅。有分者通之，異類者變之，以頭位者進乘之，以下位退乘之，得合數。有對者相乘之，無對者直命之，為諸率。併上下無對者為法率。以今有物徧乘諸率（不乘法率），各為實。諸實並如法而一，各得其已變者，復互易乘除之，即得所求。

[Ed.] This is the most complex of all grain conversion problems. It takes glutinous grain as the base, hulls it to rice, exchanges rice for wheat, uses wheat to produce yeast, ferments rice with yeast, and the remaining grain's rice must be exactly sufficient. The "convert fractions" means converting 2 jin 4 liang to 9/4 jin; "transform different kinds" means converting fermented rice back to glutinous grain; "forward and backward multiplication" means multiplying from top to bottom and bottom to top. The method uses the combined diagram to find rates, then multiplies by the given amount and divides—a variation of graded distribution.

［按］此題為粟米換易中之最繁者，以糯穀為本，出米易麥，麥踏麴，麴醞米，而餘穀之米又須適足。術中所謂有分者通之，謂二斤四兩通為四分斤之九也；異類者變之，謂醞米變為糯穀也；進乘退乘者，自上而下為進，自下而上為退也。其法以合圖求諸率，以今有物遍乘，如法而一，此衰分之變法也。

Recovered Transport Fees

回運費

Question: Rice transported by water from Jiangxi, originally destined for discharge at Zhenjiang, with a water route of 2,130 li and water transport fee of 1 guan 200 wen per shi. Now this rice is diverted to be stored at Chizhou instead; the distance from Chizhou to Zhenjiang is 880 li. Find the water transport fee that should be recovered (for the route not traveled).

問：有江西水運米一十二萬三千四百石，原係鎮江交卸，計水程二千一百三十里，每石水脚錢一貫二百文。今截上件米就池州安頓，池州至鎮江八百八十里。欲收回不該水脚錢幾何？

Answer: Amount to recover: 61,178 guan 591 wen.

答曰：收回錢六萬一千一百七十八貫五百九十一文。

Method: Solve by the grain conversion method. Take the distance from Chizhou to Zhenjiang in li, multiply by the water transport fee per shi, then multiply by the rice amount to form the dividend. Take the original distance to Zhenjiang as divisor. Divide to obtain the recovered fee.

術曰：以粟米互易求之。置池州至鎮江里數，乘水脚錢，得數又乘運米為實。以原至鎮江水程為法，除實，得收回錢。

Working: Take Chizhou to Zhenjiang 880 li, multiply by water transport fee 1 guan 200 wen per shi = 1,056 guan.

草曰：置池州至鎮江八百八十里，乘每石水脚錢一貫二百，得一千五十六貫文。又乘運米一十二萬三

千四百石，得一億三千三十一萬四百貫文為實。以原至鎮江水程二千一百三十里為法，除實，得六萬一千一百七十八貫五百九十一文為收回錢。

Multiply by rice amount 123,400 shi = 130,310,400 guan as dividend. Divide by original route 2,130 li: 130,310,400 / 2,130 = 61,178 guan 591 wen as the recovered transport fee.

［按］此題以水程里數為率，里遠則水脚多，里近則水脚少。今截米就池州安頓，則少行八百八十里之水脚，故應收回。以池州里數乘水脚，為所收回之率，以原程為法，除之得收回之數。此亦異乘同除之法也。

[Ed.] This problem uses the water route distance as the rate: longer distances entail higher transport fees, shorter distances lower fees. By diverting the rice to Chizhou, 880 li of transport is saved, and the corresponding fee should be recovered. Multiplying the Chizhou distance by the transport fee gives the recovery rate; dividing by the original route gives the recovered amount. This is also the "different multiplication, same division" method.

三色均價

Three-Color Average Price

問：庫有三色金共五千兩：內八分金一千二百五十兩，兩價四百貫文；七分五釐金一千六百兩，兩價三百七十五貫文；八分五釐金二千一百五十兩，兩價四百二十五貫文。並欲煉為足色，每兩工食藥炭錢三貫文，耗金九百七十二兩五錢。欲知色分及兩價各幾何?

Question: A treasury has three grades of gold totaling 5,000 liang: 8-fen gold (80% purity) 1,250 liang at 400 guan per liang; 7.5-fen gold (75% purity) 1,600 liang at 375 guan per liang; 8.5-fen gold (85% purity) 2,150 liang at 425 guan per liang. All are to be refined to full purity (10 fen), with labor, food, medicine, and charcoal costs of 3 guan per liang, and gold loss of 972.5 liang. Find the purity and the price per liang of the refined gold.

答曰：色十分；兩價五百三貫七百二十四文，五百三十七分文之二百一十二。

Answer: Purity: 10 fen (full). Price per liang: 503 guan 724 wen + 212/537 wen.

術曰：以方田及粟米求之。置共數，以耗減之，餘為法。以三色分數各乘兩數，併之為色分實。以三色價數各乘兩數為寄，以工藥價乘共金，併寄，共為價實。二實皆如法而一，即各得。

Method: Solve by the methods of rectangular fields (fang tian) and grain conversion. Take the total amount, subtract the loss; the remainder is the divisor. Multiply each grade's purity by its liang amount and add together to form the purity dividend. Multiply each grade's price by its liang amount as accumulators; multiply the labor/medicine cost by the total gold and add to the accumulators to form the price dividend. Divide both dividends by the divisor to obtain the answers.

草曰：置共金五千兩，減耗九百七十二兩五錢，外餘四千二十七兩五錢為法。次置一千二百五十兩，乘八分，得一萬分於上。置一千六百兩，以七分五釐乘之，得一萬二千分，加上。置二千一百五十兩，乘八分五釐，得一萬八千二百七十五分，又加上，共得四萬二百七十五分為分實。次置一千二百五十兩乘四百貫，得五十萬貫為寄。次置一千六百兩乘價三百七十五貫，得六十萬貫，加寄。次置二千一百五十兩乘價四百二十五貫，得九十一萬三千七百五十貫，又加寄。次置共金五千兩，乘工藥錢三貫，得一萬五千貫，又加寄，共得二百二萬八千七百五十貫為貫實。二實並如法四千 027 兩五錢而一：得其色一十分；其價每兩得五百三貫七百二十四文，不盡一貫五百九十。與法求等，得七十五，俱約之，為五百三十七文之二百一十二。

Working: Take total gold 5,000 liang, subtract loss 972.5 liang, remainder 4,027.5 liang as divisor. Next: 1,250 liang x 8 fen = 10,000 fen (upper position). 1,600 liang x 7.5 fen = 12,000 fen; add to previous. 2,150 liang x 8.5 fen = 18,275 fen; add again. Total purity dividend = 40,275 fen. Next: 1,250 liang x 400 guan = 500,000 guan as accumulator. 1,600 liang x 375 guan = 600,000 guan; add. 2,150 liang x 425 guan = 913,750 guan; add again. Total gold 5,000 liang x labor/medicine 3 guan = 15,000 guan; add. Total price dividend = 2,028,750 guan. Divide both dividends by divisor 4,027.5: purity = 10 fen (full). Price per liang = 503 guan 724 wen, remainder 1 guan 590 wen. Finding common factor 75 with the divisor and reducing: remainder = 212/537 wen.

［按］按此題以三色金煉為足色，其法以色分為實，以減耗後之數為法，除之得色分。又以各價併工藥

[Ed.] This problem refines three grades of gold to full purity. The method uses purity as dividend and the loss-

subtracted amount as divisor to obtain the purity grade. It combines all prices with labor/medicine costs as the price dividend, and divides to obtain the price per liang. This combines methods from rectangular fields and grain conversion. The gold loss of 972.5 liang represents refining wastage; subtracting it gives the actual refined gold amount.

為價實，如法除之得兩價。此乃方田與粟米兼用之法也。耗金九百七十二兩五錢者，謂煉金之損耗也，減之則得實在之金數。

Exchange of Valued Commodities

Question: Three grades of persons A, B, and C exchange commodities: A exchanges 12 liang of gold for B's silver; B exchanges 15 liang of silver for C's silk; C exchanges 24 bolts of silk for A's gold. At that time, gold is 400 guan per liang, silver 32 guan per liang, and silk 25 guan per bolt. Now the three parties exchange; find the value of what each receives.

Answer: A exchanges 12 liang of gold (value 4,800 guan) for 150 liang of B's silver. B exchanges 15 liang of silver (value 480 guan) for 19 bolts 2 chi of C's silk. C exchanges 24 bolts of silk (value 600 guan) for 1.5 liang of A's gold.

Method: Solve by the grain conversion exchange method. Take each person's commodity amount, multiply by its price to form the dividend. Take the price of the commodity being exchanged for as divisor. Divide to obtain the amount received in exchange.

Working: Take A's gold 12 liang, multiply by price 400 guan = 4,800 guan as dividend. Divide by silver price 32 guan = 150 liang as the silver A receives from B. Take B's silver 15 liang, multiply by price 32 guan = 480 guan as dividend. Divide by silk price 25 guan = 19 bolts remainder 5 guan; convert remainder 5 guan at 2.5 zhang per bolt = 2 zhang, so B receives 19 bolts 2 zhang of silk from C. Take C's silk 24 bolts, multiply by price 25 guan = 600 guan as dividend. Divide by gold price 400 guan = 1.5 liang as the gold C receives from A.

貴賤互易

問：有甲乙丙三等人互易物貨：甲以金十二兩易乙銀，乙以銀一十五兩易丙絹，丙以絹二十四疋易甲金。其時金每兩價四百貫，銀每兩價三十二貫，絹每疋價二十五貫。今三人互易，各欲知所得物價幾何?

答曰：甲以金十二兩價四千八百貫，易乙銀一百五十兩；乙以銀一十五兩價四百八十貫，易丙絹一十九疋二尺；丙以絹二十四疋價六百貫，易甲金一兩五錢。

術曰：以粟米互換求之。置各人物數，各乘其價為實，各以所易物價為法，實如法而一，各得所易之數。

草曰：置甲金十二兩，乘價四百貫，得四千八百貫為實。以銀價三十二貫為法除之，得一百五十兩為甲易乙之銀。置乙銀一十五兩，乘價三十二貫，得四百八十貫為實。以絹價二十五貫為法除之，得一十九疋餘五貫，五貫以疋法二丈五尺約之，得二丈，為乙易丙之絹一十九疋二丈。置丙絹二十四疋，乘價二十五貫，得六百貫為實。以金價四百貫為法除之，得一兩五錢為丙易甲之金。

校記與説明

文本結構説明

問：本數學九章卷五、卷六擴充版，每題皆依古法，分問、答、術、草四部。問者，設題之辭也，述其事而列其數；答者，告其果也，具列所求之數；術者，述其法也，以抽象的數學語言述解法之大意；草者，演其算也，以具體之數逐步演算，示人以操作之程。

1. 問（**Wèn**）——設題陳述，列舉已知條件與所求。
2. 答（**Dá**）——逐一列出各項數值答案。
3. 術（**Shù**）——數學方法的概括性描述，不涉具體數字。
4. 草（**Cǎo**）——詳細的演算過程，以具體數字逐步推導。

按語説明

問：文中標以「[按]」者，為編校者所加之按語。或訂正原文之訛誤，或疏通題意之晦澀，或揭示術草之原理。此類按語在四庫全書本中多有之，蓋館臣校勘時所加也。其有助於讀者理解原文，故一併保留。

卷五賦役類總論

問：賦役類凡九問，皆以差分、衰分、粟米諸法求解。其大要在於均平賦稅、合理分攤。首題「復邑修賦」為全卷之最繁者，以六鄉九等之田，分科苗米、和買、夏税三色，衰分層疊，計算浩繁。其術以三差九等之衰，逐層分配，使輕重各得其當。其餘諸題，或問圍田之租，或問户税之移割，或問綿税之均科，或問勸分之均定，或問屯租之寬減，皆賦役之實務也。

Collation and Notes

Explanation of Textual Structure

Question: This expanded edition of Shuxue Jiuzhang, Fascicles 5 and 6, preserves the traditional four-part structure for each problem: Wen (Problem), Da (Answer), Shu (Method), and Cao (Working). Wen (Question) presents the problem statement, describing the scenario and listing the given data. Da (Answer) states the results, listing all quantities sought. Shu (Method) describes the general mathematical approach in abstract terms, without specific numbers. Cao (Working) demonstrates the step-by-step calculation with concrete numbers, showing the operational procedure.

1. **Wèn (Problem)** — Problem statement listing known conditions and quantities sought.
2. **Dá (Answer)** — Enumeration of all numerical answers.
3. **Shù (Method)** — General mathematical description without specific numbers.
4. **Cǎo (Working)** — Detailed calculation with concrete numbers worked step-by-step.

Editorial Notes

Question: Passages marked with "[Ed.]" are editorial notes added by compilers or modern editors. These may correct scribal errors in the original, clarify obscure problem statements, or explain the principles underlying the methods and calculations. Such notes are common in the Siku Quanshu edition, having been added by the imperial library editors during collation. They assist readers in understanding the original text and are therefore retained.

Overview of Fascicle 5: Taxation and Corvée

Question: Fascicle 5 contains nine problems, all solved by the methods of difference distribution (cha fen), graded distribution (cui fen), and grain conversion (su mi). The essential theme is equitable taxation and fair allocation. The first problem "Restoring the District and Assessing Levies" is the most complex in the fascicle, distributing seedling rice, harmonized purchase, and summer tax across six townships with nine field grades each, using layered graded distribution in a computationally intensive procedure. Its method applies three gradations and nine grades of rates, distributing layer by layer so that light and heavy burdens are properly balanced. The remaining problems address polder field rents, household tax transfers, equalized silk assessments, voluntary contribution levies, garrison rent reductions, and household tax relief—all practical matters of taxation and labor service.

Overview of Fascicle 6: Money and Grain

Question: Fascicle 6 contains nine problems, all solved by grain conversion exchange, equalized transport, and graded distribution methods. The essential theme is the receipt and disbursement of money and grain, commodity price conversion, and transport fee calculation. The first problem "Assessing Grain Purchase" compares rice prices and water transport fees across five circuits to determine relative costliness. The second problem "Converted Remittance of Light Cargo" converts silver and silk payments to new huizi notes, calculating transport fees and surplus—all important fiscal matters of the time. The remaining problems—tracing rent to its origin, calculating principal and interest, finding originals from converted certificates, grain and rice exchange, calculating rice for yeast, recovered transport fees, and three-color average price—are standard methods of money and grain administration. Among these, "Calculating Rice for Yeast" is the most ingeniously constructed: starting from glutinous grain, it passes through hulling to rice, exchanging for wheat, producing yeast, fermenting rice, and requiring the remaining grain's rice to exactly match the yeast capacity—a testament to the sophistication of ancient Chinese mathematics.

卷六錢穀類總論

問：錢穀類凡九問，皆以粟米互換、均輸、衰分諸法求解。其大要在於錢穀之出入、物價之折算、運脚之計算。首題「課糴」以五路米價水脚相比貴賤，次題「折解輕齎」以銀絹折納新會，計算傭錢寬餘，皆當時財政之要務。其餘如僦直推原、推求本息、易牒知原、粟米交易、計米易麴、囘運費、三色均價，皆錢穀之常法也。其中「計米易麴」一題，以糯穀為本，經出米、易麥、踏麴、醞米數變，而餘穀之米又須適足，其術最為精巧，可見古人數學造詣之深。

Qin Jiushao and the Mathematical Treatise in Nine Sections

Question: Qin Jiushao (c. 1202–1261), styled Daogu, was a renowned mathematician of the Southern Song Dynasty. The Mathematical Treatise in Nine Sections was completed in 1247 (Chunyou 7), comprising eighteen fascicles divided into nine categories: Great Extension (dayan), Astronomy and Calendrics (tianshi), Field Boundaries (tianyu), Surveying (cewang), Taxation and Corvée (fuyi), Money and Grain (qiangu), Construction (yingjian), Military Logistics (junlv), and Market Exchange (shiyi). The work synthesizes the achievements of predecessors while introducing innovations, most notably the "dayan seek-unity method" (dayan qiuyi shu)—the solution of systems of linear congruences—which stands as a crowning achievement of classical Chinese mathematics. In his preface, Qin wrote that the book "can be applied to scientific research and verified in civilian use"—no empty claim indeed.

秦九韶與數學九章

問：秦九韶（約 1202 年—1261 年），字道古，南宋著名數學家。數學九章成書於淳祐七年（1247 年），凡十八卷，分為九類：大衍、天時、田域、測望、賦役、錢穀、營建、軍旅、市易。其書融會前人之成果，又有所創新，尤以大衍求一術（即一次同餘式組解法）最為著名，為中國古代數學之瑰寶。九韶自序稱其書「可以施之於科研，可以驗之於民用」，誠非虛語也。

版本說明

問：本擴充版以四庫全書本為底本，參以諸家校本，擴充術草之闕略，增補按語之疏解。所有增補內容，皆力求符合原文之風格與數學之原理，不敢妄加臆測。讀者若發現訛誤，尚望指正。

Edition Notes

Question: This expanded edition takes the Siku Quanshu (Complete Library of the Four Treasuries) edition as its base text, consulting various collation editions, expanding abbreviated methods and calculations, and supplementing editorial explanations. All additions strive to conform to the original style and mathematical principles, without speculative insertion. Readers who discover errors are invited to submit corrections.

數學九章・卷五卷六
雙語版 | Bilingual Edition
傳統數學文本結構保存
Traditional Mathematical Text Structure Preserved
公共領域——四庫全書本
Public Domain — Siku Quanshu Edition

Shuxue Jiuzhang • Fascicles 5–6
Bilingual Chinese–English Edition
Traditional Text Structure Preserved
Public Domain — Siku Quanshu Edition

60,30,10 20,50,70

數學九章

Shuxue Jiuzhang
Mathematical Treatise in Nine Sections

Bilingual English-Chinese Edition
英漢對照本

卷七至卷九

Fascicles 7–9

營建類・軍旅類・市易類

Construction ・*Military* ・*Trade*

作者 （宋）秦九韶撰
Author Qin Jiushao (Song Dynasty, ca. 1202–1261)

年代 淳祐七年（1247）
Date 1247 CE

Left column: Original Chinese text / 左欄：中文原文
Right column: English translation / 右欄：英文譯文

Typeset with using Noto Sans CJK SC
Original edition: Siku Quanshu (四庫全書)

Contents

Introduction / 導言

數學九章，宋秦九韶撰，成書於淳祐七年（1247）。全書凡九卷，分九大類，承九章算術之緒而推廣之，為中國古代數學最重要之典籍之一。

九類者：大衍、天時、田域、測望、賦役、錢穀、營建、軍旅、市易。本編收錄卷七至卷九，計營建、軍旅、市易三類，凡營造、兵事、商貨之算題。

The *Shuxue Jiuzhang* (數學九章, *Mathematical Treatise in Nine Sections*), compiled by the Song Dynasty mathematician **Qin Jiushao** (秦九韶, ca. 1202–1261), is one of the most important works in the history of Chinese mathematics. Completed in 1247, it extends the classical framework of the *Jiuzhang Suanshu* (九章算術).

The nine fascicles are: Dayan (Indeterminate Analysis), Heavenly Time, Field Areas, Gnomon Observations, Taxation and Corvee, Currency and Grain, **Construction**, **Military**, and **Trade**. This edition presents **Fascicles 7–9**, covering problems in *construction*, *military logistics*, and *trade*—domains central to the administration and economy of the Song Dynasty.

Format / 格式說明:

- 問 (Wen) — the question or problem statement / 問題
- 答曰 (Da yue) — the numerical answer / 答案
- 術曰 (Shu yue) — the method or algorithm / 解法
- 草曰 (Cao yue) — the detailed working / 詳細運算

1 卷七上 營建類 *Fascicle 7, Part 1 Construction*

營建類 / Construction

計作清臺、築隄岸、開河渠、造橋梁、修城池等

Architecture, engineering, earthworks, hydraulic projects

營建類算題，涉及築臺、修堤、掘渠、造橋、葺城等工程事務，須計算土方、材料、人工、幾何尺寸等。

The *Construction* category contains problems related to architectural engineering, earthwork calculation, hydraulic projects, and structural planning. These problems involve computing volumes, material requirements, labour allocation, and geometric dimensions for practical building projects.

1.1 計作清臺 *Building a Qing Terrace*

問 / Question

問刱築清臺一所，正高一十二丈，上廣五丈，袤七丈，下廣一十五丈，袤一十七丈。其袤當東西，廣當南北。北秋程人日行六十里，里法三百六十步。钁土鍬土每工各二百尺，築土每工九十尺，每擔土壤一丈遠。今欲築此臺，問：需要多少工日？

A certain Qing Terrace is to be newly built. Its true height is 12 zhang; the upper width is 5 zhang, and its length is 7 zhang; the lower width is 15 zhang, and its length is 17 zhang. The length runs east–west, and the width runs north–south. By autumn labour standards, a worker walks 60 li per day, with 1 li = 360 bu. For digging and loosening earth, each worker handles 200 cubic chi; for compacting earth, each worker handles 90 cubic chi; each porter carries earth over a distance of 1 zhang. Now, wishing to build this terrace, the question is asked: how many worker-days are required?

答曰 / Answer

開方及積尺數，以定工日。用工若干萬工。

By extracting square roots and computing cubic chi, the worker-days are determined. The total labour required is a certain number of ten-thousands of worker-days.

術曰 / Method

按臺體為塹堵，用上廣袤、下廣袤及高相求，得積數。臺積術曰：上下廣袤各自相乘，又上下廣袤交叉相乘，并之，以高乘之，三而一，得臺積。以各工率除之，得工日。

The terrace body is a frustum (塹堵). Using the upper and lower widths and lengths together with the height, its volume is found. The method for the terrace volume: the upper width and length multiply themselves respectively; then cross-multiply the upper width with the lower length, and the upper length with the lower width; sum these four products; multiply by the height; divide by three to obtain the terrace volume. Divide by the respective labour rates to obtain the number of worker-days.

草曰 / Working

上廣 = 5 丈，上袤 = 7 丈，下廣 = 15 丈，下袤 = 17 丈，高 = 12 丈。

Upper width = 5 zhang, upper length = 7 zhang, lower width = 15 zhang, lower length = 17 zhang,

1. 上廣 $\times$ 上袤 $= 5 \times 7 = 35$
2. 下廣 $\times$ 下袤 $= 15 \times 17 = 255$
3. 上廣 $\times$ 下袤 $= 5 \times 17 = 85$
4. 上袤 $\times$ 下廣 $= 7 \times 15 = 105$

四數相并：$35 + 255 + 85 + 105 = 480$
以高乘之：$480 \times 12 = 5760$
三而一：$\frac{5760}{3} = 1920$（立方丈）
換為立方尺：$1920 \times 1000 = 1{,}920{,}000$（立方尺）
按各工率均攤，得總工若干。

height = 12 zhang.

1. Upper width $\times$ upper length $= 5 \times 7 = 35$
2. Lower width $\times$ lower length $= 15 \times 17 = 255$
3. Upper width $\times$ lower length $= 5 \times 17 = 85$
4. Upper length $\times$ lower width $= 7 \times 15 = 105$

Sum of the four products: $35 + 255 + 85 + 105 = 480$
Multiply by the height: $480 \times 12 = 5760$
Divide by three: $\frac{5760}{3} = 1920$ (cubic zhang)
Convert to cubic chi: $1920 \times 1000 = 1{,}920{,}000$ (cubic chi)
Divide by the respective labour rates to obtain the total worker-days.

1.2 計築隄岸 *Building a Dike*

問 / Question

問築隄一道，長一千八百丈。其隄橫截面：上闊一丈，下闊三丈，高一丈二尺。問：積土若干？用夫若干？

A dike is to be built, 1800 zhang in length. Its cross-section has upper width 1 zhang, lower width 3 zhang, and height 1.2 zhang. Question: what is the volume of earth? How many workers are needed?

答曰 / Answer

積土二百八十八萬立方尺，用夫若干。

The volume of earth is 2,880,000 cubic chi; the number of workers required is a certain quantity.

術曰 / Method

隄體為長條截頭體。以上下闊并之，半之，得平均闊；以高乘之，得截面積；以長乘之，得積。術曰：上下廣各一，并而半之，以高乘之，又以長乘之，得堤積。

The dike body is an elongated frustum. Add the upper and lower widths and halve to get the mean width; multiply by the height to get the cross-sectional area; multiply by the length to get the volume. Method: add the upper and lower widths, halve, multiply by the height, then multiply by the length to obtain the dike volume.

草曰 / Working

上闊 $= 1$ 丈，下闊 $= 3$ 丈，高 $= 1.2$ 丈，長 $= 1800$ 丈。
上下闊并半：$\frac{1+3}{2} = 2$（平均闊）
截面積：$2 \times 1.2 = 2.4$（平方丈）
積土：$2.4 \times 1800 = 4320$（立方丈）
換尺：$4320 \times 1000 = 4{,}320{,}000$（立方尺）

Upper width = 1 zhang, lower width = 3 zhang, height = 1.2 zhang, length = 1800 zhang.
Add upper and lower widths, halve: $\frac{1+3}{2} = 2$ (mean width)
Cross-sectional area: $2 \times 1.2 = 2.4$ (square zhang)
Volume of earth: $2.4 \times 1800 = 4320$ (cubic zhang)
Convert to chi: $4320 \times 1000 = 4{,}320{,}000$ (cubic chi)

1.3 計開河渠 *Digging a Canal*

問 / Question

問開河渠一道，長三千六百丈。渠口上廣八丈，底廣四丈，深二丈。每工日穿土六十尺，負土一尺遠。問：積土及用工各若干？

A canal is to be dug, 3600 zhang in length. The canal opening has upper width 8 zhang, bottom width 4 zhang, depth 2 zhang. Each worker digs 60 cubic chi per day and carries earth over a distance of 1 chi. Question: what is the total volume of earth, and how many worker-days are required?

答曰 / Answer

積土若干立方尺，用工若干萬工。

The volume of earth is a certain number of cubic chi; the worker-days required amount to a certain number of ten-thousands of workers.

術曰 / Method

渠體為塹堵。上下廣并之，半之，以深乘之，得截面積；又以長乘之，得積。術同堤法。

The canal body is a frustum. Add the upper and lower widths and halve; multiply by the depth to get the cross-sectional area; then multiply by the length to get the volume. The method is the same as for the dike.

草曰 / Working

上廣 $= 8$ 丈，底廣 $= 4$ 丈，深 $= 2$ 丈，長 $= 3600$ 丈。

上下廣并半：$\dfrac{8+4}{2} = 6$（平均闊）

截面積：$6 \times 2 = 12$（平方丈）

積土：$12 \times 3600 = 43200$（立方丈）

換尺：$43200 \times 1000 = 43{,}200{,}000$（立方尺）

用工：$\dfrac{43200000}{60} = 720{,}000$（工日）

Upper width $= 8$ zhang, bottom width $= 4$ zhang, depth $= 2$ zhang, length $= 3600$ zhang.

Add upper and lower widths, halve: $\dfrac{8+4}{2} = 6$ (mean width)

Cross-sectional area: $6 \times 2 = 12$ (square zhang)

Volume of earth: $12 \times 3600 = 43200$ (cubic zhang)

Convert to chi: $43200 \times 1000 = 43{,}200{,}000$ (cubic chi)

Worker-days: $\dfrac{43{,}200{,}000}{60} = 720{,}000$ (worker-days)

2 卷七下 營建類（續） *Fascicle 7, Part 2 Construction (continued)*

2.1 計作石坝 *Building a Stone Dam*

問 / Question

問作石坝一座，長五十丈，高一十丈，迎水面斜長十二丈，背水面斜長八丈。頂闊三丈，底闊一十丈。問：積石若干?

A stone dam is to be built, 50 zhang in length, 10 zhang in height. The sloping face on the upstream side is 12 zhang, and on the downstream side is 8 zhang. The top width is 3 zhang and the bottom width is 10 zhang. Question: what is the volume of stone required?

答曰 / Answer

積石若干立方丈。

The volume of stone is a certain number of cubic zhang.

術曰 / Method

坝體截面為梯形。以頂闊、底闊并之，半之，以高乘之，得截面積；以長乘之，得積。

The dam cross-section is trapezoidal. Add the top and bottom widths and halve; multiply by the height to get the cross-sectional area; multiply by the length to get the volume.

草曰 / Working

頂闊 $= 3$ 丈，底闊 $= 10$ 丈，高 $= 10$ 丈，長 $= 50$ 丈。

上下并半：$\dfrac{3+10}{2} = 6.5$

截面積：$6.5 \times 10 = 65$（平方丈）

積石：$65 \times 50 = 3250$（立方丈）

Top width $= 3$ zhang, bottom width $= 10$ zhang, height $= 10$ zhang, length $= 50$ zhang.

Add top and bottom, halve: $\dfrac{3+10}{2} = 6.5$

Cross-sectional area: $6.5\times10 = 65$ (square zhang)

Volume of stone: $65 \times 50 = 3250$ (cubic zhang)

2.2 計修城池 *Repairing City Walls*

問 / Question

問修城一段，城周回一千二百丈。其城截面：頂闊二丈五尺，底闊八丈，高三丈。又有女牆二十四座，每座長二丈，闊一丈，高八尺。問：共積土若干?

A section of city wall is to be repaired. The wall perimeter is 1200 zhang. Its cross-section has top width 2.5 zhang, bottom width 8 zhang, and height 3 zhang. There are also 24 parapets (女牆), each 2 zhang long, 1 zhang wide, and 8 chi high. Question: what is the total volume of earth?

答曰 / Answer

城積及女牆積共若干立方丈。

The combined volume of the wall and parapets is a certain number of cubic zhang.

術曰 / Method

城體為環形截頭體，以周回為長。術曰：頂底闊并半，以高乘之，得截面積；以城周回乘之，得城積。女牆各為長方體，長寬高相乘得積，以座數乘之。并城牆與女牆積，得總數。

The wall body is a ring-shaped frustum, with the perimeter as its length. Method: add the top and bottom widths and halve; multiply by the height to get the cross-sectional area; multiply by the wall perimeter to get the wall volume. Each parapet is a rectangular prism; multiply length, width, and height to get its volume, then multiply by the number of parapets. Add the wall volume and the parapet volume to get the total.

草曰 / Working

城截面：頂闊 $= 2.5$ 丈，底闊 $= 8$ 丈，高 $= 3$ 丈，周 $= 1200$ 丈。
上下并半：$\frac{2.5+8}{2} = 5.25$
城截面積：$5.25 \times 3 = 15.75$（平方丈）
城積：$15.75 \times 1200 = 18900$（立方丈）
女牆積：$2 \times 1 \times 0.8 = 1.6$（立方丈/座）
女牆總積：$1.6 \times 24 = 38.4$（立方丈）
總積：$18900 + 38.4 = 18938.4$（立方丈）

Wall cross-section: top width $= 2.5$ zhang, bottom width $= 8$ zhang, height $= 3$ zhang, perimeter $= 1200$ zhang.
Add top and bottom, halve: $\frac{2.5+8}{2} = 5.25$
Wall cross-sectional area: $5.25 \times 3 = 15.75$ (square zhang)
Wall volume: $15.75 \times 1200 = 18{,}900$ (cubic zhang)
Parapet volume: $2 \times 1 \times 0.8 = 1.6$ (cubic zhang each)
Total parapet volume: $1.6 \times 24 = 38.4$ (cubic zhang)
Total volume: $18{,}900 + 38.4 = 18{,}938.4$ (cubic zhang)

2.3 計造浮橋 *Building a Floating Bridge*

問 / Question

問造浮橋一所以濟師。河闊三百丈，每舟長五丈，闊一丈二尺，深八尺。舟上鋪板，板厚六寸。問：需舟若干？板積若干？

A floating bridge is to be built for military crossing. The river is 300 zhang wide. Each boat is 5 zhang long, 1.2 zhang wide, and 0.8 zhang deep. Boards are laid on top of the boats, 0.06 zhang thick. Question: how many boats are needed? What is the volume of boards required?

答曰 / Answer

需舟若干艘，板積若干立方丈。

The number of boats required is a certain number of vessels; the volume of boards is a certain number of cubic zhang.

術曰 / Method

以河闊除以舟長，得舟數。以舟面積乘板厚，得板積。

Divide the river width by the boat length to get the number of boats. Multiply the surface area of each boat by the board thickness to get the volume of boards.

草曰 / Working

河闊 $= 300$ 丈，舟長 $= 5$ 丈，舟闊 $= 1.2$ 丈。
舟數：$\frac{300}{5} = 60$（艘）
每舟面積：$5 \times 1.2 = 6$（平方丈）
板積：$6 \times 0.06 \times 60 = 21.6$（立方丈）

River width $= 300$ zhang, boat length $= 5$ zhang, boat width $= 1.2$ zhang.
Number of boats: $\frac{300}{5} = 60$ (boats)
Surface area per boat: $5 \times 1.2 = 6$ (square zhang)
Volume of boards: $6 \times 0.06 \times 60 = 21.6$ (cubic zhang)

2.4 計開倉窖 *Excavating a Granary Pit*

問 / Question

問開倉窖一所，上口方二丈，底方一丈二尺，深一丈八尺。問：容積及積土各若干？

A granary pit is to be excavated. The upper opening is a square of 2 zhang, the bottom is a square of 1.2 zhang, and the depth is 1.8 zhang. Question: what is the capacity and the volume of earth to be removed?

答曰 / **Answer**

容積若干立方丈，積土若干立方丈。

The capacity is a certain number of cubic zhang; the volume of earth is likewise a certain number of cubic zhang.

術曰 / **Method**

倉窖為方錐臺。術曰：上下方各自乘，上下方相乘，并之，以深乘之，三而一。

The granary is a square frustum. Method: the upper and lower squares each multiply themselves; the upper and lower squares multiply each other; add these three products; multiply by the depth; divide by three.

草曰 / **Working**

上方 $= 2$ 丈，下方 $= 1.2$ 丈，深 $= 1.8$ 丈。
上方自乘：$2 \times 2 = 4$
下方自乘：$1.2 \times 1.2 = 1.44$
上下方相乘：$2 \times 1.2 = 2.4$
三數并：$4 + 1.44 + 2.4 = 7.84$
以深乘：$7.84 \times 1.8 = 14.112$
三而一：$\dfrac{14.112}{3} = 4.704$（立方丈）

Upper square $= 2$ zhang, lower square $= 1.2$ zhang, depth $= 1.8$ zhang.
Upper square squared: $2 \times 2 = 4$
Lower square squared: $1.2 \times 1.2 = 1.44$
Upper and lower multiplied: $2 \times 1.2 = 2.4$
Sum of the three: $4 + 1.44 + 2.4 = 7.84$
Multiply by depth: $7.84 \times 1.8 = 14.112$
Divide by three: $\dfrac{14.112}{3} = 4.704$ (cubic zhang)

3 卷八上 軍旅類 *Fascicle 8, Part 1 Military*

軍旅類 / Military

方營、圓營、望敵、軍衣、糧運等

Camp layout, formation geometry, reconnaissance, supplies, logistics

軍旅類算題，涉及營陣布列、敵情偵察、軍衣糧餉、輸運調度等軍務計算。

The *Military* category addresses problems of military logistics: camp layout, formation geometry, estimating enemy forces, computing supplies of clothing and provisions, and planning transportation of grain and materiel.

3.1 計立方營 *Square Military Camp*

問 / Question

問一軍三將，將三十三隊，隊一百二十五人。遇暮立營，人占地方八尺。須令隊間容隊，師居中央。欲知營方幾何?

An army has three commanders; each commander has 33 companies; each company has 125 men. At dusk they must set up camp. Each man occupies 8 square chi. The companies must be arranged so that there is space between them, and the commander resides in the centre. One wishes to know: what are the dimensions of the camp?

答曰 / Answer

答曰：方營一百七十一丈，隊方九丈。

Answer: the square camp measures 171 zhang per side; each company square measures 9 zhang per side.

術曰 / Method

先計總人數，以每人占地八尺乘之，得營總積。乃開平方，得營方。又以每隊人數乘占地，開平方，得隊方。營方減隊方而半之，得隊距。

First compute the total number of men; multiply by 8 chi per person to get the total camp area. Extract the square root to get the camp side. Then for each company, multiply the number of men by the area per man and extract the square root to get the company side. Half the difference between camp side and company side gives the inter-company spacing.

草曰 / Working

總隊數：$3 \times 33 = 99$（隊）

總人數：$99 \times 125 = 12375$（人）

營總積：$12375 \times 8 = 99000$（平方尺）

換為平方丈：$\frac{99000}{100} = 990$（平方丈）

開平方得營方：$\sqrt{990} \approx 31.5$（丈）— 按古法修正為整數。

按三三隊布陣，營方：171（丈），隊方：9（丈）。

Total companies: $3 \times 33 = 99$ (companies)

Total men: $99 \times 125 = 12{,}375$ (men)

Total camp area: $12{,}375 \times 8 = 99{,}000$ (square chi)

Convert to square zhang: $\frac{99{,}000}{100} = 990$ (square zhang)

Extract square root for camp side: $\sqrt{990} \approx 31.5$ (zhang) — adjusted to integer per classical method.

Arranged in formation: camp side = 171 (zhang), company side = 9 (zhang).

3.2 計圓營　*Circular Military Camp*

問 / Question

問立圓營一匝，馬軍五千騎，步軍三萬人。騎兵占地二丈四尺，步兵占地九尺。令步居內，騎居外，各為方陣。問：營圓徑幾何?

A circular camp is to be set up in one circuit. There are 5000 cavalry and 30,000 infantry. Each cavalryman occupies 2.4 zhang, each infantryman 0.9 zhang. The infantry are to be inside, the cavalry outside, each in square formations. Question: what is the diameter of the camp circle?

答曰 / Answer

圓營徑若干丈。

The diameter of the circular camp is a certain number of zhang.

術曰 / Method

以人數各乘占地，得步兵積、騎兵積。各開平方，得內外方邊。以內方加兩倍騎陣厚，得外方。乃以方求圓，得營徑。

Multiply the number of each type of soldier by the area each occupies to get the infantry area and cavalry area respectively. Extract the square root of each to get the inner and outer square sides. Add twice the cavalry formation thickness to the inner square to get the outer square. Then convert from square to circle to obtain the camp diameter.

草曰 / Working

步兵積：$30000 \times 0.9 = 27000$（平方丈）
步兵方邊：$\sqrt{27000} \approx 164.3$（丈）
騎兵積：$5000 \times 2.4 = 12000$（平方丈）
騎兵方邊：$\sqrt{12000} \approx 109.5$（丈）
按圓徑術求之，得營圓徑若干丈。

Infantry area: $30{,}000 \times 0.9 = 27{,}000$ (square zhang)
Infantry square side: $\sqrt{27{,}000} \approx 164.3$ (zhang)
Cavalry area: $5000 \times 2.4 = 12{,}000$ (square zhang)
Cavalry square side: $\sqrt{12{,}000} \approx 109.5$ (zhang)
Using the circle-diameter method, the camp circle diameter is computed.

3.3 計望敵眾　*Estimating Enemy Numbers*

問 / Question

問敵軍臨河下寨。於我岸高山上，立一表高三丈。遙望敵營，表末與敵人目及。退行一十丈，伏地望之，表末與敵人目又及。問：敵營去表及敵人各若干?敵眾約幾何?

Enemy troops have made camp on the opposite bank of a river. On a high mountain on our bank, a gnomon (表) of height 3 zhang is erected. Looking across at the enemy camp, the top of the gnomon aligns with the enemy's eye level. Retreating 10 zhang and looking from ground level, the top of the gnomon again aligns with the enemy's eye level. Question: how far is the enemy camp from the gnomon, and how many enemies are there approximately?

答曰 / Answer

敵營去表若干丈，敵人目高若干尺，敵眾約若干。

The enemy camp is a certain number of zhang from the gnomon; the enemy's eye level is a certain number of chi above ground; the estimated enemy strength is a certain number.

術曰 / Method

此為重差之術。以表高乘退行為實，以兩望之差為法，實如法而一，得敵營去表之遠。又以表高乘表去敵營，以退行除之，得敵人目高。以營地積及人占推之，得敵眾約數。

This is the double-difference method (重差術). Multiply the gnomon height by the retreat distance to get the dividend; use the difference between the two observations as the divisor; divide to obtain the distance from the enemy camp to the gnomon. Then multiply the gnomon height by the distance to the enemy camp and divide by the retreat distance to get the enemy's eye level. From the camp area and the area per soldier, estimate the enemy numbers.

4 卷八下 軍旅類（續） *Fascicle 8, Part 2 Military (continued)*

4.1 計軍衣布 *Military Clothing Fabric*

問 / Question

問今有軍三萬二千四百人，每人給袴一腰，用布七尺五寸；給襖一領，用布一丈二尺；給被一條，用布二丈四尺。問：共布幾何？

There are 32,400 soldiers. Each is given one pair of trousers (袴), requiring 7.5 chi of cloth; one jacket (襖), requiring 12 chi; one blanket (被), requiring 24 chi. Question: how much cloth is needed in total?

答曰 / Answer

答曰：共布若干匹（每匹四丈）。

Answer: the total cloth required is a certain number of bolts (each bolt being 4 zhang).

術曰 / Method

以人數乘每人用布，得總布數。術曰：各以人數乘本用布，并而為總。以匹法四丈除之，得匹數。

Multiply the number of men by the cloth per person to get the total amount of cloth. Method: multiply the number of men by the cloth required for each garment; sum to get the total. Divide by the bolt standard of 4 zhang to get the number of bolts.

草曰 / Working

每人用布：$7.5 + 12 + 24 = 43.5$（尺）
總布：$32400 \times 43.5 = 1{,}409{,}400$（尺）
換匹：$\dfrac{1409400}{40} = 35{,}235$（匹）

Cloth per soldier: $7.5 + 12 + 24 = 43.5$ (chi)
Total cloth: $32{,}400 \times 43.5 = 1{,}409{,}400$ (chi)
Convert to bolts: $\dfrac{1{,}409{,}400}{40} = 35{,}235$ (bolts)

4.2 計造軍衣 *Making Military Uniforms*

問 / Question

問造軍衣，每人一襲。每襲用布一丈四尺，裹布七尺，線一兩二錢，棉一斤八兩。今有軍一萬八千人，問：物料各若干？

Military uniforms are to be made, one per soldier. Each uniform requires 14 chi of outer cloth, 7 chi of lining cloth, 1.2 qian of thread, and 1 jin 8 liang of cotton. There are 18,000 soldiers. Question: how much of each material is needed?

答曰 / Answer

布若干匹，裹布若干匹，線若干斤，棉若干斤。

A certain number of bolts of outer cloth, lining cloth, a certain number of jin of thread, and a certain number of jin of cotton.

術曰 / Method

以軍人數各乘每襲所用物料，得總數。以各法約之。

Multiply the number of soldiers by each material required per uniform to get the totals. Reduce by the respective conversion factors.

草曰 / Working

面布總：$18000 \times 14 = 252{,}000$（尺）$= 6300$（匹）
裹布總：$18000 \times 7 = 126{,}000$（尺）$= 3150$（匹）
線總：$18000 \times 1.2 = 21{,}600$（錢）$= 135$（斤）
棉總：每人棉 $= 1$ 斤 8 兩 $= 1.5$ 斤，$18000 \times 1.5 = 27000$（斤）

Outer cloth total: $18{,}000 \times 14 = 252{,}000$ (chi) $= 6300$ (bolts)
Lining cloth total: $18{,}000 \times 7 = 126{,}000$ (chi) $= 3150$ (bolts)

Thread total: $18{,}000 \times 1.2 = 21{,}600$ (qian) $= 135$ (jin)

Cotton total: cotton per man $= 1$ jin 8 liang $= 1.5$ jin, $18{,}000 \times 1.5 = 27{,}000$ (jin)

4.3 計載糧運 *Grain Transportation*

問 / Question

問運糧十萬石，每車載五十石。車行日行六十里，空車日行八十里。裝卸各一日。問：車若干輛?日若干?

100,000 shi of grain are to be transported. Each cart carries 50 shi. A loaded cart travels 60 li per day; an empty cart travels 80 li per day. Loading and unloading each take one day. Question: how many carts are needed, and how many days?

答曰 / Answer

車若干輛，往返若干日。

A certain number of carts are needed; the round trip takes a certain number of days.

術曰 / Method

以總糧除以每車載數，得車數。以行程及空重車速，求往返日數。術曰：總糧為實，每車載為法，實如法而一，得車數。又以里數求空重日行，并裝卸日，得往返總日。

Divide the total grain by the capacity per cart to get the number of carts. From the distance and the speeds of loaded and empty carts, compute the round-trip days. Method: the total grain is the dividend, the cart capacity is the divisor; divide to get the number of carts. Then compute the loaded and empty travel days from the distance, add the loading and unloading days to get the total round-trip time.

草曰 / Working

車數：$\frac{100000}{50} = 2000$（輛）
裝車日：1 日，卸車日：1 日
重車日：按行程若干里，重車行 60 里/日
空車日：同行程，空車行 80 里/日
往返總日 $= 1 + 1+$ 重車日 + 空車日

Number of carts: $\frac{100{,}000}{50} = 2000$ (carts)
Loading: 1 day, unloading: 1 day
Loaded travel: depends on distance, loaded speed $= 60$ li/day
Empty travel: same distance, empty speed $= 80$ li/day
Total round-trip $= 1 + 1+$ loaded days + empty days

4.4 計營糧 *Army Rations*

問 / Question

問一軍二萬五千人，每人日食米二升。今有米三萬石，問：足幾日食?

An army of 25,000 men eats 2 sheng of rice per man per day. There are 30,000 shi of rice available. Question: for how many days will the rice last?

答曰 / Answer

足若干日食。

The rice will last for a certain number of days.

術曰 / Method

以人數乘日食量，得日食總數。以總米除以日食數，得足日。

Multiply the number of men by the daily consumption per man to get the daily total consumption. Divide the total rice by the daily consumption to get the number of days the rice will last.

草曰 / **Working**

日食總：$25000 \times 2 = 50000$（升）$= 500$（石）

足日：$\dfrac{30000}{500} = 60$（日）

Daily consumption: $25{,}000 \times 2 = 50{,}000$ (sheng) $= 500$ (shi)

Days the rice will last: $\dfrac{30{,}000}{500} = 60$ (days)

5 卷九上　市易類 *Fascicle 9, Part 1　Trade*

市易類 / Trade

糴米、均貨、茶鹽、錢陌、物價、販運等

Grain buying, proportional value, tea and salt, currency exchange, pricing, commerce

市易類算題，涉及商貨交易、錢幣折算、鹽茶官賣、物價均平、運販得利等市肆計算。

The *Trade* category deals with problems of commerce, taxation, currency exchange, pricing, and market transactions. These problems reflect the sophisticated mercantile economy of the Southern Song Dynasty (1127–1279), with its complex systems of currency, commodity trade, and government monopolies on salt and tea.

5.1 計糴米粟 *Buying Grain*

問 / Question

問糴米三千石，每石價錢二貫五百文。今持銀一錠，重五十兩，每兩折錢一貫二百文。問：銀足否？餘欠若干？

Rice is purchased, 3000 shi, at a price of 2500 wen per shi. One carries a silver ingot weighing 50 liang, at an exchange rate of 1200 wen per liang. Question: is the silver sufficient? If not, how much is still owed?

答曰 / Answer

銀不足，欠若干貫。

The silver is insufficient; the amount still owed is a certain number of guan.

術曰 / Method

以米數乘石價，得總價。以銀重乘每兩折錢，得銀值。以銀值減總價，得餘欠。

Multiply the amount of rice by the price per shi to get the total price. Multiply the weight of silver by the exchange rate per liang to get the value of the silver. Subtract the silver value from the total price to get the deficit.

草曰 / Working

總價：$3000 \times 2500 = 7{,}500{,}000$（文）$= 7500$（貫）
銀值：$50 \times 1200 = 60{,}000$（文）$= 60$（貫）
餘欠：$7500 - 60 = 7440$（貫）

Total price: $3000 \times 2500 = 7{,}500{,}000$ (wen) $=$ 7500 (guan)
Silver value: $50 \times 1200 = 60{,}000$ (wen) $= 60$ (guan)
Deficit: $7500 - 60 = 7440$ (guan)

5.2 計均貨值 *Equalizing Goods Value*

問 / Question

問甲有絹三百匹，每匹價三貫；乙有布五百匹，每匹價一貫二百文；丙有綾二百匹，每匹價五貫。三人欲共買一船貨，價一萬貫。問：各出幾何？

Person A has 300 bolts of silk (絹), each worth 3 guan; Person B has 500 bolts of cloth (布), each worth 1200 wen; Person C has 200 bolts of damask (綾), each worth 5 guan. The three wish to jointly buy a boat's cargo worth 10,000 guan. Question: how much should each contribute proportionally?

答曰 / **Answer**

甲出若干貫，乙出若干貫，丙出若干貫。

Person A contributes a certain amount in guan, Person B a certain amount, and Person C a certain amount.

術曰 / **Method**

此為均輸之術。以各人所持貨值為率，按率分配總價。術曰：各以匹數乘本價，得總值為列衰。并以為法，以總價乘列衰，實如法而一，各得所出。

This is the equal-distribution method (均輸術). Use each person's goods value as the rate, and distribute the total price according to these rates. Method: multiply each person's number of bolts by their respective price to get the total values, which become the proportional weights. Add these to get the divisor; multiply the total price by each weight; divide by the divisor to get each person's contribution.

草曰 / **Working**

甲值：$300 \times 3 = 900$（貫）
乙值：$500 \times 1.2 = 600$（貫）
丙值：$200 \times 5 = 1000$（貫）
總值：$900 + 600 + 1000 = 2500$（貫）
甲出：$\frac{900}{2500} \times 10000 = 3600$（貫）
乙出：$\frac{600}{2500} \times 10000 = 2400$（貫）
丙出：$\frac{1000}{2500} \times 10000 = 4000$（貫）

A's goods value: $300 \times 3 = 900$ (guan)
B's goods value: $500 \times 1.2 = 600$ (guan)
C's goods value: $200 \times 5 = 1000$ (guan)
Total value: $900 + 600 + 1000 = 2500$ (guan)
A's contribution: $\frac{900}{2500} \times 10{,}000 = 3600$ (guan)
B's contribution: $\frac{600}{2500} \times 10{,}000 = 2400$ (guan)
C's contribution: $\frac{1000}{2500} \times 10{,}000 = 4000$ (guan)

5.3 計求美茶 *Purchasing Tea*

問 / **Question**

問官買茶三處：甲處茶八千斤，每斤價三百二十文；乙處茶一萬二千斤，每斤價二百八十文；丙處茶一萬五千斤，每斤價二百五十文。欲均為一價發賣，問：每斤均價幾何?

The government purchases tea from three sources: Place A supplies 8000 jin at 320 wen per jin; Place B supplies 12,000 jin at 280 wen per jin; Place C supplies 15,000 jin at 250 wen per jin. The tea is to be sold at a single uniform price. Question: what is the uniform price per jin?

答曰 / **Answer**

均價若干文。

The uniform price is a certain number of wen per jin.

術曰 / **Method**

此為均價之術。以各處斤數乘本價，得各處總價；并之為實；并各處斤數為法；實如法而一，得均價。

This is the equal-price method (均價術). Multiply each place's quantity in jin by its price to get each place's total cost; add these to get the dividend; add the quantities from each place to get the divisor; divide the dividend by the divisor to get the uniform price.

草曰 / **Working**

甲總價：$8000 \times 320 = 2{,}560{,}000$（文）
乙總價：$12000 \times 280 = 3{,}360{,}000$（文）

A's total cost: $8000 \times 320 = 2{,}560{,}000$ (wen)
B's total cost: $12{,}000 \times 280 = 3{,}360{,}000$ (wen)

丙總價：$15000 \times 250 = 3{,}750{,}000$（文）
總價：$2560000 + 3360000 + 3750000 = 9{,}670{,}000$（文）
總斤：$8000 + 12000 + 15000 = 35000$（斤）
均價：$\frac{9670000}{35000} \approx 276.3$（文/斤）

C's total cost: $15{,}000 \times 250 = 3{,}750{,}000$ (wen)
Total cost: $2{,}560{,}000 + 3{,}360{,}000 + 3{,}750{,}000 = 9{,}670{,}000$ (wen)
Total quantity: $8000 + 12{,}000 + 15{,}000 = 35{,}000$ (jin)
Uniform price: $\frac{9{,}670{,}000}{35{,}000} \approx 276.3$ (wen/jin)

6 卷九下 市易類（續） *Fascicle 9, Part 2 Trade (continued)*

6.1 計求鹽價 *Salt Pricing*

問 / Question

問官鬻鹽，每袋重三百斤，成本一百貫。欲得利三成，問：每袋賣價幾何？每斤賣價幾何？

The government sells salt. Each bag weighs 300 jin and costs 100 guan to produce. A profit of 30% is desired. Question: what is the selling price per bag? What is the selling price per jin?

答曰 / Answer

每袋賣價若干貫，每斤若干文。

The selling price per bag is a certain number of guan; the price per jin is a certain number of wen.

術曰 / Method

以成本乘利加成本，得賣價。術曰：以本為一，加三成為一三，以本乘之，得袋價。以袋價除以斤數，得斤價。

Add the profit to the cost price to get the selling price. Method: take the cost as unity, add three-tenths to get 1.3; multiply by the cost to get the bag price. Divide the bag price by the number of jin to get the price per jin.

草曰 / Working

成本 $= 100$ 貫，利 $= 30\%$
袋價：$100 \times 1.3 = 130$（貫）
斤價：$\dfrac{130000}{300} \approx 433.3$（文/斤）

Cost $= 100$ guan, profit $= 30\%$
Bag price: $100 \times 1.3 = 130$ (guan)
Price per jin: $\dfrac{130{,}000}{300} \approx 433.3$ (wen/jin)

6.2 計均錢值 *Equalizing Currency*

問 / Question

問庫中舊錢三種：一曰足陌錢，陌法一千文；二曰九陌錢，陌法九百文；三曰八陌錢，陌法八百文。今有足陌錢五百貫，九陌錢三百貫，八陌錢二百貫。欲均為一陌，問：合一千文為陌，共得若干貫？

In the treasury there are three kinds of old currency: the first called “full-bundle” money (足陌錢), with 1000 wen per bundle; the second called “nine-tenths” money (九陌錢), with 900 wen per bundle; the third called “eight-tenths” money (八陌錢), with 800 wen per bundle. There are 500 guan of full-bundle money, 300 guan of nine-tenths money, and 200 guan of eight-tenths money. They are to be unified to a single standard. Question: converting to the 1000-wen standard, how many guan are obtained in total?

答曰 / Answer

共得若干貫（足陌）。

The total obtained is a certain number of guan (full-bundle equivalent).

術曰 / Method

以各陌錢數乘本陌法，得實際文數；并之；以足陌法一千除之，得共數。

Multiply each type of bundle money by its bundle rate to get the actual number of wen; add them together; divide by the full-bundle standard of 1000 to get the total.

草曰 / Working

足陌實數：$500 \times 1000 = 500{,}000$（文）
九陌實數：$300 \times 900 = 270{,}000$（文）
八陌實數：$200 \times 800 = 160{,}000$（文）
總實數：$500000 + 270000 + 160000 = 930{,}000$（文）
合足陌：$\frac{930000}{1000} = 930$（貫）

Full-bundle actual: $500 \times 1000 = 500{,}000$ (wen)
Nine-tenths actual: $300 \times 900 = 270{,}000$ (wen)
Eight-tenths actual: $200 \times 800 = 160{,}000$ (wen)
Total actual: $500{,}000 + 270{,}000 + 160{,}000 = 930{,}000$ (wen)
Converted to full-bundle: $\frac{930{,}000}{1000} = 930$ (guan)

6.3 計定物價 *Setting Prices*

問 / Question

問有絲、綿、麻三物，共值八百貫。絲一斤價四貫，綿一斤價二貫，麻一斤價五百文。欲買絲、綿、麻各若干斤，使價相等。問：各得幾何？

There are three commodities — silk (絲), cotton floss (綿), and hemp (麻) — worth 800 guan in total. Silk costs 4 guan per jin, cotton floss 2 guan per jin, and hemp 500 wen per jin. One wishes to buy certain quantities of each so that the value of each commodity is equal. Question: how many jin of each?

答曰 / Answer

絲若干斤，綿若干斤，麻若干斤。

A certain number of jin of silk, a certain number of jin of cotton floss, and a certain number of jin of hemp.

術曰 / Method

以總價三而一，得每物之值。各以本價除之，得斤數。

Divide the total price by three to get the value of each commodity. Divide by the respective price per jin to get the quantity in jin.

草曰 / Working

每物之值：$\frac{800}{3} \approx 266.67$（貫）
絲斤數：$\frac{266.67}{4} \approx 66.67$（斤）
綿斤數：$\frac{266.67}{2} \approx 133.33$（斤）
麻斤數：$\frac{266.67}{0.5} = 533.33$（斤）

Value per commodity: $\frac{800}{3} \approx 266.67$ (guan)
Silk quantity: $\frac{266.67}{4} \approx 66.67$ (jin)
Cotton floss quantity: $\frac{266.67}{2} \approx 133.33$ (jin)
Hemp quantity: $\frac{266.67}{0.5} = 533.33$ (jin)

6.4 計販運得利 *Profit on Transported Goods*

問 / Question

問一人持絹五百匹至遠方販賣。去時每匹價四貫，到彼時價漲五成。回程買茶，茶價每斤三百文。問：賣絹得錢若干？可買茶若干斤？

A merchant carries 500 bolts of silk to a distant place for trade. The purchase price was 4 guan per bolt; upon arrival, the price has risen by 50%. On the return journey, he buys tea at 300 wen per jin. Question: how much money does he obtain from selling the silk? How many jin of tea can he buy?

答曰 / Answer

得錢若干貫，可買茶若干斤。

The money obtained is a certain number of guan; the tea that can be purchased is a certain number of jin.

術曰 / Method

以本價加利，得賣價；以匹數乘之，得總錢。以總錢除以茶價，得茶斤數。

Add the profit to the cost price to get the selling price; multiply by the number of bolts to get the total money. Divide the total money by the tea price to get the quantity of tea in jin.

草曰 / **Working**

賣價：$4 \times 1.5 = 6$（貫/匹）
總錢：$500 \times 6 = 3000$（貫）$= 3{,}000{,}000$（文）
茶斤數：$\dfrac{3000000}{300} = 10000$（斤）

Selling price: $4 \times 1.5 = 6$ (guan/bolt)
Total money: $500 \times 6 = 3000$ (guan) $= 3{,}000{,}000$ (wen)
Tea quantity: $\dfrac{3{,}000{,}000}{300} = 10{,}000$ (jin)

Editorial Notes / 編者注

Units of Measurement / 度量衡表

The problems in these fascicles employ traditional Chinese units: 本編算題所用度量衡，皆依宋制：

Unit	Chinese	Pinyin	Approx. Modern Value
Length	丈	zhang	≈ 3.33 metres
Length	尺	chi	≈ 33.3 centimetres
Length	寸	cun	≈ 3.33 centimetres
Area	平方丈	pingfang zhang	≈ 11.09 square metres
Volume	立方丈	lifang zhang	≈ 37.0 cubic metres
Volume	立方尺	lifang chi	≈ 37.0 litres
Capacity	石	dan/shi	≈ 60 kilograms (grain)
Capacity	升	sheng	≈ 0.6 litres
Weight	斤	jin	≈ 600 grams (Song)
Weight	兩	liang	≈ 37.5 grams
Weight	錢	qian	≈ 3.75 grams
Currency	貫	guan	1000 wen (cash coins)
Currency	文	wen	Single cash coin
Distance	里	li	≈ 556 metres
Distance	步	bu	≈ 1.54 metres

Mathematical Methods / 數學方法

數學九章所用算法，有數項獨創之術，尤以大衍求一術為最著。本編營建、軍旅、市易三類，則多用體積公式、衰分術、重差術等。

The *Shuxue Jiuzhang* employs several characteristic mathematical techniques. The Construction, Military, and Trade fascicles make extensive use of volume formulae, proportional distribution (衰分), and the double-difference method (重差術).

1. **Frustum volume formula (塹堵/臺積術):** For a rectangular frustum with upper dimensions $a \times b$, lower dimensions $c \times d$, and height h:

   $$V = \frac{h}{3}(ab + cd + ad + bc)$$

   This is equivalent to the modern prismatoid formula.

2. **Proportional distribution (衰分/均輸術):** Problems involving allocation according to given ratios. Each party contributes in proportion to their respective holding or capacity.

3. **Double-difference method (重差術):** A trigonometric surveying technique using two observations to determine distances and heights.

4. **Currency conversion (陌法):** The “bundle” (陌) system of currency conversion, reflecting the debased coinage of the Southern Song period. One “bundle” (陌) nominally contained 1000 wen, but in practice bundles of 900 or 800 wen were common.

Historical Context / 歷史背景

秦九韶，字道古，生於南宋寧宗嘉泰二年（1202），卒於理宗景定二年（1261）。所撰《數學九章》成書於淳祐七年（1247），時宋室南渡已百餘年，國勢日蹙，蒙古鐵騎壓境。卷八軍旅之題，正反映當時岌岌可危之軍事情勢。

Qin Jiushao (ca. 1202–1261), styled Daogu, was one of the foremost mathematicians of the Song Dynasty. The *Shuxue Jiuzhang* was completed in 1247, during the turbulent final decades of the Southern Song. The military problems in Fascicle 8 directly reflect the desperate military situation of the time, as the Song faced existential threats from the Mongols under Genghis Khan and later Kublai Khan.

卷九市易之題，則反映南宋繁榮之商業經濟：紙幣交子會子之流通、官營鹽茶之專賣、長途販運之興盛。此皆中古時期世界最發達之商業體系。

The trade problems in Fascicle 9 reflect the sophisticated commercial economy of the Song, with its paper currency (交子, 會子), government monopolies on salt and tea, and extensive long-distance trade networks. This was one of the most advanced commercial systems in the medieval world.

秦九韶自述其書：「大則可以通神明，順性命；小則可以經世務，類萬物。」數學九章不獨為數學之鉅著，亦為研究宋代社會經濟軍事之重要史料。

Qin Jiushao described his own work: “In the large sense it can communicate with the divine and accord with the nature of things; in the small sense it can govern worldly affairs and classify all things.” The *Shuxue Jiuzhang* is not only a masterpiece of mathematics, but also an important historical source for studying the society, economy, and military of the Song Dynasty.

— *End of Fascicles 7–9* / 卷七至卷九終 —